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MASTER THESIS

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**Pretentious approach to analytic
number theory**

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Title: Pretentious approach to analytic number theory

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Abstract: The goal of this thesis is to present the pretentious approach to analytic number theory recently developed by Granville, Soundararajan, and others. In the first four chapters, we show the classical proof of the prime number theorem. We then develop the pretentious approach, explain its differences, advantages, and disadvantages and present another proof of the prime number theorem based on Halász's theorem. This theorem is then proven using new techniques of Granville, Harper, and Soundararajan, which are substantially easier than the previous proofs. In the last chapter, we show how pretentious techniques can be used to obtain more intuitive proofs of other classical theorems or obtain new results.

Keywords: Analytic number theory, distribution of prime numbers, pretentious approach, Halász's Theorem

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Zásady pro vypracování:

The recently developed "pretentious approach" to analytic number theory studies the asymptotic behavior of arithmetic functions using Halasz's theorem from 1975. This allows one to prove a number of important results in analytic number theory without relying on the analytic continuation of the Riemann zeta function. The goal of the thesis is to study and compare the classical and pretentious proofs of such theorems, such as the prime number theorem, Siegel-Walfisz theorem or Halasz theorem, and possibly try to further generalize some of these results.

Seznam odborné literatury:

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D. Koukoulopoulos: Pretentious multiplicative functions and the prime number theorem for arithmetic progressions, Compos. Math., 149 (2013), no. 7, 1129-1149

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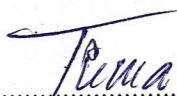
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Děkan

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Introduction

The purpose of this thesis is to give an overview of the pretentious approach to analytic number theory, a new theory developed in the last 10 years by Andrew Granville, Kannan Soundararajan, and their collaborators. Their goal is to re-state and prove all of the classical theorems in a new way, where the main notion is when two arithmetic functions *pretend* to be one another. This can be expressed using some sort of distance function, which is not a norm nor a metric in the classical sense, but has some nice properties, for example, satisfies the triangle inequality. The first advantage of this approach is that the statements and the proofs become more intuitive and elementary. The second advantage is that the results often refer to general multiplicative functions, and can be therefore used in different settings. Moreover, these techniques do not use the theory of analytic continuation, which plays a key role in the classical analytic number theory, and can, therefore, be used even when an analytic continuation is not known to exist or does not exist at all.

The first motivation that started the development of this theory was trying to find a new proof of Linnik's Theorem, which states that in any arithmetic progression where there can be primes, there is a relatively small one. The proof of this theorem was very technical and was considered as one of the hardest proof in number theory. A new proof was found by Friedlander and Iwaniec, which was more elementary and avoided many difficulties with zeros of L -functions. Granville and Soundararajan had an idea of their own version and succeeded in obtaining a shorter and more motivated proof of this theorem themselves. They realized that their techniques are more general and were soon able to use them to prove other classical theorems. They have however faced some difficulties, for example with obtaining a good error term in the prime number theorem, which were mostly solved by their collaborators.

The classical approach heavily depends on analytic continuation and the location of zeros of L -functions. For example, the prime number theorem is equivalent to the statement that the analytic continuation of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ does not have any zeros on the line $\operatorname{Re}(s) = 1$. The unsolved Riemann hypothesis states that all non-trivial zeros of $\zeta(s)$ have $\operatorname{Re}(s) = \frac{1}{2}$, and the wider zero-free region to the left of the line $\operatorname{Re}(s) = 1$ one obtains, the better becomes the error in the estimate of the number of primes up to x . The known zero-free regions are however very narrow and haven't been significantly improved in the last 50 years. For example, we do not even know whether there is an $\varepsilon > 0$ such that all zeros of ζ have $\operatorname{Re}(s) < 1 - \varepsilon$. Moreover, the known zero-free regions were obtained by extrapolating our knowledge about $\zeta(s)$ in the half-plane with $\operatorname{Re}(s) > 1$ a tiny bit to the left of the line $\operatorname{Re}(s) = 1$, it makes, therefore, sense to try not to pass the 1-line at all.

The motivation of the new approach was, therefore, to try to reprove the classical results without referring to zeros or the analytic continuation of L -functions at all. The prime number theorem can be stated in terms of the Möbius function, in particular, that its average up to x tends to 0 as x goes to infinity, and many other results can be equivalently expressed in similar terms about other multiplicative functions.

Halász gave in 1971 a characterization of multiplicative functions on the unit

disc whose average tends to zero. He noticed that examples of functions whose averages do not tend to zero are n^{it} for any real t and showed that these are the only such examples, up to little perturbations. Granville and Soundararajan expressed this in terms of *adistance* of multiplicative functions and say that the mean value of f is 0 unless $f(n)$ *pretends* to be n^{it} for some real number t , whence the name *pretentious*. This notion of distance allows us to express various difficult concepts intuitively and without referring to zeros of L -functions.

The aim of their effort was of course not to reprove old results, but to develop a new theory which would be more flexible and intuitive. They successfully used it to give new results as well, for example results about large character sums [GS3], large values of L -functions [GS4], least quadratic non-residue [XL] and the completion of the proof of Arithmetic Unique Quantum Ergodicity [So1],[So2],[HS].

To put the new techniques into context together with the classical methods, the first part of the thesis gives an overview of the classical techniques, starting with basic notions and elementary ways to estimate various quantities and ending with a demonstration of the use of complex analysis in the proof of the prime number theorem.

In the first chapter, we begin by introducing simple but important concepts like multiplicative functions and Dirichlet convolution. We also show basic methods how to estimate these functions and their sums, which are essential in analytic number theory and we use these results all over the thesis.

The second chapter is devoted to Dirichlet series and their properties. Because we would like to refrain talking about analytic continuations and their zeros, we give only their most important properties which are essential in the classical proof of the prime number theorem. We will be mainly interested in the Riemann zeta function $\zeta(s)$, show how various other L -functions can be expressed in terms of ζ and show how it can be analytically continued to the half-plane $\operatorname{Re}(s) > 0$. We also prove some general theorems about convergence of Dirichlet series which are essential in further work.

In the third chapter, we finally start to investigate prime numbers. We begin with the elementary estimates of Chebyshev and their consequences and show several equivalent formulations of the prime number theorem. We further proceed to Perron's formula, which enables us to evaluate discrete sums, such as the number of primes up to a given value, in terms of integrals of their Dirichlet series, which are usually easier to estimate.

In the fourth chapter, we use the so far developed methods to present a classical proof of the prime number theorem. The main purpose is to illustrate how complex analytic methods are used in proving statements about prime numbers, not to give the best results possible. We will therefore not talk about some important classical topics, such as the functional equation of the Riemann zeta function or the estimate of the number of zeros of $\zeta(s)$ up to a given height, as they are not required for the proof.

This brings us to the second part of the thesis, where we finally demonstrate the pretentious approach.

In the fifth chapter, we introduce the main ideas of the new approach. We define the distance function, show two forms of the triangle inequality that it satisfies and show its relation to the size of the Dirichlet series. We also formulate

Halász's Theorem and show how it can be used to give a short proof of the prime number theorem.

The sixth chapter contains the full proof of Halász's Theorem. It is based on the recent article [GHS1] by Granville, Harper, and Soundararajan, where the authors present a more intuitive and less technical proof.

The last chapter is aimed to show how the pretentious techniques can be used further to prove other hard results in analytic number theory. First, we talk about the Riemann hypothesis and explain how it can be stated without the need to talk about its analytic continuation. We then proceed to the classical theorems of Siegel, Walfisz, and Linnik. These results are about the distribution of prime numbers in arithmetic progressions, so we introduce Dirichlet characters, show their main properties and how they are used in the proof of Dirichlet's Theorem. We also explain how to translate this proof into the pretentious notions, where it shows up that both methods lead to the same difficulty – to show that $L(\chi, 1) \neq 0$ for any Dirichlet character χ . We finally show how to state the Siegel-Walfisz's Theorem and the Linnik's Theorem and see that the pretentious notions allow us to talk about these very intuitively.

1. Preliminaries

In this chapter, we are going to define the very basic notions of analytic number theory, such as arithmetic functions or the asymptotic notation. We also introduce the principal tools in investigating sums of arithmetic functions, i.e., partial summation and Dirichlet's hyperbola method.

The content of this chapter can be found in any introductory text on analytic number theory, for example, [Ap], [Ko1] or [Mu].

1.1 The ring of arithmetic functions

In number theory, we are interested in various properties of integers, for example, whether they are primes or in how many ways they can be written as a sum of 4 squares. Most of these interesting properties can be described in terms of arithmetic functions. By an arithmetic function, we mean any function $f : \mathbb{N} \rightarrow \mathbb{C}$.

All of the following are important examples of arithmetic functions:

- Euler's totient function $\varphi(n) := \#\{1 \leq k \leq n : (k, n) = 1\}$.
- The number of divisors function $\tau(n) := \#\{d \geq 1 : d|n\}$.
- The Möbius function $\mu(n) = \begin{cases} 0 & \text{if } n \text{ is not squarefree,} \\ (-1)^k & \text{if } n \text{ is a product of } k \text{ distinct primes.} \end{cases}$
- The Von Mangoldt's function $\Lambda(n) = \begin{cases} 0 & \text{if } n \text{ is not a prime power,} \\ \log p & \text{if } n = p^k \text{ for a prime } p. \end{cases}$
- By 1 we denote the function which is equal to 1 for every n , i.e., $1(n) = 1$.
- By δ we denote the function which is equal to 1 if $n = 1$ and 0 otherwise.
- The characteristic function of primes $p(n)$, so $p(n) = \begin{cases} 1 & \text{if } n \text{ is a prime,} \\ 0 & \text{otherwise.} \end{cases}$

An important class of arithmetic functions are *multiplicative* functions:

Definition. An arithmetic function f is called *multiplicative* if it satisfies $f(mn) = f(m)f(n)$ for each pair of positive coprime integers m, n .

The function f is called *totally multiplicative* if the identity $f(mn) = f(m)f(n)$ holds for all positive integers m, n .

From the examples above, the Möbius function, Euler's totient function, the number of divisors function, 1 and δ are multiplicative. Other examples of multiplicative functions are the functions $(n \mapsto n^{it})$ (for simplification denoted by n^{it}) for any real t , which will be particularly important in the part of the thesis devoted to the pretentious approach.

It follows from the fundamental theorem of arithmetic that multiplicative functions are determined by their values at prime powers and totally multiplicative functions are determined only by their values at primes.

Arithmetic functions can be point-wise added or multiplied together to form other arithmetic functions.

Arithmetic functions satisfy various interesting formulas, one of the most important is stated in the following lemma:

Lemma 1.1. *If $n \geq 1$ is an integer, then*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The lemma holds for $n = 1$. Suppose that $n > 1$ and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is the prime factorization of n with different primes. Then, eliminating the zero terms, i.e., those divisors of n which are divisible by a square, and grouping together the divisors with the same number of prime factors, we get

$$\begin{aligned} \sum_{d|n} \mu(d) &= 1 + \sum_{d|p_1 \dots p_k} \mu(d) = \sum_{i=1}^k \mu(p_i) + \sum_{1 \leq i < j \leq k} \mu(p_i p_j) + \dots + \mu(p_1 \dots p_k) = \\ &= 1 - \binom{k}{1} + \binom{k}{2} - \dots + (-1)^{k-1} \binom{k}{k-1} + (-1)^k = (1-1)^k = 0. \end{aligned}$$

□

There are other similar formulas which can be proved in an elementary way:

- $\sum_{d|n} \Lambda(d) = \log n$
- $\sum_{d|n} 1(d) = \sum_{d|n} 1 = \tau(n)$
- $\sum_{d|n} \varphi(d) = n$

These formulas provide a motivation for the following definition.

Definition. *Let f and g be two arithmetic functions. Their Dirichlet convolution is defined as the arithmetic function*

$$h = \sum_{d|n} f(d) \cdot g\left(\frac{n}{d}\right).$$

We denote this by $h = f * g$.

We can also write

$$(f * g)(n) = \sum_{ab=n} f(a)g(b),$$

where the sums runs over all pairs of positive integers (a, b) satisfying $ab = n$. This shows that Dirichlet convolution is a multiplicative analogy of the classical convolution, used for example in Fourier analysis. An advantage of this formula is also that it can be generalized to convolutions of more functions, so that

$$(f_1 * f_2 * \dots * f_k)(n) = \sum_{a_1 a_2 \dots a_k = n} f_1(a_1) f_2(a_2) \dots f_k(a_k).$$

Note that the sum of an arithmetic function over its divisors is just its convolution with 1. Using this fact, we can rewrite the above formulas using our new notation:

- $\mu * 1 = \delta$,
- $\Lambda * 1 = \log$,
- $1 * 1 = \tau$,
- $\varphi * 1 = n$, where by n we mean the identity function.

It can be easily verified that the set of arithmetic functions together with (point-wise) addition and Dirichlet convolution forms a ring. Moreover, it follows directly from the definition that Dirichlet convolution is commutative.

It can be easily seen that the function δ is a unity in the ring of arithmetic functions and Lemma 1.1 tells us that μ is the inverse of 1 with respect to Dirichlet convolution. This is the content of the following theorem.

Theorem 1.2 (Möbius inversion). *Let f, g be arithmetic functions. Then $f = g * 1$ if and only if $g = f * \mu$.*

Proof. First suppose that $f = g * 1$. Convolving both sides with μ , using associativity of Dirichlet convolution and Lemma 1.1, we get $f * \mu = (g * 1) * \mu = g * (1 * \mu) = g * \delta = g$.

For the other implication, it suffices to convolute both sides with 1 instead of μ . $\square$

Möbius inversion can be used to establish other relations between arithmetic functions. The following are derived from the formulas above using a direct application of Möbius inversion.

- $\Lambda = \mu * \log$,
- $\varphi = n * \mu$,
- $1 = \tau * \mu$.

The first identity will be important when counting primes and it can be rewritten in another way:

Lemma 1.3. *For each n , it holds that*

$$\Lambda(n) = - \sum_{d|n} \mu(d) \log d,$$

in other words

$$\Lambda = -(\mu \log) * 1.$$

Proof. For $n = 1$, the lemma is true. If $n > 1$, we can use the formula for Λ which we already know to get

$$\begin{aligned} \Lambda(n) &= \sum_{d|n} \mu(d) \log \frac{n}{d} = \sum_{d|n} \mu(d) (\log n - \log d) = \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d = \\ &= - \sum_{d|n} \mu(d) \log d \end{aligned} ,$$

where the last equality follows from Lemma 1.1. $\square$

1.2 Asymptotic notation and basic estimates

The purpose of analytic number theory is counting various quantities – the number of primes up to x or in a given interval, the total number of twin primes, or the average number of divisors of the first n positive integers. It is hard to express these quantities exactly so what we are really interested in are good estimates – good in the sense that the error is as small as possible. For this purpose, we introduce the following notation, which simplifies a lot of computations.

Definition. Let f and g be two complex or real functions (defined at least on a set of the form $\{z : |z| \geq K\}$ for some real number K).

1. We say that $f = O(g)$ if there exist positive constants C and A such that if $|x| > A$, then $|f(x)| \leq C \cdot |g(x)|$. The constant C is called the implied constant.
2. We write that $f = o(g)$ if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$.
3. We write that $f \asymp g$ if $f = O(g)$ and $g = O(f)$.
4. We write that $f \sim g$ if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$.

If $f = O(g)$, we also write $f \ll g$ or $g \gg f$.

As the statement get more complicated, this notation is used with less care, often leaving the reader with a not easy task to find out what the writer meant. This has, however, a good reason – it would be simply too complicated and nasty to try to write everything formally correctly including all the details. For example, we will see in later chapters that the functions f, g can depend on more than 1 variable. In this case, the implied constant is meant to be the same independent of all the variables if not indicated otherwise. For example, the notation $f = O_\varepsilon(g)$ or $f \ll_\varepsilon g$ means that the implied constant can depend on ε .

Let us now get to the topic of estimating sums. The next theorem is the most basic, but often powerful tool. The idea is to express a sum we would like to estimate in terms of an integral, which are usually easier to deal with. We can think of the following theorem as trying to estimate the sum of a sequence a_n weighted with a smooth function f . If we have good estimates on $\sum_{n \leq x} a_n$, this gives us a good information on the weighted sum $\sum_{n \leq x} a_n f(n)$. There are various other versions which improve on the estimate when we assume more differentiability on f , e.g., the Euler-Maclaurin summation formula, but we will get along with the basic version.

Theorem 1.4 (Partial summation). Let $(a_n)_{n=1}^\infty$ be a sequence of complex numbers and $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ a differentiable function. Let $A(x) := \sum_{n \leq x} a_n$. Then for every $x > 1$, we have

$$\sum_{n \leq x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt. \quad (1.1)$$

Proof. Let $S(x) := \sum_{n \leq x} a_n f(n)$.

First suppose that x is an integer. Write a_n as $A(n) - A(n-1)$. Then

$$\begin{aligned} S(x) &= a_1 f(1) + \sum_{2 \leq n \leq x} (A(n) - A(n-1)) f(n) = \\ &= a_1 f(1) + \sum_{2 \leq n \leq x} A(n) f(n) - \sum_{2 \leq n \leq x} A(n-1) f(n). \end{aligned}$$

Shifting the index in the second sum yields

$$S(x) = a_1 f(1) + A(x) f(x) - a_1 f(2) + \sum_{2 \leq n \leq x-1} A(n) (f(n) - f(n+1)).$$

Since

$$a_1 f(1) - a_1 f(2) = A(1) (f(1) - f(2))$$

and

$$A(n) (f(n) - f(n+1)) = -A(n) \int_n^{n+1} f'(t) dt = -\int_n^{n+1} A(t) f'(t) dt,$$

we get the result

$$S(x) = A(x) f(x) - \int_1^x A(t) f'(t) dt.$$

When x is not an integer, we can write $x = \lfloor x \rfloor + \{x\}$. Then $S(x) = S(\lfloor x \rfloor)$. Let $y := \lfloor x \rfloor$ and denote by $R(x)$ the right-hand side of (1.1). We need to show that $R(x) = R(y)$. Lets compute:

$$\begin{aligned} R(x) &= A(x) f(x) - \int_1^x A(t) f'(t) dt = \\ &= A(y) f(y) - A(y) (f(y) - f(x)) - \int_1^y A(t) f'(t) dt - \int_y^x A(t) f'(t) dt = \\ &= R(y) + A(y) (f(x) - f(y)) - A(y) \int_y^x A(t) f'(t) dt = R(y), \end{aligned}$$

where in the third equality, we used the fact that the function $A(t)$ is constant and equal to $A(y)$ on the interval $[y, x]$. □

Corollary 1.5. Let $(a_n)_{n=1}^\infty$ be a sequence of complex numbers, $1 \leq x < y$ and $f : [x, y] \rightarrow \mathbb{R}$ a differentiable function. Let $A(x) := \sum_{n \leq x} a_n$. Then we have

$$\sum_{x < n \leq y} a_n f(n) = A(y) f(y) - A(x) f(x) - \int_x^y A(t) f'(t) dt. \quad (1.2)$$

Note that the theorem and corollary hold even when f is not defined at the point x or y , with the integrals replaced by $\int_{x^+}^y$ or $\int_x^{y^-}$ respectively.

Proof. Note that

$$\sum_{y < n \leq x} a_n f(n) = \sum_{n \leq x} a_n f(n) - \sum_{n \leq y} a_n f(n),$$

and use partial summation on both sums on the right hand side. □

As we said, partial summation is used when we have some information about the sum $\sum_{n \leq x} a_n$ and would like to obtain estimate of the sum $\sum_{n \leq x} a_n f(n)$ weighted with a differentiable function f . We can then replace the sequence (a_n) by the sequence $(a_n f(n))$ to get further estimates.

We can start by letting $a_n = 1$, since we know that $\sum_{n \leq x} a_n = \lfloor x \rfloor = x + O(1)$. This allows us to evaluate $\sum_{n \leq x} f(n)$ for various differentiable functions f .

Proposition 1.6. *There exists a constant γ such that for all $x > 1$,*

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + O\left(\frac{1}{x}\right).$$

Proof. We use partial summation with $a_n := 1$ and $f(x) := \frac{1}{x}$. Then $A(x) = \lfloor x \rfloor$ and

$$\sum_{n \leq x} a_n f(n) = \sum_{n \leq x} \frac{1}{n} = \frac{\lfloor x \rfloor}{x} + \int_1^x \frac{\lfloor t \rfloor}{t^2} dt.$$

We can estimate $\frac{\lfloor x \rfloor}{x} = 1 + O\left(\frac{1}{x}\right)$ and write $\frac{\lfloor t \rfloor}{t^2} = \frac{t - \{t\}}{t^2} = \frac{1}{t} - \frac{\{t\}}{t^2}$. Hence

$$\int_1^x \frac{\lfloor t \rfloor}{t^2} dt = \log x - \int_1^x \frac{\{t\}}{t^2} dt.$$

Now let $\tilde{\gamma} := \int_1^\infty \frac{\{t\}}{t^2} dt$. Note that $\tilde{\gamma} \leq \int_1^\infty \frac{1}{t^2} dt < \infty$. Then

$$\int_1^x \frac{\lfloor t \rfloor}{t^2} dt = \log x - \tilde{\gamma} + \int_x^\infty \frac{\{t\}}{t^2} dt = \log x - \tilde{\gamma} + O\left(\frac{1}{x}\right).$$

Altogether, we get

$$\sum_{n \leq x} \frac{1}{n} = \log x + 1 - \tilde{\gamma} + O\left(\frac{1}{x}\right)$$

and setting $\gamma := 1 - \tilde{\gamma}$ gives the right constant from the statement of the proposition. $\square$

The constant γ which appeared in the proof above is called the *Euler-Mascheroni* constant and is equal to $\lim_{x \rightarrow \infty} \sum_{n \leq x} \frac{1}{n} - \log x$. It arises often in number theory or other branches of mathematics and is roughly equal to $\gamma \doteq 0.577\dots$, it is however an open problem whether it is a rational or irrational number.

Proposition 1.7. *For all $x > 1$,*

$$\sum_{n \leq x} \log n = x \log x - x + O(\log x).$$

Proof. We use partial summation with $a_n := 1$ and $f(x) := \log x$. Then

$$\begin{aligned} \sum_{n \leq x} \log n &= \lfloor x \rfloor \log x - \int_1^x \frac{\lfloor t \rfloor}{t} dt = (x + O(1)) \log x - \int_1^x \frac{t - \{t\}}{t} dt = \\ &= x \log x + O(\log x) - (x - 1) - \int_1^x \frac{\{t\}}{t} dt = x \log x - x + O(\log x). \end{aligned}$$

$\square$

Corollary 1.8. *For $x > 1$, it holds that*

$$\sum_{n \leq x} \log \frac{x}{n} = x + O(\log x).$$

Proof. We have

$$\begin{aligned} \sum_{n \leq x} \log \frac{x}{n} &= \sum_{n \leq x} (\log x - \log n) = \lfloor x \rfloor \log x - \sum_{n \leq x} \log n = \\ &= x \log x + O(\log x) - x \log x + x + O(\log x) = x + O(\log x). \end{aligned}$$

□

1.3 Dirichlet's hyperbola method

In this section, we are going to introduce a method invented by Dirichlet. It is useful when dealing with sums of the form $\sum_{n \leq x} f(n)$, where $f = g * h$ and we already have some knowledge about the rate of growth of g and h . The basic idea is to write

$$\sum_{n \leq x} f(n) = \sum_{n \leq x} \sum_{d|n} g(d)h\left(\frac{n}{d}\right) = \sum_{de \leq x} g(d)h(e),$$

and now rearrange the last sum in a suitable way. The name of Dirichlet's hyperbola method comes from the fact that we are summing over points under a hyperbola.

To see how Dirichlet's method is useful, let us compare it with partial summation. Suppose that our goal was to estimate $\sum_{n \leq x} \tau(n)$. We already know that $\tau = 1 * 1$, so

$$\sum_{n \leq x} \tau(n) = \sum_{n \leq x} \sum_{d|n} 1.$$

Interchanging summation and using the estimate from Proposition 1.6 we get:

$$\sum_{n \leq x} \tau(n) = \sum_{d \leq x} \sum_{n \leq \frac{x}{d}} 1 = \sum_{d \leq x} \left\lfloor \frac{x}{d} \right\rfloor = \sum_{d \leq x} \left(\frac{x}{d} + O(1) \right) = x \log x + O(x).$$

Note that in the result, we got the leading term, but the error term is not very satisfying. The poor error arises from the fact that in the last two equalities, where we used an estimate for $\lfloor \frac{x}{d} \rfloor$, we added the term $O(1)$ x -times which led to the poor error $O(x)$.

The next theorem shows a clever way to rearrange the terms in the sum to get a better estimate.

Theorem 1.9 (Dirichlet's hyperbola method). *Let f , g and h be arithmetic functions satisfying $f(n) = (g * h)(n)$. Let $H(x) := \sum_{n \leq x} h(x)$ and $G(x) := \sum_{n \leq x} g(x)$.*

Then for every $1 < y < x$, it holds that

$$\sum_{n \leq x} f(n) = \sum_{d \leq y} g(d)H\left(\frac{x}{d}\right) + \sum_{d \leq \frac{x}{y}} h(d)G\left(\frac{x}{d}\right) - G(y)H\left(\frac{x}{y}\right).$$

Proof. Let $1 < y < x$ be arbitrary. Then

$$\begin{aligned}
\sum_{n \leq x} f(n) &= \sum_{de \leq x} g(d)h(e) = \sum_{\substack{de \leq x \\ d \leq y}} g(d)h(e) + \sum_{\substack{de \leq x \\ d > y}} g(d)h(e) = \\
&= \sum_{d \leq y} \sum_{e \leq \frac{x}{d}} g(d)h(e) + \sum_{d > y} \sum_{e \leq \frac{x}{d}} g(d)h(e) = \\
&= \sum_{d \leq y} g(d)H\left(\frac{x}{d}\right) + \sum_{e \leq \frac{x}{y}} h(e) \sum_{y < d \leq \frac{x}{e}} g(d) = \\
&= \sum_{d \leq y} g(d)H\left(\frac{x}{d}\right) + \sum_{e \leq \frac{x}{y}} h(e) \left(G\left(\frac{x}{e}\right) - G(y)\right) = \\
&= \sum_{d \leq y} g(d)H\left(\frac{x}{d}\right) + \sum_{e \leq \frac{x}{y}} h(e)G\left(\frac{x}{e}\right) - G(y) \sum_{e \leq \frac{x}{y}} h(e) = \\
&= \sum_{d \leq y} g(d)H\left(\frac{x}{d}\right) + \sum_{e \leq \frac{x}{y}} h(e)G\left(\frac{x}{e}\right) - G(y)H\left(\frac{x}{y}\right),
\end{aligned}$$

which is what we wanted. $\square$

Let us now use this theorem to estimate the sum of the number of divisors. In this case we have $h = g = 1$, so $H\left(\frac{x}{d}\right) = \left\lfloor \frac{x}{d} \right\rfloor = G\left(\frac{x}{d}\right)$. Then

$$\sum_{n \leq x} \tau(n) = \sum_{d \leq y} \left\lfloor \frac{x}{d} \right\rfloor + \sum_{d \leq \frac{x}{y}} \left\lfloor \frac{x}{d} \right\rfloor - \lfloor y \rfloor \left\lfloor \frac{x}{y} \right\rfloor,$$

and since the expression is symmetric in terms of y and $\frac{x}{y}$, to get the best error term, we choose $y = \sqrt{x}$. Then by Proposition 1.6:

$$\begin{aligned}
\sum_{n \leq x} \tau(n) &= 2x \sum_{d \leq \sqrt{x}} \frac{1}{d} - \lfloor \sqrt{x} \rfloor^2 + O(\sqrt{x}) = \\
&= 2x \left(\log \sqrt{x} + \gamma + O\left(\frac{1}{\sqrt{x}}\right) \right) - x + O(\sqrt{x}) = \\
&= x \log x + (2\gamma - 1)x + O(\sqrt{x}),
\end{aligned}$$

which gives a better error term than the previous use of partial summation.

Note that when we used the estimate for $\left\lfloor \frac{x}{d} \right\rfloor$, we only added the error $O(1)$ $2\sqrt{x}$ -times as opposed to the previous case when we added it x -times.

Another way of explaining this is that the approximation $\left\lfloor \frac{x}{d} \right\rfloor = \frac{x}{d} + O(1)$ is good when d is small, but becomes worse and worse as d approaches x . In the Dirichlet's hyperbola method, we only have to use this estimate for $d \leq \sqrt{x}$, which leads to the better error.

We have proved the following proposition:

Proposition 1.10. *For each $x > 0$, we have*

$$\sum_{n \leq x} \tau(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}).$$

It is an important open problem (called the Dirichlet divisor problem) to find the smallest c such that

$$\left| x \log x + (2\gamma - 1)x - \sum_{n \leq x} \tau(n) \right| \ll x^c.$$

The best known value of c is $131/416 \doteq 0.31$ and was obtained by M.N. Huxley in 2003[Hu], but it is widely believed that the optimal value is $c = 1/4$.

2. Dirichlet series

2.1 Definition and convergence

Dirichlet series provide a key connection between arithmetic functions and complex analysis and allow us to use rich analytic methods to study properties of integers. The main relation between arithmetic functions and their Dirichlet series will be given in Section 3.3 in the form of Perron's formula. For additional information, we refer the reader to [Ap], [Ko1], [Mu] or other introductory texts on analytic number theory.

Let us start with the definition of a Dirichlet series.

Definition. Let f be an arithmetic function. Then the Dirichlet series of f is the function of complex variable s defined as

$$L(f, s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s},$$

for these s where the series converges.

The main example of Dirichlet series is when $f(n) = 1$ for every n . Then

$$L(f, s) = \sum_{n=1}^{\infty} \frac{1}{n^s} =: \zeta(s)$$

is the famous Riemann zeta function.

Dirichlet series are in some ways similar to Taylor series or power series in real analysis. Similarly to the radius of convergence and absolute convergence of a Taylor series, Dirichlet series have half-plane of convergence or absolute convergence bounded by a vertical line – this is because the absolute value of n^s only depends on the real part of s . We prove these properties formally in the following propositions.

Proposition 2.1. Let f be an arithmetic function whose Dirichlet series converges absolutely for some $s \in \mathbb{C}$. Then there exists a real number (or possibly $-\infty$) σ_a such that $L(f, s)$ converges absolutely for $\operatorname{Re}(s) > \sigma_a$ and $L(f, s)$ does not converge absolutely for $\operatorname{Re}(s) < \sigma_a$.

Proof. Let $\sigma_a := \inf \{r \in \mathbb{R} : L(f, r) \text{ converges}\}$. For any complex number s , we have $|n^s| = |n|^{\operatorname{Re}(s)}$, therefore for every s with $\operatorname{Re}(s) > \sigma_a$, $L(f, s)$ converges absolutely.

On the other hand, if $\operatorname{Re}(s) < \sigma_a$, since $L(f, \operatorname{Re}(s))$ does not converge absolutely, neither does $L(f, s)$. $\square$

The number σ_a is called the abscissa of absolute convergence of the Dirichlet series.

Proposition 2.2. Let f be an arithmetic function whose Dirichlet series converges for some $s \in \mathbb{C}$. Then there exists a real number (or possibly $-\infty$) σ_c such that $L(f, s)$ converges whenever $\operatorname{Re}(s) > \sigma_c$ and $L(f, s)$ does not converge when $\operatorname{Re}(s) < \sigma_c$.

Proof. Let $\sigma_c := \inf\{\operatorname{Re}(s) : L(f, s) \text{ converges}\}$, let s be such that $\operatorname{Re}(s) > \sigma_c$ and let $s_0 := \frac{\operatorname{Re}(s) - \sigma_c}{2}$. Then the series $L(f, s_0)$ converges and $\operatorname{Re}(s - s_0) > 0$. If we denote by $q = s - s_0$ by partial summation, for any $1 < M < N$, we have

$$\begin{aligned} \sum_{M < n \leq N} \frac{a_n}{n^s} &= \sum_{M < n \leq N} \frac{a_n}{n^{s_0+q}} = \\ &= \frac{1}{N^q} \sum_{n \leq N} \frac{a_n}{n^{s_0}} - \frac{1}{M^q} \sum_{m \leq M} \frac{a_m}{m^{s_0}} + q \int_M^N \frac{1}{t^{q+1}} \sum_{n \leq t} \frac{a_n}{n^{s_0}} dt. \end{aligned}$$

If we use the equality

$$\sum_{n \leq t} \frac{a_n}{n^{s_0}} = \sum_{m \leq M} \frac{a_m}{m^{s_0}} + \sum_{M < n \leq t} \frac{a_n}{n^{s_0}},$$

we get

$$\begin{aligned} \sum_{M < n \leq N} \frac{a_n}{n^s} &= \frac{1}{N^q} \sum_{n \leq N} \frac{a_n}{n^{s_0}} - \frac{1}{M^q} \sum_{m \leq M} \frac{a_m}{m^{s_0}} + q \sum_{m \leq M} \frac{a_m}{m^{s_0}} \int_M^N \frac{1}{t^{q+1}} dt + \\ &+ q \int_M^N \frac{1}{t^{q+1}} \sum_{M < n \leq t} \frac{a_n}{n^{s_0}} dt = \\ &= \frac{1}{N^q} \sum_{n \leq N} \frac{a_n}{n^{s_0}} - \frac{1}{M^q} \sum_{m \leq M} \frac{a_m}{m^{s_0}} + \sum_{m \leq M} \frac{a_m}{m^{s_0}} \left(\frac{1}{M^q} - \frac{1}{N^q} \right) + \\ &+ q \int_M^N \frac{1}{t^{q+1}} \sum_{M < n \leq t} \frac{a_n}{n^{s_0}} dt = \\ &= \frac{1}{N^q} \sum_{M < n \leq N} \frac{a_n}{n^{s_0}} + q \int_M^N \frac{1}{t^{q+1}} \sum_{M < n \leq t} \frac{a_n}{n^{s_0}} dt. \end{aligned}$$

Let us now choose an arbitrary $\varepsilon > 0$. Because $L(f, s_0)$ converges, by Cauchy's criterion there exists an N_0 such that if $N_0 \leq M \leq N$, then

$$\sum_{M < n \leq N} \frac{a_n}{n^{s_0}} < \varepsilon.$$

Therefore for such M, N , we get

$$\begin{aligned} \left| \sum_{M < n \leq N} \frac{a_n}{n^s} \right| &\leq \frac{\varepsilon}{N^{\operatorname{Re}(q)}} + \varepsilon q \int_M^N \frac{1}{t^{q+1}} dt \leq \varepsilon + \varepsilon |q| \int_1^\infty \frac{1}{t^{\operatorname{Re}(q)+1}} dt = \\ &= \varepsilon \left(1 + \frac{|q|}{\operatorname{Re}(q)} \right), \end{aligned} \tag{2.1}$$

so by Cauchy's criterion, the series converges.

If $\operatorname{Re}(s) < \sigma_c$, the series $L(f, s)$ diverges by the definition of σ_c . $\square$

The number σ_c is called the abscissa of convergence of the Dirichlet series and the half-plane $\{s : \operatorname{Re}(s) > \sigma_c\}$ is called the half-plane of convergence.

Next we show that the convergence is locally uniform.

Proposition 2.3. *Let f be an arithmetic function and $\sigma_c \neq \infty$ be the abscissa of convergence of its Dirichlet series. Then the series converges locally uniformly in the half-plane $\operatorname{Re}(s) > \sigma_c$.*

Proof. Let K be a compact subset of the half-plane of convergence of $L(f, s)$. Then there are positive numbers R and δ such that for any $s \in K$, $|s| < R$ and $\operatorname{Re}(s) > \sigma_c + \delta$. Let $s_0 = \sigma_c + \frac{\delta}{2}$. Then the series $L(f, s_0)$ converges and by Cauchy's criterion, for any $\varepsilon > 0$, there exists an N_0 such that if $N_0 \leq M < N$, then

$$\left| \sum_{M < n \leq N} \frac{a_n}{n^{s_0}} \right| \leq \varepsilon.$$

If we choose an arbitrary $s \in K$, and $N_0 \leq M < N$, we can use the same estimates as in (2.1) to get

$$\left| \sum_{M < n \leq N} \frac{a_n}{n^s} \right| \leq \varepsilon \left(1 + \frac{|s - s_0|}{\operatorname{Re}(s - s_0)} \right) \leq \varepsilon \left(1 + \frac{(2K + |s_0|)}{\delta} \right).$$

Since the constant $1 + \frac{(2K + |s_0|)}{\delta}$ does not depend on the particular value of $s \in K$, by Cauchy's criterion for uniform convergence, the Dirichlet's series of f converges uniformly in K . Since K was an arbitrary compact subset of the half-plane of convergence of $L(f, s)$, the series converges locally uniformly in its half-plane of convergence. $\square$

Due to the fact that the convergence is locally uniform, we are free to interchange limits, sums and integrals in the half-plane of absolute convergence.

Corollary 2.4. *The series $L(f, s)$ defines a holomorphic function on its half-plane of convergence.*

Proof. By Proposition 2.3, $L(f, s)$ is a uniform limit of holomorphic functions, hence it is holomorphic itself. $\square$

Dirichlet series behave nicely with respect to operations on arithmetic functions, especially to Dirichlet's convolution.

Proposition 2.5. *Let f, g be two arithmetic functions and $s \in \mathbb{C}$. Then*

1. *If both $L(f, s)$ and $L(g, s)$ converge (absolutely), then $L(f + g, s)$ converges (absolutely) and $L(f + g, s) = L(f, s) + L(g, s)$.*
2. *If both $L(f, s)$ and $L(g, s)$ converge absolutely, then $L(f * g, s)$ also converges absolutely and $L(f * g, s) = L(f, s)L(g, s)$.*

Proof. 1. This is a direct consequence of related general theorems on convergence of series.

2. We have

$$L(f, s)L(g, s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s} \cdot \sum_{m=1}^{\infty} \frac{g(m)}{m^s}$$

and because both series converge absolutely, we can multiply the series term by term. Moreover, the resulting series will also converge absolutely. Collecting together the terms with the same denominator, we get

$$L(f, s)L(g, s) = \sum_{k=1}^{\infty} \frac{\sum_{mn=k} f(n)g(m)}{k^s} = \sum_{k=1}^{\infty} \frac{(f * g)(k)}{k^s} = L(f * g, s).$$

$\square$

We will see in section 2.3 that the formula $L(f * g, s) = L(f, s)L(g, s)$ allows us to express a lot of interesting L -functions in terms of the Riemann zeta function $\zeta(s)$.

2.2 Euler products

One of the first appearances of Dirichlet's series is due to Euler, who used them to give an alternative proof of the infinitude of primes and in fact, showed a stronger theorem that the series $\sum_{p \text{ prime}} \frac{1}{p}$ diverges.

Let us show his observation about the zeta function. He observed that if we have the sum

$$\sum_{n=1}^{\infty} \frac{1}{n^s},$$

it can be written like

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \dots \right) = \prod_p \left(\frac{1}{1 - p^{-s}} \right).$$

Here the first equality is a consequence of the fundamental theorem of arithmetic – after multiplying together various factors of the infinite products taking only finitely many terms not equal to 1, in the denominators, we get all the different products of powers of primes, so by the fundamental theorem of arithmetic we get each natural number as a denominator exactly once. The second equality is just the sum of a geometric series.

Euler's observation was the following – if the sum of reciprocals of primes was convergent, then the limit of the right-hand side as $s \rightarrow 1^+$ would be finite, which would be a contradiction with the divergence of the harmonic series.

If f is a multiplicative function, we can apply Euler's argument as above and express $L(f, s)$ as a product over the primes as follows:

$$L(f, s) = \prod_p \left(\sum_{k=0}^{\infty} 1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \dots \right).$$

In the case when f is completely multiplicative, the sum is a geometric series and can be simplified as

$$L(f, s) = \prod_p \left(\frac{1}{1 - \frac{f(p)}{p^s}} \right).$$

This expression is called an *Euler product* of $L(f, s)$ and can be very useful. For instance, one of the main questions of analytic number theory, closely related to the distribution of prime numbers, is where are the zeros of the Riemann zeta function (or rather its analytic continuation, see section 2.4) located. The existence of the Euler product and some basic facts about convergence of infinite products tell us that $\zeta(s)$ does not have any zeros with $\text{Re}(s) > 1$.

2.3 Examples

We have already introduced the most important example of a Dirichlet series, which is the Riemann zeta function. Using Proposition 2.5, we can express the Dirichlet series of many important arithmetic functions in terms of $\zeta(s)$. For instance, we have the following (the equalities hold in an appropriate half-plane of convergence):

- $\tau(n) = (1 * 1)(n)$, so $L(\tau, s) = \zeta(s)^2$.
- For the function δ , we have $L(\delta, s) = 1$.
- For the Möbius function μ , since $\mu * 1 = \delta$, it must hold that $L(\mu, s) \cdot \zeta(s) = 1$, so $L(\mu, s) = \frac{1}{\zeta(s)}$.
- If n denotes the identity function, then

$$L(n, s) = \sum_{n=1}^{\infty} \frac{n}{n^s} = \zeta(s-1).$$

- Since $\varphi = n * \mu$, we have $L(\varphi, s) = \frac{\zeta(s-1)}{\zeta(s)}$.

If we have a Dirichlet series

$$L(f, s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s},$$

its derivative $L'(f, s)$ has the same abscissa of absolute convergence and in that half-plane, it holds that

$$L'(f, s) = - \sum_{n=1}^{\infty} \frac{f(n) \log n}{n^s},$$

and in general, for the k^{th} derivative, we have

$$L^{(k)}(f, s) = (-1)^k \sum_{n=1}^{\infty} \frac{f(n) (\log n)^k}{n^s}.$$

Important in the context of estimating the number of primes will be the Dirichlet series of the Von Mangoldt's function Λ . This is the topic of the following proposition.

Proposition 2.6. *For $\text{Re}(s) > 1$, it holds that*

$$L(\Lambda, s) = - \frac{\zeta'(s)}{\zeta(s)}$$

Proof. Note that $-\frac{\zeta'}{\zeta}(s) = -(\log \zeta)'(s)$. We use the Euler product of ζ to compute the right-hand side. We have

$$-(\log \zeta)'(s) = \left[\sum_p \log \left(1 - \frac{1}{p^s} \right) \right]'$$

and using the Taylor series of logarithm $\log(1 - x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}$, we get

$$-(\log \zeta)'(s) = \left[-\sum_p \sum_{n=1}^{\infty} \frac{1}{np^{ns}} \right]' = \sum_p \sum_{n=1}^{\infty} \frac{\log p}{p^{ns}} = \sum_p \frac{\log p}{p^s} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where the second equality follows from differentiating the series term by term (which is justified from the fact that the convergence is locally uniform). $\square$

2.4 Meromorphic continuation of Riemann zeta function

In section 3.3, we will see that it is possible to express the sum $\sum_{n \leq x} \Lambda(n)$ in terms of an integral involving the Dirichlet series $L(\Lambda, s)$. In order to be able to use the full strength of complex analysis, we need to define $\zeta(s)$ to the left of the line $\operatorname{Re}(s) = 1$. In his article in 1859, Riemann showed how to analytically continue $\zeta(s)$ to the whole complex plane with the exception of 1, where it has a simple pole with residue 1. It will be sufficient for us to show how to define $\zeta(s)$ in the half-plane $\operatorname{Re}(s) > 0$.

We will first prove a useful formula for the Riemann zeta function.

Theorem 2.7. *For any s with $\operatorname{Re}(s) > 1$, and for any integer $N > 1$, it holds that*

$$\zeta(s) = \sum_{n=1}^{N-1} \frac{1}{n^s} - \frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} - s \int_N^{\infty} \frac{t - [t] - \frac{1}{2}}{t^{s+1}} dt. \quad (2.2)$$

Proof. Let us choose an arbitrary s with $\operatorname{Re}(s) > 1$, integer N and $x > N$. By partial summation, we have

$$\sum_{n \leq x} \frac{1}{n^s} = \sum_{n=1}^N \frac{1}{n^s} + \sum_{N \leq n \leq x} \frac{1}{n^s} = \sum_{n=1}^N \frac{1}{n^s} + \frac{[x]}{x^s} - \frac{1}{N^{s-1}} - s \int_N^x \frac{[t]}{t^{s+1}} dt.$$

Sending x to ∞ yields

$$\zeta(s) = \sum_{n=1}^N \frac{1}{n^s} - \frac{1}{N^{s-1}} - s \int_N^{\infty} \frac{[t]}{t^{s+1}} dt.$$

To finish the proof, we show that

$$-\frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} + \frac{1}{N^{s-1}} - s \int_N^{\infty} \frac{t - \frac{1}{2}}{t^{s+1}} dt = 0, \quad (2.3)$$

because then

$$\begin{aligned} \zeta(s) &= \sum_{n=1}^N \frac{1}{n^s} - \frac{1}{N^{s-1}} - s \int_N^{\infty} \frac{[t]}{t^{s+1}} dt - \frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} + \frac{1}{N^{s-1}} - s \int_N^{\infty} \frac{t - \frac{1}{2}}{t^{s+1}} dt = \\ &= \sum_{n=1}^{N-1} \frac{1}{n^s} - \frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} - s \int_N^{\infty} \frac{t - [t] - \frac{1}{2}}{t^{s+1}} dt, \end{aligned}$$

as was to be proved.

We directly evaluate the integrals in (2.3):

$$-s \int_N^\infty \frac{1}{t^s} dt = -s \left[\frac{-1}{(s-1)t^{s-1}} \right]_N^\infty = \frac{-sN^{1-s}}{s-1},$$

$$\frac{s}{2} \int_N^\infty \frac{1}{t^{s+1}} dt = \frac{s}{2} \left[\frac{-1}{st^s} \right]_N^\infty = \frac{N^{-s}}{2},$$

hence

$$-\frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} + \frac{1}{N^{s-1}} - s \int_N^\infty \frac{t - \frac{1}{2}}{t^{s+1}} dt =$$

$$-\frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} + \frac{1}{N^{s-1}} + \frac{N^{-s}}{2} - \frac{sN^{1-s}}{s-1} = 0.$$

□

Note that in formula (2.2), the integral on the right-hand side converges for every s with $\operatorname{Re}(s) > 0$. We can therefore define $\zeta(s)$ for s in this half-plane except when $s = 1$. Let us now see what happens at the point $s = 1$.

Proposition 2.8. *The function $\zeta(s)$ defined by formula (2.2) has a simple pole at $s = 1$ with residue 1.*

Proof. To prove the proposition, it suffices to show that $\zeta(s) - \frac{1}{s-1}$ is holomorphic at $s = 1$. Let $N = 2$. Then we have

$$\zeta(s) - \frac{1}{s-1} = 1 - \frac{1}{2^{s+1}} + \frac{2^{s-1} - 1}{s-1} - s \int_2^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} ds.$$

This function is clearly holomorphic when $s \neq 1$. Moreover, the limit as $s \rightarrow 1$ of the right-hand side exists and is finite, so we can conclude that the function is also holomorphic around $s = 1$ □

3. The distribution of prime numbers

Mathematicians have been fascinated by prime numbers ever since people started counting. Tables of primes have been discovered in most ancient civilizations. It is then natural to ask the question how often does one encounter primes in the set of integers.

The first documented attempts to estimate the number of primes up to a given x were made by Gauss, Legendre, and Dirichlet and it took about 100 years before these estimates were proven true independently by Hadamard and de la Vallée Poussin.

Let $\pi(x)$ denote the number of primes up to x , so

$$\pi(x) = |\{1 \leq p \leq x; p \text{ is a prime}\}|.$$

Gauss estimated that

$$\pi(x) \sim \int_2^x \frac{1}{\log t} dt \sim \frac{x}{\log x},$$

where the second asymptotic follows after evaluating the integral by parts. After being proved at the end of 19th century, this asymptotic became known as the prime number theorem.

The first expression involving integral has two advantages and a disadvantage. Its drawback is that the integral cannot be evaluated exactly using elementary functions, therefore we will denote it by $\int_2^x \frac{1}{\log t} dt = \text{Li}(x)$. However, integration by parts gives us a good idea about the rate of growth of this function.

The first advantage of the formulation with integral is that it gives us a better intuition of the statement. It basically says that if we choose an integer t at random, it is prime with probability about $\frac{1}{\log t}$. This is of course not a rigorous statement, but it is useful in many heuristics and it often enables us to make accurate predictions before we are able to prove them rigorously.

The asymptotic with integral is also more precise than the other estimate which involves only elementary functions.

We refer the reader to [Ap], [Ko1] or [Mu] for additional information.

3.1 Elementary estimates

The first steps towards the proof of the prime number theorem were made by Chebyshev in 1850, who used elementary methods to show that $\pi(x) \asymp \frac{x}{\log x}$. The following theorem shows his estimates, even though the proof is due to Erdős, which can be found in [K1].

Theorem 3.1 (Chebyshev). *There exist constants $0 < A < B$ such that for all $x > 1$,*

$$\frac{Ax}{\log x} \leq \pi(x) \leq \frac{Bx}{\log x}.$$

Proof. The proof is based on two estimates of the binomial coefficient $\binom{2n}{n}$.

The first estimates are

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n.$$

These follow directly from binomial theorem used for $(1+1)^{2n}$, where $\binom{2n}{n}$ is the largest of the $2n+1$ coefficients.

To get another estimate, we use some properties of the primes dividing $\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$. The first fact is that each prime p satisfying $n < p \leq 2n$ divides this number, so

$$\prod_{n < p \leq 2n} p \leq \binom{2n}{n}.$$

A less obvious property is that if a power of any prime $p^a \mid \binom{2n}{n}$, then $p^a \leq 2n$. This can be deduced from the formula expressing the power of a prime which divides a factorial: this formula tells us that if p^a is the highest power of p which divides $\binom{2n}{n}$, then

$$a = \sum_{i=1}^{\infty} \left\lfloor \frac{2n}{p^i} \right\rfloor - 2 \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$$

(note that both of the sums are in fact finite). Because for each real number α , we have $0 \leq \lfloor 2\alpha \rfloor - 2\lfloor \alpha \rfloor \leq 1$, it follows that

$$a \leq \sum_{\substack{i, \\ p^i \leq 2n}} 1,$$

so indeed $p^a \leq 2n$.

Since there are at most $\pi(2n)$ possible primes dividing the binomial coefficient, we get the estimate

$$\prod_{n < p \leq 2n} p \leq \binom{2n}{n} \leq (2n)^{\pi(2n)}.$$

Combining the two estimates together in one way, we get $(2n)^{\pi(2n)} \geq \frac{4^n}{2n+1}$, so taking logarithms yields

$$\pi(2n) \geq \frac{2n}{\log 2n} \log 2 - \frac{\log(2n+1)}{\log 2n} > \frac{2n}{\log 2n} \log 2 - 2,$$

which gives the first estimate for even numbers. For odd numbers, just realize that for $n \geq 2$, $\pi(2n-1) = \pi(2n)$.

To get the upper bound of Chebyshev, we combine the estimates of the binomial coefficients in the other way. We have

$$\prod_{n < p \leq 2n} p \leq 4^n,$$

so taking logarithms, we get

$$\sum_{n < p \leq 2n} \log p \leq n \log 4.$$

To get Chebyshev's upper bound, take an arbitrary $x > 0$ and integer m such that $2^m \leq x$ and $2^{m+1} > x$. Then

$$\sum_{p \leq x} \log p = \sum_{i=0}^m \sum_{2^i < p \leq 2^{i+1}} \log p \leq \sum_{i=0}^m 2^i \log 4 \leq 2^{m+1} \log 4 \leq 2 \log 4 \cdot x, \quad (3.1)$$

therefore

$$\begin{aligned} 2 \log 4 \cdot x &\geq \sum_{p \leq x} \log p \geq \sum_{\sqrt{x} < p \leq x} \log p \geq (\pi(x) - \pi(\sqrt{x})) \log \sqrt{x} \geq \\ &\geq (\pi(x) - \sqrt{x}) \log \sqrt{x} \end{aligned}$$

so

$$\pi(x) \leq 4 \log 4 \frac{x}{\log x} + \sqrt{x},$$

and because $\sqrt{x} = O\left(\frac{x}{\log x}\right)$, we can choose the constant B large enough so that the inequality holds for all $x > 1$. $\square$

The following is a direct corollary of Chebyshev's estimates:

Corollary 3.2. *For $x > 1$, we have*

$$\pi(x) = O\left(\frac{x}{\log x}\right)$$

Chebyshev's estimates allow us to make other interesting deductions about primes. It is often easier to count the primes with weight $\log p$, but we would like to get the results about the actual number of primes, i.e., primes counted with the weight 1. The following proposition gives us a way to go back and forth between these.

Let us denote by $\theta(x)$ the number of primes up to x , where each prime is counted with the weight $\log p$, so

$$\theta(x) = \sum_{p \leq x} \log p.$$

Proposition 3.3. *For $x > 1$,*

$$\theta(x) = \log x \cdot \pi(x) + O\left(\frac{x}{\log x}\right).$$

Moreover, $\theta(x) = O(x)$.

Proof. By partial summation used with the sequence $a_n = p(n)$, where $p(n)$ is the characteristic function of primes, we have

$$\sum_{p \leq x} \log p = \log x \cdot \pi(x) - \int_2^x \frac{\pi(t)}{t} dt.$$

By Chebyshev's estimates, the integrand is $O\left(\frac{1}{\log x}\right)$, so the whole integral is $O(\text{Li}(x)) = O\left(\frac{x}{\log x}\right)$.

Using Corollary 3.2, we get that $\theta(x) = O(x)$ $\square$

We will see that it is easier not only to count primes with weight $\log p$ but to include the powers of primes as well. Let us denote the number of prime powers counted with weight $\log p$ by $\psi(x)$, so

$$\psi(x) = \sum_{p^n \leq x} \log p = \sum_{n \leq x} \Lambda(n).$$

The reason why it is more convenient to include the prime powers is that the Von Mangoldt's function has some nice multiplicative properties, in particular, it satisfies the formula

$$\log n = \sum_{d|n} \Lambda(d).$$

The following proposition shows us that the error we get by counting also the prime powers is relatively small.

Proposition 3.4. *For $x > 1$, we have*

$$|\psi(x) - \theta(x)| = O(\sqrt{x}).$$

Therefore it holds that

$$\psi(x) = O(x).$$

Proof. First we prove the bound on the difference between ψ and θ . We have

$$\begin{aligned} |\psi(x) - \theta(x)| &= \sum_{n=2}^{\infty} \sum_{p^n \leq x} \log p = \sum_{p \leq \sqrt{x}} \left(\left\lfloor \frac{\log x}{\log p} \right\rfloor - 1 \right) \log p = \\ &= \sum_{p \leq \sqrt{x}} (\log x - \log p) + O \left(\sum_{p \leq \sqrt{x}} \log p \right) \leq \\ &\leq \sum_{p \leq \sqrt{x}} \log x + O \left(\sum_{p \leq \sqrt{x}} \log p \right) \leq \\ &\leq \pi(\sqrt{x}) \log x + O(\theta(\sqrt{x})) = O(\sqrt{x}), \end{aligned}$$

where in the second equality, we interchanged summation (note that the sums are in fact finite) and in the last inequality we used Chebyshev's estimates and Proposition 3.3.

Combining this result with Proposition 3.3, we see that $\psi(x) = O(x)$. $\square$

With these results, we can now prove some elementary estimates of the rate of growth of various sums over primes and the prime powers. In the following results, we count the prime or their powers with a weight which gets smaller as the primes are larger (e.g. $\frac{1}{\log p}$ or even $\frac{1}{p}$), which makes it easier to find the main rate of growth than in the case of the functions $\pi(x)$, $\theta(x)$ or $\psi(x)$.

The following estimates are often called after Mertens.

Proposition 3.5. *For every $x > 1$, we have*

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + O(1).$$

Proof. We will use the formula $\log = \Lambda * 1$. Then

$$\sum_{n \leq x} \log n = \sum_{n \leq x} \sum_{d|n} \Lambda(d) = \sum_{d \leq x} \Lambda(d) \left\lfloor \frac{x}{d} \right\rfloor = x \sum_{d \leq x} \frac{\Lambda(d)}{d} + O\left(\sum_{d \leq x} \Lambda(d)\right).$$

Dividing both sides by x and subtracting the error term, we have

$$\sum_{d \leq x} \frac{\Lambda(d)}{d} = \frac{1}{x} \sum_{n \leq x} \log n + O\left(\frac{\psi(x)}{x}\right).$$

By propositions 1.7 and 3.4, we get the result

$$\sum_{d \leq x} \frac{\Lambda(d)}{d} = \log(x) + O(1).$$

□

Corollary 3.6. *For $x > 1$, we have*

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1).$$

Proof. We have

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} - \sum_{p \leq x} \frac{\log p}{p} = \sum_{\substack{p \leq x \\ m \geq 2}} \frac{\log p}{p^m} \leq \sum_{\substack{n \geq 1, \\ m \geq 2}} \frac{\log n}{n^m} = O(1),$$

so the result follows from Proposition 3.5. □

Proposition 3.7. *For $x > 1$, we have*

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + O(1).$$

Proof. We use partial summation with $a_n = \frac{p(n) \log n}{n}$ and $f(x) = \frac{1}{\log x}$. By Corollary 3.6, $A(t) = \log t + O(1)$, so by partial summation, we have

$$\begin{aligned} \sum_{p \leq x} \frac{1}{p} &= \sum_{n \leq x} a_n f(n) = (\log x + O(1)) \frac{1}{\log x} + \int_2^x \frac{\log t + O(1)}{t \log^2 t} dt = \\ &= \int_2^x \frac{1}{t \log t} dt + \int_2^x \frac{O(1)}{t \log^2 t} dt + O(1) = \log \log x + O(1), \end{aligned}$$

where the last equality holds because

$$\int_2^x \frac{O(1)}{t \log^2 t} dt \ll \int_2^\infty \frac{1}{t \log^2 t} dt \ll 1.$$

□

At the end of this chapter, let us state the Brun-Titchmarsh Theorem about the distribution of prime numbers in short intervals. We will omit the proof as it would require introducing the whole machinery of sieve methods, but it is fairly elementary and can be found for example in the third chapter of [Ko1]. The theorem has many possible statements, e.g., about primes in arithmetic progressions or with particular constants, but we will not need it in such generality. It also has an equivalent formulation when counting primes weighted with their logarithms, i.e., for the function $\psi(x)$.

Theorem 3.8 (Brun-Titchmarsh). *Let $1 < y < x$. Then*

$$\pi(x) - \pi(x - y) \ll \pi(y).$$

3.2 Equivalent formulations of the prime number theorem

In this section, we use elementary methods to provide some equivalent formulations of the prime number theorem.

Theorem 3.9. *The following assertions are equivalent:*

- (1) $\pi(x) \sim \frac{x}{\log x}$ as $x \rightarrow \infty$,
- (2) $\theta(x) \sim x$ as $x \rightarrow \infty$,
- (3) $\psi(x) \sim x$ as $x \rightarrow \infty$.

Proof. The equivalence of (2) and (3) follows from Proposition 3.4.

The equivalence of (1) and (2) follows from Proposition 3.3. $\square$

Our next goal is to show a relation between the prime number theorem and the Möbius function. This formulation will be especially important for us later in the pretentious approach because the Möbius function is multiplicative. Set

$$M(x) := \sum_{n \leq x} \mu(n).$$

Proposition 3.10. *For any $x \geq 1$, it holds that*

$$\left| \sum_{n \leq x} \frac{\mu(n)}{n} \right| \leq 1.$$

Proof. First check that for $x \leq 4$ the proposition is true.

Next write 1 as

$$1 = \sum_{n \leq x} \sum_{d|n} \mu(d) = \sum_{ab \leq x} \mu(a) = \sum_{a \leq x} \mu(a) \left\lfloor \frac{x}{a} \right\rfloor = x \sum_{a \leq x} \frac{\mu(a)}{a} - \sum_{a \leq x} \mu(a) \left\{ \frac{x}{a} \right\}.$$

From the last equality we get that

$$\left| \sum_{n \leq x} \frac{\mu(n)}{n} \right| = \left| \frac{1}{x} + \frac{1}{x} \sum_{a \leq x} \mu(a) \left\{ \frac{x}{a} \right\} \right| \leq \frac{1}{x} + \frac{1}{x} \sum_{a \leq x} |\mu(a)| \leq 1,$$

where the last inequality holds because $x > 4$ so $\sum_{a \leq x} |\mu(a)| \leq x - 1$. $\square$

Theorem 3.11. *The following are equivalent:*

- (1) $\psi(x) \sim x$ as $x \rightarrow \infty$,
- (2) $M(x) = o(x)$ as $x \rightarrow \infty$.

Proof. (1) $\Rightarrow$ (2) : First we prove that $\sum_{n \leq x} \mu(n) \log n = o(x \log x)$ as $x \rightarrow \infty$ implies (2). By partial summation, we have

$$\sum_{n \leq x} \mu(n) \log n = M(x) \log x - \int_1^x \frac{M(t)}{t} dt.$$

Since the integral is $O(x) = o(x \log x)$, we get

$$\frac{M(x)}{x} = \frac{1}{x \log x} \sum_{n \leq x} \mu(n) \log n + o(1).$$

Thus it suffices to show that

$$\sum_{n \leq x} \mu(n) \log n = o(x \log x).$$

Assume that (1) holds. By Lemma 1.3, $\Lambda = (-\mu \log) * 1$, so $-\mu \log = \Lambda * \mu$. By Dirichlet's hyperbola method for a suitable $y > 0$ to be chosen later, we obtain

$$\begin{aligned} - \sum_{n \leq x} \mu(n) \log(n) &= \sum_{n \leq x} (\Lambda * \mu)(n) = \\ &= \sum_{k \leq y} \Lambda(k) M\left(\frac{x}{k}\right) + \sum_{k \leq \frac{x}{y}} \mu(k) \psi\left(\frac{x}{k}\right) - \psi(y) M\left(\frac{x}{y}\right). \end{aligned} \quad (3.2)$$

Fix an arbitrary $\varepsilon > 0$ and fix y such that for $x \geq y$, it holds that $1 - \varepsilon \leq |\psi(x)/x| \leq 1 + \varepsilon$ (such a y exists because we assume (1)). In order to show that all the terms of (3.2) are $o(x \log x)$, we show that for x large enough, all terms on the right hand side are $\leq K \cdot \varepsilon \cdot x \log x + O(x)$ for a fixed constant K independent of ε :

- For the first term, if x large in terms of y , e.g., $x \geq y^{\frac{1}{\varepsilon}}$, using that $\left| M\left(\frac{x}{k}\right) \right| \leq \frac{x}{k}$ and Proposition 3.5, we have

$$\left| \sum_{k \leq y} \Lambda(k) M\left(\frac{x}{k}\right) \right| \leq x \sum_{k \leq y} \frac{\Lambda(k)}{k} = x \log y + O(x) \leq \varepsilon x \log x + O(x).$$

- For the second term, since we sum over terms where $\frac{x}{k} \geq y$, we can split the sum depending on the sign of $\mu(k)$ and appropriately estimate $\left| \psi\left(\frac{x}{k}\right) \right|$

using our choice of y . We get:

$$\begin{aligned}
\left| \sum_{k \leq \frac{x}{y}} \mu(k) \psi\left(\frac{x}{k}\right) \right| &\leq \left| \sum_{\substack{k \leq \frac{x}{y} \\ \mu(k) > 0}} \mu(k) \left(\frac{x}{k} + \varepsilon \frac{x}{k}\right) + \sum_{\substack{k \leq \frac{x}{y} \\ \mu(k) < 0}} \mu(k) \left(\frac{x}{k} - \varepsilon \frac{x}{k}\right) \right| \leq \\
&\leq \left| (x + \varepsilon x) \sum_{\substack{k \leq \frac{x}{y} \\ \mu(k) > 0}} \frac{1}{k} + (\varepsilon x - x) \sum_{\substack{k \leq \frac{x}{y} \\ \mu(k) < 0}} \frac{1}{k} \right| = \\
&= \varepsilon x \sum_{k \leq \frac{x}{y}} \frac{1}{k} + x \sum_{k \leq \frac{x}{y}} \frac{\mu(k)}{k} \leq \varepsilon \cdot x \log x + O(x)
\end{aligned}$$

where the last inequality holds by Proposition 1.6 and Proposition 3.10.

- For the third term, we have $\psi(y)M\left(\frac{x}{y}\right) \leq x \frac{\psi(y)}{y} \leq x + \varepsilon x = O(x)$.

Altogether, we showed that there is an absolute constant K such that for an arbitrary $\varepsilon > 0$ and x large enough, it holds that $-\sum_{n \leq x} \mu(n) \log n \leq K \cdot \varepsilon \cdot x \log x$, which means that the left-hand side of (3.2) is $o(x \log x)$, as we wanted to prove.

(2) $\Rightarrow$ (1) : Assume that (2) holds. We would like to prove that

$$\frac{1}{x} \sum_{n \leq x} (\Lambda(n) - 1) \rightarrow 0$$

as $x \rightarrow \infty$. Since $\Lambda = \mu * \log$ and $\tau = 1 * 1$, we can write

$$\Lambda - 1 = \mu * \log - \mu * \tau = \mu * (\log + c - \tau) - c,$$

where c is an arbitrary constant arithmetic function. If we set $c := 2\gamma$ and let $f(n) := \log n + c - \tau(n)$, then by propositions 1.7 and 1.10, we get for each $x > 0$ that

$$\left| \sum_{n \leq x} f(n) \right| = \left| x \log x - x + 2\gamma [x] - (x \log x + (2\gamma - 1)x) + O(\sqrt{x}) \right| \leq K \cdot \sqrt{x}$$

for some constant K . By Dirichlet's hyperbola method, if we let $F(x) := \sum_{n \leq x} f(n)$, then for each $y > 0$:

$$\begin{aligned}
\left| \sum_{n \leq x} (\Lambda(n) - 1) \right| &= \left| \sum_{n \leq x} (\mu * f)(n) \right| + O(1) = \\
&= \left| \sum_{d \leq y} \mu(d) F\left(\frac{x}{d}\right) + \sum_{\substack{d \leq \frac{x}{y}}} f(d) M\left(\frac{x}{d}\right) - M(y) F\left(\frac{x}{y}\right) \right| + O(1).
\end{aligned}$$

Fix an arbitrary $\varepsilon > 0$. By the assumptions, there exists a y such that if $x \geq y$, then $|M(x)| < \varepsilon \cdot x$. We finish the proof by estimating the last sum for this y and showing that the absolute value of each summand is $\leq C \cdot \varepsilon \cdot x$ with a fixed constant C independent of ε and x large enough.

- For the first sum, we get:

$$\left| \sum_{d \leq y} \mu(d) F\left(\frac{x}{d}\right) \right| \leq \sum_{d \leq y} \left| F\left(\frac{x}{d}\right) \right| \leq K\sqrt{x} \sum_{d \leq y} \frac{1}{\sqrt{d}} \leq K\sqrt{x} \cdot \lfloor y \rfloor \leq K\varepsilon x,$$

where the last estimate holds if x is large enough.

- For the second sum, since $d \leq \frac{x}{y}$ implies $\frac{x}{d} \geq y$, by our choice of y , we have

$$\left| \sum_{d \leq \frac{x}{y}} f(d) M\left(\frac{x}{d}\right) \right| \leq \varepsilon x \left| \sum_{d \leq \frac{x}{y}} \frac{f(d)}{d} \right|.$$

If we use partial summation for this sum, we get

$$\begin{aligned} \left| \sum_{d \leq \frac{x}{y}} \frac{f(d)}{d} \right| &= \left| F\left(\frac{x}{y}\right) \cdot \frac{y}{x} + \int_1^{\frac{x}{y}} \frac{F(t)}{t^2} dt \right| \leq \left| K\sqrt{\frac{x}{y}} \cdot \frac{y}{x} \right| + \left| \int_1^\infty \frac{K}{t^{\frac{3}{2}}} dt \right| = \\ &= K\sqrt{\frac{y}{x}} + O(1) = O(1), \end{aligned}$$

where the implied constant does not depend on y . Hence

$$\left| \sum_{d \leq \frac{x}{y}} f(d) M\left(\frac{x}{d}\right) \right| \leq C' \cdot \varepsilon x$$

for some constant C' independent of y and ε .

- For the last term, if x is large enough, we get

$$\left| M(y) F\left(\frac{x}{y}\right) \right| \leq \varepsilon \cdot y \cdot K\sqrt{\frac{x}{y}} = (K\varepsilon\sqrt{y})\sqrt{x} \leq K\varepsilon x.$$

Putting all the estimates together, we have

$$\sum_{n \leq x} (\Lambda - 1) \leq C\varepsilon x,$$

where C is a fixed constant which does not depend on epsilon. Since $\varepsilon > 0$ was arbitrary, this finishes the proof. $\square$

3.3 Complex analysis and Perron's formula

In this section, we begin by proving Perron's formula which will be our analytic tool used to estimate sums of arithmetic functions. It is a beautiful example of a link between two initially completely different fields of mathematics – complex analysis and number theory. One to observe and realize this amazing connection was Riemann, who wrote his first and only article about number theory in 1859. In this paper, he gave an overall strategy towards the proof of the prime number theorem, and it took more than three additional decades before all the details of the proof were finally carried out.

The main ingredient to Perron's formula is provided by the following lemma. It shows how to distinguish numbers greater than 1 from these smaller than 1 using a complex integral.

Throughout the thesis, by the integral

$$\int_{c-i\infty}^{c+i\infty} f(z)dz,$$

we mean Cauchy's principal value of the integral, i.e.,

$$\int_{c-i\infty}^{c+i\infty} f(z)dz = \lim_{T \rightarrow \infty} \int_{c-iT}^{c+iT} f(z)dz.$$

Lemma 3.12. *Let $x > 0$. Then for every $c > 0$, it holds that*

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^s}{s} ds = \begin{cases} 0 & \text{if } 0 < x < 1, \\ \frac{1}{2} & \text{if } x = 1, \\ 1 & \text{if } x > 1. \end{cases}$$

Proof. We divide the proof into three cases, depending on x .

$0 < x < 1$: Let $T > 0$ and denote by C_T the semicircle centered at c with radius T passing through the points $c - iT$, $c + T$ and $c + iT$ (oriented from $c + iT$ to $c - iT$). Since the function $\frac{x^s}{s}$ is holomorphic in the half-plane $\text{Re}(s) > 0$ and because $c > 0$, by Cauchy's theorem

$$\int_{c-iT}^{c+iT} \frac{x^s}{s} ds + \int_{C_T} \frac{x^s}{s} ds = 0.$$

To proof the theorem, it therefore suffices to show that the integral over C_T vanishes as T tends to ∞ . Parametrize the semicircle C_T as $\varphi(t) = c + Te^{-it}$, $t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and estimate the integral as follows:

$$\begin{aligned} \left| \int_{C_T} \frac{x^s}{s} ds \right| &= \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^{c+Te^{-it}}}{c + Te^{-it}} T i e^{-it} dt \right| \leq x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{|x^{Te^{-it}}|}{\left| \frac{c}{T} + e^{-it} \right|} dt \leq \\ &\leq x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^{T \text{Re}(e^{-it})} dt = x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^{T \cos t} dt. \end{aligned}$$

Let us fix an arbitrary $1 > \varepsilon > 0$. We can then write the last integral as follows:

$$\begin{aligned} x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^{T \cos t} dt &= x^c \int_{-\frac{\pi}{2}+\varepsilon}^{\frac{\pi}{2}-\varepsilon} x^{T \cos t} dt + x^c \int_{-\frac{\pi}{2}}^{-\frac{\pi}{2}+\varepsilon} x^{T \cos t} dt + x^c \int_{\frac{\pi}{2}-\varepsilon}^{\frac{\pi}{2}} x^{T \cos t} dt \leq \\ &\leq \pi x^c x^{T \cos(\frac{\pi}{2}-\varepsilon)} + 2x^c \varepsilon. \end{aligned}$$

Because $\cos(\frac{\pi}{2} - \varepsilon) > 0$ and $0 < x < 1$, $x^{T \cos(\frac{\pi}{2}-\varepsilon)}$ vanishes as T goes to ∞ . Hence if T is large enough, we get

$$x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^{T \cos t} dt \leq \varepsilon(1 + 2x^c).$$

Since ε was arbitrary, it follows that

$$\left| \int_{C_T} \frac{x^s}{s} ds \right| \leq x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^{T \cos t} dt \rightarrow 0,$$

so the first part is done.

$x > 1$: This case will be very similar to the previous one, except that we change the contour C_T to be the semicircle passing through the points $c - iT$, $c + iT$ and $c - T$. Hence, for $T > c$, the semicircle together with the line segment joining the points $c - iT$ and $c + iT$ contains the origin. The function $\frac{x^s}{s}$ has a pole of order 1 with residue 1 at $s = 0$, hence by Cauchy's residue theorem

$$\frac{1}{2\pi i} \left(\int_{c-iT}^{c+iT} \frac{x^s}{s} ds + \int_{C_T} \frac{x^s}{s} ds \right) = 1,$$

so to proof the lemma, it suffices to show that the integral over C_T vanishes as T tends to ∞ . We can parametrize C_T as $\varphi(t) = c + Te^{it}$ for $t \in [\frac{\pi}{2}, \frac{3\pi}{2}]$ so we get

$$\begin{aligned} \left| \int_{C_T} \frac{x^s}{s} ds \right| &= \left| \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{x^{c+Te^{it}}}{c + Te^{it}} Te^{it} dt \right| \leq x^c \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left| \frac{x^{Te^{it}}}{c + Te^{it}} T dt \right| \leq \\ &\leq x^c \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} x^{T \cos t} dt, \end{aligned}$$

and we can finish the proof similarly as in the previous case.

$x = 1$: We are going to directly compute the integral:

$$\int_{c-iT}^{c+iT} \frac{1}{s} ds = \int_{-T}^T \frac{i}{c + it} dt = \int_{-T}^T \frac{ic + t}{c^2 + t^2} dt = ic \int_{-T}^T \frac{1}{c^2 + t^2} dt + \int_{-T}^T \frac{t}{c^2 + t^2} dt.$$

Since $\frac{t}{c^2 + t^2}$ is an odd function, the second integral is equal to 0. Hence it suffices to evaluate the first integral and take the limit as T tends to infinity:

$$ic \int_{-T}^T \frac{1}{c^2 + t^2} dt = i \left[\arctan \left(\frac{t}{c} \right) \right]_{-T/c}^{T/c} = 2i \arctan \left(\frac{T}{c} \right) \xrightarrow{T \rightarrow \infty} \pi i.$$

Hence, indeed,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{1}{s} ds = \frac{1}{2}.$$

□

We use this lemma to prove the first version of Perron's formula, which shows the relation between counting discrete objects like prime numbers and evaluating complex integrals of Dirichlet's series. In the next theorem, we have some technical assumptions such as $x \notin \mathbb{Z}$, which will be omitted later in the truncated version of Perron's formula with error term.

Theorem 3.13. *Let f be an arithmetic function, $c > 0$ and suppose that there exists an $\varepsilon > 0$ such that $L(f, s)$ converges absolutely for $\operatorname{Re}(s) > c - \varepsilon$. Then for any $x > 0$, $x \notin \mathbb{Z}$*

$$\sum_{n < x} f(n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} L(f, s) \cdot \frac{x^s}{s} ds.$$

Proof. Let us define an indicator function $I(n)$, which is 1 if $n < x$ and 0 otherwise. The condition can be rewritten as

$$I(n) = \begin{cases} 1, & \text{if } \frac{x}{n} > 1, \\ 0, & \text{if } \frac{x}{n} < 1. \end{cases}$$

We can now use Lemma 3.12 with $\frac{x}{n}$ in the role of x to get

$$\begin{aligned} \sum_{n < x} f(n) &= \sum_{n=1}^{\infty} f(n) I(n) = \sum_{n=1}^{\infty} f(n) \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\left(\frac{x}{n}\right)^s}{s} ds = \\ &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \sum_{n=1}^{\infty} \frac{f(n)}{n^s} \cdot \frac{x^s}{s} ds = \\ &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} L(f, s) \cdot \frac{x^s}{s} ds, \end{aligned}$$

which was to be proved. The interchange of summation and integral is justified by uniformity of the convergence of $L(f, s)$ on the line $\operatorname{Re}(s) = c$. $\square$

Of particular importance will be the previous theorem used for the function $f = \Lambda$.

Corollary 3.14. *For any $x > 1$, $x \notin \mathbb{Z}$ and $c > 1$, we have*

$$\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds.$$

Proof. Since $L(\Lambda, s) = -\frac{\zeta'}{\zeta}(s)$ is absolutely convergent for $\operatorname{Re}(s) > 1$, the assertion follows directly from the previous theorem. $\square$

Note that the above result is indeed an exact formula for $\psi(x)$, not just an approximation!

Instead of evaluating complex integrals over an infinite vertical line, it is usually easier to truncate it at some height T and estimate the error term which arises because of that. This is the subject of the following lemma.

Lemma 3.15. *There exists a positive constant K such that for any $c > 0$, $T \geq 1$*

$$\text{and } x > 0, \text{ if we denote by } \delta(x) = \begin{cases} 0 & \text{if } 0 < x < 1, \\ \frac{1}{2} & \text{if } x = 1, \\ 1 & \text{if } x > 1, \end{cases} \quad \text{then}$$

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds - \delta(x) \right| \leq \begin{cases} \frac{Kx^c}{T|\log x|+1} & \text{if } x > 0, x \neq 1, \\ \frac{Kc}{T} & \text{if } x=1. \end{cases}$$

Proof. Let $x > 0$. We divide the proof into three cases.

$0 < x < 1$: Let $T \geq 1$. First suppose that $T|\log x| \geq 1$, let $k > c$ and consider the rectangle with vertices $c - iT$, $c + iT$, $k + iT$, $k - iT$. By Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds = \frac{1}{2\pi i} \int_{c-iT}^{k-iT} \frac{x^s}{s} ds + \frac{1}{2\pi i} \int_{k-iT}^{k+iT} \frac{x^s}{s} ds - \frac{1}{2\pi i} \int_{c+iT}^{k+iT} \frac{x^s}{s} ds,$$

so it suffices to estimate the integrals on the right-hand side. We can make the following estimates:

$$\frac{1}{2\pi} \left| \int_{c-iT}^{k-iT} \frac{x^s}{s} ds - \int_{c+iT}^{k+iT} \frac{x^s}{s} ds \right| \leq \frac{1}{\pi} \int_c^k \frac{x^s}{T} ds = \frac{x^c - x^k}{T|\log x|} \leq \frac{x^c}{T|\log x|},$$

and

$$\left| \frac{1}{2\pi i} \int_{k-iT}^{k+iT} \frac{x^s}{s} ds \right| \leq \frac{x^k}{2\pi} \int_{-T}^T \frac{1}{k} ds \xrightarrow{k \rightarrow \infty} 0.$$

Hence letting $k \rightarrow \infty$, we get

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds \leq \frac{1}{\pi} \cdot \frac{x^c}{T|\log x|}.$$

Since we assumed $T|\log x| \geq 1$, for $K' > 2$, we have $\frac{x^c}{T|\log x|} \leq \frac{K'x^c}{T|\log x|+1}$, so for this K' ,

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds \right| \leq \frac{K'}{\pi} \cdot \frac{x^c}{T|\log x|+1}.$$

Now suppose that $T|\log x| \leq 1$, so $T|\log x| + 1 \leq 2$. Let γ denote the part of the circle centered at 0, passing through the points $c+iT$ and $c-iT$ to the right of the line $\text{Re}(s) = c$. Then by Cauchy's theorem

$$\begin{aligned} \left| \frac{1}{2\pi} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds \right| &= \frac{1}{2\pi} \left| \int_{\gamma} \frac{x^s}{s} ds \right| \leq \frac{1}{2\pi\sqrt{c^2+T^2}} \int_{\gamma} x^c ds \leq \\ &\leq x^c \leq \frac{Kx^c}{T|\log x|+1}, \end{aligned}$$

where the last inequality holds for any $K > 2$.

$x > 1$: This case will be very similar to the previous one, except that we will shift the contours to the left, so we cross the singularity at $s = 0$, which is a simple pole of residue 1.

First suppose that $T|\log x| \geq 1$ and pick $k < 0$. By Cauchy's theorem, we have

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds = 1 - \frac{1}{2\pi i} \int_{c+iT}^{k+iT} \frac{x^s}{s} ds - \frac{1}{2\pi i} \int_{k+iT}^{k-iT} \frac{x^s}{s} ds - \frac{1}{2\pi i} \int_{k-iT}^{c-iT} \frac{x^s}{s} ds.$$

Hence

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds - 1 \right| \leq \frac{1}{2\pi} \int_{k+iT}^{c+iT} \left| \frac{x^s}{s} \right| ds + \frac{1}{2\pi} \int_{k-iT}^{c-iT} \left| \frac{x^s}{s} \right| ds + \frac{1}{2\pi} \int_{k-iT}^{k+iT} \left| \frac{x^s}{s} \right| ds,$$

and we can estimate these integrals as in the previous case:

$$\frac{1}{2\pi} \int_{k+iT}^{c+iT} \left| \frac{x^s}{s} \right| ds + \frac{1}{2\pi} \int_{k-iT}^{c-iT} \left| \frac{x^s}{s} \right| ds \leq \frac{1}{\pi T} \int_k^c x^s ds = \frac{x^c - x^k}{\pi T \log x} \leq \frac{x^c}{\pi T \log x},$$

and

$$\frac{1}{2\pi} \int_{k-iT}^{k+iT} \left| \frac{x^s}{s} \right| ds \leq \frac{x^k}{2\pi} \int_{-T}^T \frac{1}{|k|} ds \xrightarrow{k \rightarrow -\infty} 0,$$

so sending k to ∞ , we get for any $K' > 2$

$$\left| \frac{1}{2\pi} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds - 1 \right| \leq \frac{1}{\pi} \cdot \frac{x^c}{T \log x} \leq \frac{K'}{\pi} \cdot \frac{x^c}{T \log x + 1}.$$

If $T \log x \leq 1$, let γ be the part of the circle centered at 0, passing through the points $c - iT$, $c + iT$ and containing 0. Then by Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds = 1 - \frac{1}{2\pi i} \int_{\gamma} \frac{x^s}{s} ds,$$

so subtracting 1 and taking absolute values, we get

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s} ds - 1 \right| \leq \frac{1}{2\pi} \int_{\gamma} \left| \frac{x^s}{s} \right| ds \leq \frac{1}{2\pi T} \int_{\gamma} x^c ds \leq x^c \leq \frac{K' x^c}{T \log x + 1},$$

where the last inequality holds for any $K' > 2$.

This finishes the proof where $y \neq 1$ and we see that it holds for any constant $K > 2$.

$x = 1$: It remains to handle the last case, where we can compute the integral directly. We have,

$$\begin{aligned} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{1}{s} ds &= \frac{1}{2\pi} \int_{-T}^T \frac{1}{c + is} ds = \frac{1}{2\pi} \int_0^T \frac{1}{c + is} + \frac{1}{c - is} ds = \\ &= \frac{1}{\pi} \int_0^T \frac{c}{c^2 + s^2} ds = \frac{1}{\pi} \int_0^{\frac{T}{c}} \frac{1}{1 + t^2} dt, \end{aligned}$$

and the last integral is equal to

$$\frac{1}{\pi} \int_0^{\infty} \frac{1}{1 + t^2} dt - \frac{1}{\pi} \int_{\frac{T}{c}}^{\infty} \frac{1}{1 + t^2} dt \leq \frac{1}{2} + \int_{\frac{T}{c}}^{\infty} \frac{1}{t^2} dt = \frac{1}{2} + O\left(\frac{c}{T}\right).$$

□

We are now ready to state and prove Perron's formula.

Theorem 3.16 (Perron's formula). *Let f be an arithmetic function, $T \geq 1$ and c such the Dirichlet series $L(f, s)$ converges absolutely for $\text{Re}(s) > c - \varepsilon$ for some $\varepsilon > 0$. Then for any $x > 1$, we have*

$$\sum_{n \leq x} f(n) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} L(f, s) \cdot \frac{x^s}{s} ds + O\left(f(x) + \sum_{n=1}^{\infty} \frac{f(n)}{n^c} \cdot \frac{x^c}{T |\log \frac{x}{n}| + 1}\right),$$

$$\text{where } f(x) = \begin{cases} 0, & \text{if } x \text{ is not an integer,} \\ f(n), & \text{if } x=n \text{ is an integer.} \end{cases}$$

Proof. First suppose that x is not an integer. we will proceed similarly as in the case of the non-truncated version. Let

$$I(n) = \begin{cases} 1, & \text{if } \frac{x}{n} > 1, \\ 0, & \text{if } \frac{x}{n} < 1. \end{cases}$$

Then by Lemma 3.15, we have

$$\begin{aligned}\sum_{n < x} f(n) &= \sum_{n=1}^{\infty} f(n) I(n) = \sum_{n=1}^{\infty} f(n) \left[\int_{c-iT}^{c+iT} \frac{\left(\frac{x}{n}\right)^s}{s} ds + O\left(\frac{\left(\frac{x}{n}\right)^c}{T|\log \frac{x}{n}| + 1}\right) \right] = \\ &= \int_{c-iT}^{c+iT} L(f, s) \cdot \frac{x^s}{s} ds + O\left(\sum_{n=1}^{\infty} \frac{f(n)}{n^c} \cdot \frac{x^c}{T|\log \frac{x}{n}| + 1}\right),\end{aligned}$$

which we wanted. Again, the interchange of summation and integral is justified by uniform convergence of the Dirichlet series on the line $\operatorname{Re}(s) = c$.

If x is an integer, we must add to the error term $\frac{1}{2}f(x)$, so we are left with $O\left(f(x) + \sum_{n=1}^{\infty} \frac{f(n)}{n^c} \cdot \frac{x^c}{T|\log \frac{x}{n}| + 1}\right)$, as desired. $\square$

4. The classical proof of the prime number theorem

In this chapter, we provide a classical proof of the prime number theorem in the following form:

Theorem 4.1. *There exists a positive constant K , such that for $x > 1$,*

$$\psi(x) = x + O\left(xe^{-K(\log x)^{1/10}}\right).$$

We already know that this is equivalent to

Theorem 4.2. *There exists a positive constant K , such that for $x > 1$,*

$$\pi(x) = \text{Li}(x) + O\left(xe^{-K(\log x)^{1/10}}\right).$$

Our proof can be found in [Mu], other variants are presented in [Ko1] or [Ap].

The error term looks rather complicated, but writing it as a single exponential we can see that it is better than $O\left(\frac{x}{(\log x)^A}\right)$ for any A , but worse than $O(x^{1-\varepsilon})$ for any $\varepsilon > 0$.

Let us first describe the method we are going to use and give some intuitive outline of the proof.

The basic idea presented by Riemann is to use Perron's formula and estimate the integral using Cauchy's residue theorem. To do so, we need to shift the contour of integration to the left, where the term $\frac{x^s}{s}$ becomes small, and then count the contribution of residues of the poles inside the contour.

The poles of the integrand $-\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s}$ are at the following points:

- At $s = 0$ with residue $-\frac{\zeta'(0)}{\zeta(0)}$.
- At the poles of ζ . The only pole of ζ is at $s = 1$, which is simple with residue 1, so the residue of the integrand at $s = 1$ is x .
- At zeros of ζ . If ρ is a zero of multiplicity m , the residue of $-\zeta'/\zeta$ at ρ is $-m$, hence the residue of the integrand at ρ is $-m \cdot \frac{x^\rho}{\rho}$.

The initial line of integration must be as close as possible to the line $\text{Re}(s) = 1$ because of x^c in the error term, but also far enough so that $\zeta(s)$ is not too large. A good choice is usually $c = 1 + \frac{1}{\log x}$. If we enclose this line into a contour C which contains the pole of ζ at 1 and does not contain 0, we get

$$\psi(x) = x - \sum_{\substack{\rho \text{ inside } C \\ \zeta(\rho)=0}} \frac{x^\rho}{\rho} + \text{error}, \quad (4.1)$$

where we sum over each zero as many times as its multiplicity, and the error comes from the truncated Perron's formula and an estimate on the integrand $-\frac{\zeta'}{\zeta} \cdot \frac{x^s}{s}$ along C without the line $\text{Re}(s) = 1 + \frac{1}{\log x}$.

From (4.1) we see that the main term comes from the pole at $s = 1$ and assuming that “error” is small, we see that it is important to count the contribution of the zeros of ζ .

We can estimate the difference taking absolute values:

$$|\psi(x) - x| \ll \left| \sum_{\substack{\rho \text{ inside } C \\ \zeta(\rho)=0}} \frac{x^\rho}{\rho} \right| \ll \sum_{\substack{\rho \text{ inside } C \\ \zeta(\rho)=0}} \frac{x^{\operatorname{Re}(\rho)}}{|\rho|},$$

so we see that the size of the error mostly depends on the real part of the zeros of the zeta function.

In his memoir, Riemann stated his famous and still unproven hypothesis that all the non-trivial zeros lie on the so-called critical line $\operatorname{Re}(s) = 1/2$ (the trivial zeros are in the negative odd integers and can be easily detected using the functional equation).

Most analytic proofs of the prime number theorem depend on finding a zero-free region of the zeta function. The converse is also true, so good error terms in the prime number theorem give zero-free regions of the Riemann zeta function. This is the reason why most mathematicians thought that the prime number theorem cannot be proved using only “elementary methods”, e.g., without the use of complex analysis. They were proven wrong when Selberg and Erdős gave a completely elementary proof in 1948, and more different proofs were found since then.

The Riemann hypothesis is equivalent to the best possible error term $O(\sqrt{x})$, because it can be proved that the zeros of $\zeta(s)$ are distributed symmetrically about the critical line, so if $\zeta(a+bi) = 0$ with $0 \leq a \leq 1/2$, then also $\zeta((1-a)+bi) = 0$. To illustrate how far we are from proving the conjecture, note that it is not even known whether there exists an $\varepsilon > 0$ such that $\operatorname{re}(\rho) < 1 - \varepsilon$. The best error term so far was proved by Korobov and Vinogradov in 1958 and has not been significantly improved for almost 60 years.

The following proof can be found in chapter 4 of [Mu]

We already know that the zeta function does not have any zeros with $\operatorname{Re}(s) > 1$. At the time of Hadamard and de la Vallée Poussin, it was already known that to prove the prime number theorem, it is enough to show that no zeros lie on the line $\operatorname{Re}(s) = 1$. We will prove this using a nice (though not very enlightening) formula of Mertens.

Lemma 4.3. *For any $\theta \in \mathbb{R}$, $3 + 4\cos(\theta) + \cos(2\theta) \geq 0$*

Proof. We can use the basic trigonometric identities to get

$$\begin{aligned} 3 + 4\cos(\theta) + \cos(2\theta) &= 3 + 4\cos(\theta) + \cos^2(\theta) - \sin^2(\theta) = \\ &= 2 + 4\cos(\theta) + 2\cos^2(\theta) = 2(1 + \cos(\theta))^2 \geq 0. \end{aligned}$$

□

Theorem 4.4 (Mertens Formula). *Let $\sigma > 1$. Then for any $t \in \mathbb{R}$, it holds that*

$$|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| \geq 1.$$

Proof. Using the Euler product and the Taylor expansion of logarithm, we get for $s = \sigma + it$ with $\sigma > 1$:

$$\begin{aligned}\log \zeta(\sigma + it) &= - \sum_p \left(\log \left(1 - \frac{1}{p^s} \right) \right) = \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{ms}} = \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\sigma}} p^{-it} = \\ &= \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\sigma}} (\cos(\theta_p) + i \sin(\theta_p)),\end{aligned}$$

where $\theta_p = -t \log p$. Therefore

$$\log |\zeta(\sigma + it)| = \operatorname{Re}(\log \zeta(\sigma + it)) = \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\sigma}} \cos(\theta_p),$$

so for any $t \in \mathbb{R}$, we have

$$\begin{aligned}3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| &= \\ &= \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\sigma}} (3 + 4 \cos(\theta_p) + \cos(2\theta_p)) \geq 0,\end{aligned}$$

where the last inequality holds because by Lemma 4.3, all the summands are nonnegative. To finish the proof, it suffices to exponentiate the last inequality. $\square$

Theorem 4.5. *The Riemann zeta function does not have any zero ρ with $\operatorname{Re}(\rho) = 1$.*

Proof. Suppose for contradiction that for some $t \in \mathbb{R}$, $\zeta(1 + it) = 0$. Then $t \neq 0$ so $\zeta(1 + 2it) \neq \infty$. Since the pole of ζ at 1 is simple

$$\lim_{\sigma \rightarrow 1^+} |\zeta(\sigma)\zeta(\sigma + it)| < \infty,$$

so

$$\lim_{\sigma \rightarrow 1^+} |\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| = 0.$$

This is in contradiction with Theorem 4.4, so such a zero cannot exist. $\square$

We will use Mertens formula to further widen the zero-free region beyond the 1-line. To do so, we will need some estimates on ζ and ζ' in this area. To do so, we will use formula (2.2).

For the rest of this section, for any $T > e^2$, we denote by R_T the rectangle with vertices $2 + iT, 2 - iT, \sigma_T - iT, \sigma_T + iT$, where $\sigma_T = 1 - \frac{1}{\log T}$

Proposition 4.6. *For $s \in R_T$, we have*

$$\zeta(s) - \frac{1}{s-1} = O(\log T)$$

Proof. Since we already know that ζ is meromorphic with only one simple pole at $s = 1$ with residue 1, $\zeta(s) - \frac{1}{s-1}$ is continuous, hence bounded, in some region around 1, say $|s - 1| \leq \frac{1}{\log T}$. For the rest of the proof, we may therefore assume that $|s - 1| > \frac{1}{\log T}$.

By Theorem 2.7 we have

$$\zeta(s) - \frac{1}{s-1} = \sum_{n=1}^{N-1} \frac{1}{n^s} + \frac{N^{-s}}{2} + \frac{N^{1-s} - 1}{s-1} - s \int_N^\infty \frac{t - \lfloor t \rfloor - \frac{1}{2}}{t^{s+1}} dt.$$

We choose $N = \lfloor T \rfloor$. Then

$$\frac{N^{1-s} - 1}{s-1} = \int_1^N \frac{1}{t^s} dt \leq \int_1^T \frac{1}{t^s} dt,$$

and

$$\sum_{n=1}^{N-1} \frac{1}{n^s} \leq 1 + \int_1^T \frac{1}{t^s} ds,$$

so

$$\left| \zeta(s) - \frac{1}{s-1} \right| \leq \left| 1 + \frac{N^{-s}}{2} + 2 \int_1^T \frac{1}{t^s} dt - s \int_N^\infty \frac{t - \lfloor t \rfloor - \frac{1}{2}}{t^{s+1}} dt \right|.$$

The term $1 + \frac{N^{-s}}{2} = 1 + \frac{1}{2\lfloor T \rfloor^s} \leq 1 + \frac{1}{2T^{1-\frac{1}{\log T}}} = O(1)$, because $T^{\frac{1}{\log T}} = e$. Hence we are left with estimating the two integrals.

Write $s = \sigma + iu$. Then for the first integral, we have

$$2 \int_1^T \left| \frac{1}{t^s} \right| dt \leq 2 \int_1^T \frac{1}{t^\sigma} dt \leq 2 \int_1^T \frac{1}{t^{\sigma_T}} dt = 2 \left(\frac{T^{1-\sigma_T} - 1}{\sigma_T - 1} \right) = O(\log T).$$

For the second integral, note that $|t - \lfloor t \rfloor - \frac{1}{2}| \leq \frac{1}{2}$, so

$$\begin{aligned} \left| s \int_N^\infty \frac{t - \lfloor t \rfloor - \frac{1}{2}}{t^{s+1}} dt \right| &\leq \frac{|s|}{2} \int_{\lfloor T \rfloor}^\infty \frac{1}{t^{\sigma+1}} dt \leq \frac{|s|}{2} \cdot \frac{1}{\sigma(T-1)^\sigma} \leq \frac{|s|}{2\sigma_T(T-1)^{\sigma_T}} \leq \\ &\leq \frac{T+2}{T-1} \cdot \frac{e}{1 - \frac{1}{\log T}}, \end{aligned}$$

where we used that $|s| \leq T+2$ for $s \in R_T$. Note that the last function is bounded as $T \rightarrow \infty$, which finishes the proof. $\square$

Proposition 4.7. *For s on the boundary of R_T , we have $\zeta(s) = O(\log T)$.*

Proof. For s on the boundary of R_T , we have $|s-1| \geq \min(\frac{1}{\log T}, T)$, so $\left| \frac{1}{s-1} \right| \leq \max(\log T, \frac{1}{T}) = O(\log T)$. Hence we can write $|\zeta(s)| \leq \left| \zeta(s) - \frac{1}{s-1} \right| + \left| \frac{1}{s-1} \right| = \left| \zeta(s) - \frac{1}{s-1} \right| + O(\log T)$ and use similar estimates as in Proposition 4.6 to get $\left| \zeta(s) - \frac{1}{s-1} \right| = O(\log T)$ for s on the boundary of R_T . $\square$

Now, we prove a similar estimate for the derivative of the zeta function.

Proposition 4.8. *For $s \in R_T$, we have*

$$\zeta'(s) + \frac{1}{(s-1)^2} = O((\log T)^2).$$

Proof. Differentiating formula (2.2), we get for any $N \in \mathbb{N}$

$$\begin{aligned} \zeta'(s) = & - \sum_{n=1}^{N-1} \frac{\log n}{n^s} - \frac{\log N}{2N^s} + \frac{(1-s) \log N \cdot N^{1-s} - N^{1-s}}{(s-1)^2} - \\ & - \int_N^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} dt + s \int_N^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} \log t dt \end{aligned}$$

We note that

$$\frac{(1-s) \log N \cdot N^{1-s} - N^{1-s} + 1}{(s-1)^2} = \int_1^N \frac{\log t}{t^s} dt,$$

so

$$\begin{aligned} \zeta'(s) + \frac{1}{(s-1)^2} = & - \sum_{n=1}^{N-1} \frac{\log n}{n^s} - \frac{\log N}{2N^s} + \int_1^N \frac{\log t}{t^s} dt - \\ & - \int_N^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} dt + s \int_N^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} \log t dt. \end{aligned}$$

Let $N = [T]$. We can estimate the first, second and fourth term similarly as in 4.6, so we are left with estimating two integrals.

- For the first integral, we have:

$$\left| \int_1^N \frac{\log t}{t^s} dt \right| \leq \int_1^T \frac{\log t}{t^{\sigma_T}} dt \leq \log T \cdot T^{1/\log T} \int_1^T \frac{1}{t} dt \ll (\log T)^2.$$

- For the last integral, we have:

$$\begin{aligned} \left| s \int_N^\infty \frac{t - [t] - \frac{1}{2}}{t^{s+1}} \log t dt \right| & \leq \left| \frac{s}{2} \right| \int_N^\infty \frac{\log t}{t^{\sigma_T+1}} dt \ll \\ & \ll \frac{T+2}{T} \cdot \log T + (T+2) \int_N^\infty \frac{1}{t^{\sigma_T+1}} dt \ll \\ & \ll \log T. \end{aligned}$$

□

Proposition 4.9. *For s on the boundary of R_T , $\zeta'(s) = O((\log T)^2)$.*

Proof. The proof is a modification of the above proof similar to the proof of 4.7. □

Now we use Merten's formula and the mean value theorem to find a wider zero-free region for $\zeta(s)$.

Theorem 4.10. *There are positive constants a, b such that if $1 \leq |\operatorname{Im}(s)| \leq T$ and $1 - \frac{a}{(\log T)^9} \leq \operatorname{Re}(s) \leq 2$, then*

$$|\zeta(s)| > \frac{b}{(\log T)^7}.$$

Proof. Mertens formula tells us that $|\zeta(\sigma)^3\zeta(\sigma+it)^4\zeta(\sigma+2it)| \geq 1$ for $\sigma > 1$. Therefore

$$|\zeta(\sigma+it)|^4 \geq \zeta(\sigma)^{-3}\zeta(\sigma+2it)^{-1}.$$

Since $(\sigma-1)\zeta(\sigma)$ is continuous for $1 \leq \sigma \leq 2$, it can be bounded by a constant K' as

$$|\zeta(\sigma)(\sigma-1)| \leq K',$$

therefore

$$|\zeta(\sigma)^{-3}| \geq K'' |(\sigma-1)^3|$$

for a constant K'' .

By Proposition 4.6, there is a constant K_1 such that

$$|\zeta(\sigma+2it)|^{-1} \geq K_1(\log T)^{-1}.$$

Putting these estimates together, we have

$$|\zeta(\sigma+it)|^4 \geq K(\sigma-1)^3(\log T)^{-1}$$

for some constant K and $1 < \sigma \leq 2$.

If $1 + \frac{a}{(\log T)^9} \leq \sigma \leq 2$, for a constant a to be chosen later, then $(\sigma-1)^3 \gg (\log T)^{-27}$, so for such σ , we have

$$|\zeta(\sigma+it)|^4 \gg (\log T)^{-28}.$$

Therefore for a positive constant b' ,

$$|\zeta(\sigma+it)| \geq \frac{b'}{(\log T)^7}.$$

To finish the proof, we will use the mean value theorem to widen the region where the last inequality holds.

Let $1 - \frac{a}{(\log T)^9} \leq \sigma \leq 1 + \frac{a}{(\log T)^9}$, $1 < t \leq T$ and $s = \sigma + it$. Let $\sigma' = 1 + \frac{a}{(\log T)^9}$ and $s' = \sigma' + it$. By the mean value theorem for complex functions, there exist two points u, v on the line segment joining s and s' such that

$$\begin{aligned} \operatorname{Im}(\zeta(s') - \zeta(s)) &= (\sigma' - \sigma) \operatorname{Im}(\zeta'(u)), \\ \operatorname{Re}(\zeta(s') - \zeta(s)) &= (\sigma' - \sigma) \operatorname{Re}(\zeta'(u)). \end{aligned}$$

By Proposition 4.8, there exists a constant $C > 0$ such that

$$|\zeta(s') - \zeta(s)| \leq C((\sigma' - \sigma)(\log T)^2).$$

Hence

$$|\zeta(s)| \geq |\zeta(s')| - C((\sigma' - \sigma)(\log T)^2) \geq \frac{b' - 2Ca}{(\log T)^7}.$$

Therefore if we choose a small enough such that $b = b' - 2Ca > 0$, the theorem is proved. $\square$

Corollary 4.11. *There is a positive constant a such that for all $s = \sigma + it$ with $1 \leq |t| \leq T$ and $1 - \frac{a}{(\log T)^9} \leq \sigma \leq 2$,*

$$\frac{\zeta'}{\zeta}(s) \ll (\log T)^9.$$

Proof. We can take a as in Theorem 4.10 and use Proposition 4.8. Then for each s satisfying the assumptions, we have that

$$\frac{\zeta'}{\zeta}(s) \ll \frac{(\log T)^2}{(\log T)^{-7}} \ll (\log T)^9.$$

□

In the next proposition, we estimate the error term given by Perron's formula used for the von Mangoldt's function.

Proposition 4.12. *Let $c = 1 + \frac{a}{(\log T)^9}$ with a from Corollary 4.11. Then for any $T, x > 1$, we have*

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^c} \cdot \frac{x^c}{T \left| \log \frac{x}{n} \right| + 1} = O \left(x^c \frac{(\log T)^9}{T} + \frac{x(\log x)^2}{T} \right).$$

Proof. First note that by partial summation and Proposition 3.5, we have for any $k > 1$ that

$$\sum_{n \leq x} \frac{\Lambda(n)}{n^k} = \sum_{n \leq x} \frac{\Lambda(n)}{n} \cdot n^{1-k} = \frac{O(\log x)}{x^{k-1}} + (k-1) \int_1^x \frac{O(\log t)}{t^k} dt.$$

Sending x to ∞ and computing the integral by parts, we have

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^k} = O \left(\frac{1}{k-1} \right).$$

Now we will split the sum we are computing according to the size of $\left| \log \frac{x}{n} \right|$.

First, if $n \leq \frac{x}{2}$ or $n \geq \frac{3x}{2}$, then $\left| \log \frac{x}{n} \right| \geq \log \frac{3}{2}$, so the summation over these n is bounded by

$$\frac{x^c}{T \log \frac{3}{2} + 1} \cdot \sum_{\substack{n \leq \frac{x}{2} \\ n \geq \frac{3x}{2}}} \frac{\Lambda(n)}{n^c} = O \left(\frac{x^c (\log T)^9}{T} \right),$$

where we used the preceding estimate on the sum $\sum_n \frac{\Lambda(n)}{n^c}$.

For $\frac{x}{2} < n < \frac{3x}{2}$, let $z = 1 - \frac{n}{x}$. Then $|z| < \frac{1}{2}$ and

$$\log \frac{x}{n} = -\log(1-z) = z + \frac{z^2}{2} + \dots,$$

so for $|z| \leq \frac{1}{2}$, we have $\left| \log \frac{x}{n} \right| \geq |z|(1 + \frac{z}{2}) \geq \frac{3}{4}|z|$. Therefore we get the estimate

$$\begin{aligned} \sum_{\frac{x}{2} < n < \frac{3x}{2}} \left(\frac{x}{n} \right)^c \cdot \frac{\Lambda(n)}{T \left| \log \frac{x}{n} \right| + 1} &\ll \frac{2^c \log \left(\frac{3x}{2} \right)}{T} \cdot \sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{\left| \log \frac{x}{n} \right| + 1} \ll \\ &\ll \left(\frac{\log x}{T} \cdot \sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{\left| \log \frac{x}{n} \right| + 1} \right), \end{aligned}$$

so it suffices to show that the last sum is $O(x \log x)$. We use the estimate $\left| \log \frac{x}{n} \right| \geq \frac{3}{4} \left| 1 - \frac{n}{x} \right|$ to get

$$\sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{\left| \log \frac{x}{n} \right| + 1} \leq \sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{\frac{3}{4} \left| 1 - \frac{n}{x} \right| + 1} = x \cdot \sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{|x - n| + x}.$$

Now note that as n ranges from $\frac{x}{2}$ to $\frac{3x}{2}$, $|x - n|$ ranges through $\left\lfloor \frac{x}{2} \right\rfloor + \{x\} - 1, \left\lfloor \frac{x}{2} \right\rfloor + \{x\} - 2, \dots, \{x\}, 1 - \{x\}, \dots, \left\lfloor \frac{x}{2} \right\rfloor - \{x\}$. Hence

$$\sum_{\frac{x}{2} < n < \frac{3x}{2}} \frac{1}{|x - n| + x} \leq 2 \sum_{n \leq \frac{3x}{2} + 1} \frac{1}{n} = O(\log x),$$

which finishes the proof. $\square$

We are finally ready to prove the prime number theorem.

proof of theorem 4.1: Let $x > 1$ and $c = 1 + \frac{a}{(\log T)^9}$ with a from Corollary 4.11 and T to be chosen later. By Perron's formula and Proposition 4.12, we get that

$$\psi(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds + O\left(\frac{x^c(\log T)^9}{T} + \frac{x(\log x)^2}{T}\right).$$

Let $b = 1 - \frac{a}{(\log T)^9}$. The only singularity of the integrand in the rectangle with vertices $c + iT$, $c - iT$, $b - iT$, and $b + iT$ is the simple pole at 1 with residue x , we can therefore use Cauchy's Residue Theorem to obtain

$$x + \frac{1}{2\pi i} \left(\int_{c+iT}^{b+iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds + \int_{b+iT}^{b-iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds + \int_{b-iT}^{c-iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds \right).$$

We estimate the integrals using Corollary 4.11:

$$\frac{1}{2\pi i} \left| \int_{b+iT}^{c+iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds \right| \ll (\log T)^9 \frac{2a}{(\log T)^9} \cdot \frac{x^c}{T} \ll \frac{x^c(\log T)^9}{T},$$

a similar estimate holds for the other horizontal integral

$$\frac{1}{2\pi i} \left| \int_{b-iT}^{c-iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds \right| \ll (\log T)^9 \frac{2a}{(\log T)^9} \cdot \frac{x^c}{T} \ll \frac{x^c(\log T)^9}{T}.$$

For the last integral, we have

$$\frac{1}{2\pi i} \left| \int_{b-iT}^{b+iT} -\frac{\zeta'}{\zeta}(s) \cdot \frac{x^s}{s} ds \right| \ll (\log T)^9 x^b \int_{-T}^T \frac{1}{|b + is|} ds \ll x^b (\log T)^{10}.$$

Hence we get

$$\psi(x) = x + O\left(\frac{x^c(\log T)^9}{T} + x^b(\log T)^{10} + \frac{x(\log x)^2}{T}\right)$$

and it suffices to choose the right T .

We will show that T such that $2a \log x = (\log T)^{10}$ gives the right error term. For such T , we have $\log T = (2a)^{1/10}(\log x)^{1/10}$ so we can estimate the error terms.

For the first term,

$$\begin{aligned} \frac{x^c(\log T)^9}{T} &= x \cdot \exp \left(\frac{a}{(\log T)^9} \log x + \log(9 \log T) - \log T \right) = \\ &= x \cdot \exp \left(\frac{a}{(2a)^{9/10}} (\log x)^{1/10} + \log(9 \log T) - (2a)^{1/10} (\log x)^{1/10} \right) \ll \\ &\ll x \cdot \exp \left(\frac{-a}{(2a)^{9/10}} (\log x)^{1/10} \right), \end{aligned}$$

For the second term, we have

$$x^b(\log T)^{10} \ll x^b \log x \ll x^{b+\varepsilon}$$

for any $\varepsilon > 0$. Finally for the third term,

$$\frac{x(\log x)^2}{T} = x \cdot \exp(\log(2 \log x) - \log T) \ll x \cdot \exp(-c(\log x)^{1/10})$$

for any constant $c > (2a)^{1/10}$. Combining all the estimates together, we get for some constant $k > 0$ that

$$\psi(x) = x + O\left(x \cdot e^{-k(\log x)^{1/10}}\right),$$

which was to be proved. □

5. The pretentious approach

5.1 Introduction

We are finally ready to begin with the main part of the thesis – to give an overview of the *pretentious* approach to analytic number theory. Its goal is to provide a more intuitive theory which would be strong enough to prove all of the classical results about the distribution of prime numbers, such as the prime number theorem or deep results about the distribution of primes in arithmetic progressions like Dirichlet Theorem, Siegel-Walfisz Theorem or Linnik’s Theorem.

The motivation behind this theory is that the classical proofs were very hazy, it was hard to gain an idea of what is going on in the proof besides the nasty computations or whether similar ideas could be applied in other circumstances. Moreover, the use of the analytic continuation of L -functions was often artificial and the authors of the new theory thought that it might be omitted completely.

For example, we saw in the proof of the prime number theorem that due to the wider zero-free region of $\zeta(s)$, it was possible to shift the contour of integration to the left and obtain satisfactory estimates on the error terms. However, to get a zero-free region, we used Mertens formula, which is true in the region $\operatorname{Re}(s) > 1$, and the mean value theorem to extrapolate some information a tiny bit to the left of the line $\operatorname{Re}(s) = 1$. If this is our only method to obtain zero-free regions, why should we cross the 1-line at all? It might be more efficient to use our knowledge of $\zeta(s)$ in the region $\operatorname{Re}(s) > 1$ directly.

The reason why the new theory is called pretentious, and also why it is more intuitive than the classical approach, is, that it is based on the notion of two functions pretending to be one another. This is defined using some sort of a distance function, which is not a metric in the classical sense but satisfies the triangle inequality.

We saw in Section 3.2 that the prime number theorem can be stated in terms of the Möbius function, in particular, that it is equivalent to the statement that the mean value of $\mu(n)$ tends to 0. Halász in 1969 gave a characterization of all multiplicative functions with values in the unit disc whose mean value goes to 0. He noticed that examples of multiplicative functions whose mean does not approach zero are n^{it} with $t \in \mathbb{R}$, and also their small perturbations, i.e., functions which pretend to be n^{it} (like the function defined as $f(p^k) = p^{kit}$ for any prime p except $p = 17$), which are defined in terms of the distance function in Section 5.3. He then proved that these were the only examples, so that if $f(n)$ is a multiplicative function with values in the unit disc which does not pretend to be n^{it} for any $t \in \mathbb{R}$, then $\sum_{n \leq x} f(n) = o(x)$.

To deduce the prime number theorem from Halász’s result, it suffices to show that $\mu(n)$ does not pretend to be n^{it} for any $t \in \mathbb{R}$. This is easy to show intuitively (here the symbol $\approx$ does not have any exact meaning), because if $\mu(n) \approx n^{it}$, then $(\mu(n))^2 \approx n^{2it}$. But $(\mu(n))^2 \approx 1$, because $(\mu(p))^2 = 1$ for every prime p , so the only way is $t = 0$. But then it would have to hold that $\mu(n) \approx 1$, which is absurd.

In a similar way, results about the distribution of prime numbers can be stated in pretentious terms, with $\mu(n)$ replaced by appropriate functions involving

Dirichlet characters, as we will see in Chapter 7.

Halász's Theorem has very few assumptions and can be therefore used in a very general setting. Since we do not need an analytic continuation, this theorem can give us results about some cases when the analytic continuation is not known to exist or does not exist at all. A drawback of this universality is that even though it is the strongest possible result in such a general case, which is enough to give the main term in the prime number theorem, it does not give a very good error term. We will, however, explain that the pretentious results can be combined with some ideas from sieve theory to produce the prime number theorem in the strongest known form.

There is also another inconvenience in this approach – the proof of Halász's Theorem itself. For a long time, this proof was very difficult and unmotivated. Even though it did not use analytic continuation, it used difficult analytic tools like the Plancherel formula so the pretentious proof of the prime number theorem cannot be considered elementary. Fortunately, Andrew Granville, Adam Harper, and Kannan Soundararajan came up with a new proof, which is shorter and the techniques are in a way similar to the circle method of Hardy and Littlewood. We will present this proof in Chapter 6.

Most of the content of this chapter can be found in [GS1] or [GS2]. For a beautiful general introduction into the subject, see [Gr].

5.2 The generalized Von Mangoldt function

In the pretentious approach, we often work with general multiplicative functions and then deduce results about those of particular interest for us. To do so, in this section we define the generalized Von Mangoldt function $\Lambda_f(n)$ associated to a multiplicative function f and show some of its useful properties.

Definition. Let f be a multiplicative function with values inside the unit disc and let $F(s) = L(f, s)$ for $\operatorname{Re}(s) > 1$. Then we define the Von Mangoldt function associated to f , denoted by $\Lambda_f(n)$, as the n^{th} coefficient of the Dirichlet series $-\frac{F'}{F}(s)$, so that

$$-\frac{F'}{F}(s) = \sum_{n=1}^{\infty} \frac{\Lambda_f(n)}{n^s}.$$

Proposition 5.1. Let f be a multiplicative function with $|f(n)| \leq 1$ for every $n \in \mathbb{N}$. Then Λ_f has the following properties:

(1) for every positive integer n , we have

$$f(n) \log n = (f * \Lambda_f)(n) = \sum_{d|n} f(d) \Lambda_f\left(\frac{n}{d}\right).$$

(2) $\Lambda_f(1) = 0$ and if p is a prime, then

$$\Lambda_f(p) = f(p) \log p.$$

(3) Λ_f is supported on prime powers, i.e., $\Lambda_f(n) = 0$ unless n is a power of a prime.

(4) For any prime p and $k \geq 1$, we have

$$|\Lambda_f(p^k)| \leq \log p (2^k - 1).$$

Proof. (1): If $F(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$, then $-F'(s) = \sum_{n=1}^{\infty} \frac{f(n) \log n}{n^s}$, so comparing coefficients of the series $-F'(s)$ and $F(s) \cdot \left(-\frac{F'}{F}(s)\right)$ and using Proposition 2.5, we get the desired identity.

(2): Using (1) with $n = 1$, $0 = f(1) \log 1 = f(1) \Lambda_f(1)$. Since f is multiplicative, $f(1) = 1$ so $\Lambda_f(1) = 0$.

If p is a prime, again by (1) we get $f(p) \log p = f(1) \Lambda_f(p) + f(p) \Lambda_f(1) = \Lambda_f(p)$.

(3): First suppose that $n = pq$ for some different primes p, q . Then by (1) and (2), we have

$$\begin{aligned} f(pq) \log(pq) &= \Lambda_f(pq) + f(p) \Lambda_f(q) + f(q) \Lambda_f(p) = \\ &= \Lambda_f(pq) + f(p) f(q) \log q + f(q) f(p) \log p = \\ &= \Lambda_f(pq) + f(pq) \log(pq), \end{aligned}$$

so it follows that $\Lambda_f(pq) = 0$.

Now we will show that for each $n > 1$ and prime number p not dividing n , it holds that $\Lambda_f(p^k n) = 0$ for any $k \geq 1$, from which the assertion follows. We will prove this by induction over k .

If $k = 1$, we need that $\Lambda_f(pn) = 0$. We will prove this by induction over n .

The cases $n = 2$ for $p \neq 2$ and $n = 3$ for $p \neq 3$ are already done. Suppose that $\Lambda_f(pm) = 0$ for each $m < n$. Then by (1), we have

$$f(pn) \log(pn) = \sum_{d|pn} f(d) \Lambda_f\left(\frac{pn}{d}\right) = \sum_{d|n} f(pd) \Lambda_f\left(\frac{n}{d}\right) + \sum_{d|n} f(d) \Lambda_f\left(\frac{pn}{d}\right),$$

where we split the sum into two parts, depending on whether the divisor d is divisible by p . By the induction hypothesis, all terms of the second sum are 0 except when $d = 1$ and $d = n$. Using multiplicativity of f and (1) and (2), we get

$$\begin{aligned} f(pn) \log(pn) &= f(p) \sum_{d|n} f(d) \Lambda_f\left(\frac{n}{d}\right) + \Lambda_f(pn) + f(n) \Lambda_f(p) = \\ &= f(p) f(n) \log n + \Lambda_f(pn) + f(n) f(p) \log p = \\ &= f(pn) \log(pn) + \Lambda_f(pn), \end{aligned}$$

so indeed $\Lambda_f(pn) = 0$ and the first induction over n is completed.

Now suppose that the assertion is true for every $l < k$ and we would like to prove it for k . We proceed similarly as in the previous part by induction over n . If n is 2 or 3, the assertion follows from what we have already proved (by possibly interchanging the roles of p and n .)

Next suppose that the claim holds for every $m < n$. We again split the sum over divisors d of $p^k n$ depending on the power of p which divides d . We get

$$\begin{aligned} f(p^k n) \log(p^k n) &= \sum_{d|p^k n} f(d) \Lambda_f \left(\frac{p^k n}{d} \right) = \\ &= \sum_{d|n} f(d) \Lambda_f \left(\frac{p^k n}{d} \right) + \sum_{d|n} f(pd) \Lambda_f \left(\frac{p^{k-1} n}{d} \right) + \dots \\ &\dots + \sum_{d|n} f(p^k d) \Lambda_f \left(\frac{n}{d} \right). \end{aligned}$$

By multiplicativity of f and (1), the last sum equals $f(p^k n) \log n$. By the induction hypothesis for n , the terms of the first sum are zero unless $d = n$ or $d = 1$. For the other sums, we use the induction hypothesis for k , so we see that their terms also vanish unless $d = n$. Putting all this together, we get

$$\begin{aligned} f(p^k n) \log(p^k n) &= \\ &= \Lambda_f(p^k n) + f(n) \Lambda_f(p^k) + f(pn) \Lambda_f(p^{k-1}) + \dots \\ &\dots + f(p^{k-1} n) \Lambda_f(p) + f(p^k n) \log n = \\ &= \Lambda_f(p^k n) + f(n) \sum_{d|p^k} f(d) \Lambda_f \left(\frac{p^k}{d} \right) + f(p^k n) \log n = \\ &= \Lambda_f(p^k n) + f(n) f(p^k) \log(p^k) + f(p^k n) \log n = \\ &= \Lambda_f(p^k n) + f(p^k n) \log(p^k n), \end{aligned}$$

where in the third equality, we used (1). It follows that $\Lambda_f(p^k n) = 0$ and we are done.

(4): We will prove this by induction over k . If $k = 1$, the assertion follows from (2). Let us now suppose that it holds for every $l < k$ and let us prove it for k . By (1), we have

$$f(p^k) \log(p^k) = \sum_{l=0}^k \Lambda_f(p^l) f(p^{k-l}),$$

so

$$\Lambda_f(p^k) = f(p^k) \log(p^k) - \sum_{l=0}^{k-1} \Lambda_f(p^l) f(p^{k-l}).$$

Therefore by the induction hypothesis, we have

$$\begin{aligned} |\Lambda_f(p^k)| &\leq k \log p + \sum_{l=0}^{k-1} |\Lambda_f(p^l)| \leq \log p \cdot \left(k + \sum_{l=0}^{k-1} (2^l - 1) \right) = \\ &= \log p \cdot \left(k - k + \sum_{l=0}^{k-1} 2^l \right) = \log p (2^k - 1). \end{aligned}$$

□

The following proposition will be helpful when we will want to prove some relation between the distance function and the size of the Dirichlet series.

Proposition 5.2. *Let f be a multiplicative function with $|f(s)| \leq 1$ and $F(s) := L(f, s)$ for $\operatorname{Re}(s) > 1$. Then*

$$\log F(s) = \sum_p \frac{f(p)}{p^s} + O(1).$$

Proof. Because

$$-(\log F)'(s) = -\frac{F'}{F}(s) = \sum_{n=1}^{\infty} \frac{\Lambda_f(n)}{n^s},$$

we have

$$\log F(s) = \sum_{n=1}^{\infty} \frac{\Lambda_f(n)}{\log n \cdot n^s}.$$

Therefore by Proposition 5.1, we have

$$\log F(s) = \sum_p \frac{\Lambda_f(p)}{\log p \cdot p^s} + \sum_{p^k, k \geq 2} \frac{\Lambda_f(p^k)}{\log p \cdot p^{ks}} = \sum_p \frac{f(p)}{p^s} + \sum_{p^k, k \geq 2} \frac{\Lambda_f(p^k)}{\log p \cdot p^{ks}}.$$

It therefore suffices to show that the later sum over prime powers is bounded.

Again by Proposition 5.1, we have

$$\begin{aligned} \left| \sum_{p^k, k \geq 2} \frac{\Lambda_f(p^k)}{\log p \cdot p^{ks}} \right| &\leq \sum_{p^k, k \geq 2} \frac{2^k - 1}{p^{ks}} \leq \sum_p \sum_{k \geq 2} \left(\frac{2}{p^s} \right)^k = \\ &= \sum_p \frac{4}{p^{2s} - 2p^s} < \infty, \end{aligned}$$

where the last equality holds after summing the geometric series. □

5.3 The distance function

The key notion of the pretentious approach is the distance of two multiplicative functions. Using this distance, we can say when one function pretends to be another, which allows us to talk about difficult concepts very intuitively.

In this section, we define the distance function and prove some of its key properties – mainly that it satisfies the triangle inequality and that it is closely related to the size of the Dirichlet series.

The following definition is very general, we will use it only for some specific coefficients a_n .

Definition. *Let f, g be two multiplicative functions with $|f(n)|, |g(n)| \leq 1$ for all n , $A = (a_n)$ a sequence of nonnegative real numbers and $x \in \mathbb{R}$ or $x = \infty$. Then we define*

$$\mathbb{D}_A^2(f, g, x) = \sum_{n \leq x} a_n \left(1 - \operatorname{Re} \left(f(n) \overline{g(n)} \right) \right).$$

In particular, if $a_n = \begin{cases} 0, & \text{if } n \text{ is not a prime,} \\ \frac{1}{p}, & \text{if } n=p \text{ is a prime,} \end{cases}$ then we define the distance

$$\mathbb{D}^2(f, g, x) := \mathbb{D}_A^2(f, g, x) = \sum_{p \leq x} \frac{1 - \operatorname{Re}(f(p)\overline{g(p)})}{p}.$$

Definition. Let f, g be two multiplicative functions with values inside the unit disc. We say that f pretends to be g or that f is g -pretentious, if

$$\mathbb{D}(f, g, x) = O(1), \text{ as } x \rightarrow \infty.$$

The following lemma follows directly from the definition and will be useful for us.

Lemma 5.3. If f, g, h are multiplicative functions with values in the unit disc and $|h(n)| = 1$ for each n , then for any sequence of nonnegative numbers $A = (a_n)$ and $x \in \mathbb{R}$ or $x = \infty$, we have

$$\mathbb{D}_A(f, g, x) = \mathbb{D}_A(fh, gh, x).$$

Proof. Directly by the definition, we have

$$\begin{aligned} \mathbb{D}_A^2(fh, gh, x) &= \sum_{n \leq x} a_n \left(1 - \operatorname{Re}(f(n)h(n)\overline{h(n)g(n)}) \right) = \\ &= \sum_{n \leq x} a_n \left(1 - \operatorname{Re}(f(n)\overline{g(n)}) \right) = \mathbb{D}_A^2(f, g, x). \end{aligned}$$

Because both sides are nonnegative, we can take square roots to obtain the result. $\square$

The intuition behind the definition above is clear: if f, g are two multiplicative functions with values on the unit disc, then $1 - \operatorname{Re}(f(n)\overline{g(n)}) = 0$ if and only if $f(n) = g(n)$ and $|f(n)| = 1$. To illustrate that this definition is indeed the correct one, we will show in the following lemma and proposition that it is closely related to the size of the Dirichlet's series.

Lemma 5.4. If $x > 1$, and a_p is a sequence indexed by prime numbers such that $|a_p| \leq 2$ for every prime p , then for any real numbers t and $x > 1$, we have

$$\sum_{p \leq x} \frac{a_p}{p^{1+it}} = \sum_p \frac{a_p}{p^{1+\frac{1}{\log x}+it}} + O(1).$$

Proof. To prove the lemma, we will show that

$$\sum_{p \leq x} \frac{a_p}{p^{1+\frac{1}{\log x}+it}} = \sum_{p \leq x} \frac{a_p}{p^{1+it}} + O(1) \tag{5.1}$$

and

$$\left| \sum_{p > x} \frac{a_p}{p^{1+\frac{1}{\log x}+it}} \right| \leq \sum_{p > x} \frac{2}{p^{1+\frac{1}{\log x}}} = O(1). \tag{5.2}$$

For the first assertion, we notice that for $p \leq x$, we have

$$p^{-\frac{1}{\log x}} = e^{-\frac{\log p}{\log x}} = 1 - \frac{\log p}{\log x} + \frac{1}{2} \left(\frac{\log p}{\log x} \right)^2 - \dots = 1 + O \left(\frac{\log p}{\log x} \right),$$

where the implied constant does not depend on p . Plugging this into (5.1), we get

$$\sum_{p \leq x} \frac{a_p}{p^{1+it}} p^{-\frac{1}{\log x}} = \sum_{p \leq x} \frac{a_p}{p^{1+it}} + O \left(\frac{1}{\log x} \sum_{p \leq x} \frac{|a_p| \log p}{p} \right).$$

To estimate the error term, we use Proposition 3.6. We obtain

$$\frac{1}{\log x} \sum_{p \leq x} \frac{|a_p| \log p}{p} \leq \frac{2}{\log x} \sum_{p \leq x} \frac{\log p}{p} \ll 1,$$

so the first assertion is proved.

For the second assertion, by the assumption about the sequence a_p , we have

$$\sum_{p > x} \frac{a_p}{p^{1+\frac{1}{\log x}+it}} \ll \sum_{p > x} \frac{1}{p^{1+\frac{1}{\log x}}}.$$

Let $\alpha = 1 + \frac{1}{\log x}$ and choose an arbitrary $Y > x$. By partial summation, we have

$$\sum_{Y \geq p > x} \frac{1}{p^\alpha} = \frac{\pi(Y)}{Y^\alpha} - \frac{\pi(x)}{x^\alpha} + \alpha \int_x^Y \frac{\pi(t)}{t^{1+\alpha}} dt,$$

so letting $Y \rightarrow \infty$, we can use Chebyshev's estimates and the fact that $\alpha > 1$ to get

$$\sum_{p > x} \frac{1}{p^\alpha} = -\frac{\pi(x)}{x^\alpha} + \alpha \int_x^\infty \frac{\pi(t)}{t^{1+\alpha}} dt = O \left(\frac{1}{x^{\alpha-1} \log x} \right) + O \left(\int_x^\infty \frac{1}{t^\alpha \log t} dt \right),$$

where in the last equality, we used Chebyshev's estimates. We see that as $x \rightarrow \infty$, the first term is $o(1)$, so it suffices to evaluate the integral. To do so, first let us substitute $u = \log t$, after which (and the definition of α) we get

$$\int_x^\infty \frac{1}{t^\alpha \log t} dt = \int_{\log x}^\infty \frac{1}{u \cdot e^{\frac{u}{\log x}}} du.$$

Now substituting $v = \frac{u}{\log x}$, we get

$$\int_{\log x}^\infty \frac{1}{u \cdot e^{\frac{u}{\log x}}} du = \int_1^\infty \frac{1}{v \cdot e^v} dv,$$

which converges and does not depend on x .

Altogether, $\sum_{p > x} \frac{1}{p^\alpha} = O(1)$ as $x \rightarrow \infty$, so we are done. $\square$

Before showing the connection between the distance function and the size of the Dirichlet series, we define the truncated Dirichlet series and show its relation to the non-truncated one.

Definition. Let f be a multiplicative function with $|f(n)| \leq 1$ for each n and let $x > 1$. Then we define its truncated Dirichlet series $F_x(s)$ via its Euler product as

$$F_x(s) := \prod_{p \leq x} \left(\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} \right).$$

Corollary 5.5. Let f be a multiplicative function with $|f(n)| \leq 1$ for each n . Let $F(s) := L(f, s)$ and $F_x(s)$ be the truncated Dirichlet series of f . Then for any real t and $x > 1$, we have

$$F_x(1+it) \asymp F\left(1 + \frac{1}{\log x} + it\right).$$

Proof. Note that F_x is the Dirichlet series of the multiplicative function that is equal to f on primes $\leq x$ and 0 otherwise. Hence, taking logarithms and using Proposition 5.2, we have

$$\log F_x(1+it) = \sum_{p \leq x} \frac{f(p)}{p^{1+it}} + O(1).$$

Using Lemma 5.4 and Proposition 5.2 again, we obtain

$$\sum_{p \leq x} \frac{f(p)}{p^{1+it}} = \sum_p \frac{f(p)}{p^{1+\frac{1}{\log x}+it}} + O(1) = \log F\left(1 + \frac{1}{\log x} + it\right) + O(1).$$

Hence $\log F_x(1+it) = \log F\left(1 + \frac{1}{\log x} + it\right) + O(1)$. Exponentiating gives us the result. $\square$

The following proposition finally shows that the distance function is related to the size of the Dirichlet series.

Proposition 5.6. Let f be a multiplicative function with $|f(n)| \leq 1$, and for s with $\operatorname{Re}(s) > 1$, let $F(s) := L(f, s)$ and $F_x(s)$ be the truncated Dirichlet series of f . Then for any real t and $x > 1$, we have

$$|F_x(1+it)| \asymp \left| F\left(1 + \frac{1}{\log x} + it\right) \right| \asymp \log x \cdot e^{-\mathbb{D}^2(f, n^{it}, x)}.$$

Proof. The first asymptotic was already proved in Corollary 5.5, so it suffices to prove the second one.

Taking logarithm of the left-hand side and using Proposition 5.2, we get

$$\begin{aligned} \log \left| F\left(1 + \frac{1}{\log x} + it\right) \right| &= \operatorname{Re} \left(\log F\left(1 + \frac{1}{\log x} + it\right) \right) = \\ &= \operatorname{Re} \left(\sum_p \frac{f(p)p^{-it}}{p^{1+\frac{1}{\log x}}} \right) + O(1). \end{aligned}$$

Using Lemma 5.4 yields

$$\log \left| F\left(1 + \frac{1}{\log x} + it\right) \right| = \operatorname{Re} \left(\sum_{p \leq x} \frac{f(p)p^{-it}}{p} \right) + O(1).$$

We can adjust the right-hand side using Proposition 3.7 to get

$$\begin{aligned} \operatorname{Re} \left(\sum_{p \leq x} \frac{f(p)p^{-it}}{p} \right) &= \log \log x - \sum_{p \leq x} \frac{1 - \operatorname{Re}(f(p)p^{-it})}{p} + O(1) = \\ &= \log \log x - \mathbb{D}^2(f, n^{it}, x) + O(1). \end{aligned}$$

Taking exponentials, we see that

$$\left| F \left(1 + \frac{1}{\log x} + it \right) \right| = \log x \cdot e^{-\mathbb{D}^2(f, n^{it}, x)} \cdot e^{O(1)},$$

which gives the result. $\square$

Recall that Perron's formula enables us to evaluate sums of multiplicative functions in terms of integrals of the corresponding Dirichlet's series. The Proposition 5.6 shows that the modulus of the Dirichlet's series is closely related to the distance between f and the functions n^{it} , so knowledge about these distances really provides us some information about multiplicative functions. we will explore this connection further when we will discuss Halász's Theorem.

Now we will prove the triangle inequality. We will begin with the following lemma, whose proof is due to Eric Naslund and can be found in [Gr]:

Lemma 5.7. *Let x, y, z be complex numbers inside the unit disc. Then*

$$\sqrt{1 - \operatorname{Re}(x\bar{y})} \leq \sqrt{1 - \operatorname{Re}(x\bar{z})} + \sqrt{1 - \operatorname{Re}(z\bar{y})}$$

Proof. Let $\eta(a, b) = \sqrt{1 - \operatorname{Re}(a\bar{b})}$ and take three complex numbers x, y, z from the unit disc. Let $r := |z|$. Then $r \leq 1$ and we can write $x\bar{z} = r(a + bi)$, $z\bar{y} = r(c + di)$, where $a + bi, c + di$ are from the unit disc. Then $x\bar{z}z\bar{y} = r^2(a + bi)(c + di)$, so $x\bar{y} = (a + bi)(c + di)$. Moreover, $1 + ra \leq 2$, $1 + rc \leq 2$, so $\sqrt{(1 + ra)(1 + rc)} \leq 2$. Hence we have

$$\begin{aligned} 2\sqrt{1 - ra}\sqrt{1 - rc} &\geq \sqrt{(1 + ra)(1 + rc)}\sqrt{1 - ra}\sqrt{1 - rc} = \sqrt{1 - r^2a^2}\sqrt{1 - r^2c^2} \geq \\ &\geq \sqrt{1 - a^2}\sqrt{1 - c^2} \geq |bd|, \end{aligned}$$

where the last inequality follows from the fact that $|a + bi|, |c + di| \leq 1$.

Now we have two options depending on the sign of $a + c$.

- If $a + c \leq 0$, then $-r(a + c) \geq 0$, so $1 - ra - rc + ac \geq 1 + ac \geq 0$.
- If $a + c > 0$, then $r(-a - c) \geq -a - c$, so again $1 - ra - rc + ac \geq 1 - a - c + ac = (1 - a)(1 - c) \geq 0$.

In both cases, $1 - ra - rc + ac \geq 0$, so $1 - ra + 1 - rc \geq 1 - ac$. Putting it altogether, we get

$$\begin{aligned} (\eta(x, z) + \eta(z, y))^2 &= \left(\sqrt{1 - ra} + \sqrt{1 - rc} \right)^2 = \\ &= (1 - ra + 1 - rc) + 2\sqrt{1 - ra}\sqrt{1 - rc} \geq \\ &\geq 1 - ac + bd = \eta^2(x, y). \end{aligned}$$

Taking square roots gives us the result. $\square$

Corollary 5.8 (Triangle inequality 1). *Let f, g, h be multiplicative functions with values inside the unit disc, $A = (a_n)$ be a sequence of nonnegative real numbers and $x > 0$ or $x = \infty$. Then*

$$\mathbb{D}_A(f, g, x) \leq \mathbb{D}_A(f, h, x) + \mathbb{D}_A(h, g, x).$$

Proof. Let $n \leq x$. For complex numbers a, b , let $\eta(a, b) = 1 - \operatorname{Re}(a\bar{b})$. Then, since all of $f(n), g(n), h(n)$ lie in the unit disc, we can use the previous lemma to obtain

$$\sqrt{\eta(f(n), g(n))} \leq \sqrt{\eta(f(n), h(n))} + \sqrt{\eta(h(n), g(n))}.$$

Hence

$$\eta(f(n), h(n)) \leq \eta(f(n), h(n)) + \eta(h(n), g(n)) + 2\sqrt{\eta(f(n), h(n))}\sqrt{\eta(h(n), g(n))}.$$

By the definition of $\mathbb{D}_A$, we have

$$\begin{aligned} \mathbb{D}_A^2(f, g, x) &= \sum_{n \leq x} a_n \cdot \eta(f(n), g(n)) \leq \\ &\leq \sum_{n \leq x} a_n \left(\eta(f(n), h(n)) + \eta(h(n), g(n)) \right) + \\ &+ 2 \sum_{n \leq x} a_n \sqrt{\eta(f(n), h(n))} \sqrt{\eta(h(n), g(n))} = \\ &= \mathbb{D}_A^2(f, h, x) + \mathbb{D}_A^2(h, g, x) + 2 \sum_{n \leq x} a_n \sqrt{\eta(f(n), h(n))} \sqrt{\eta(h(n), g(n))} \leq \\ &\leq \mathbb{D}_A^2(f, h, x) + \mathbb{D}_A^2(h, g, x) + \\ &+ 2 \sqrt{\sum_{n \leq x} a_n \cdot \eta(f(n), h(n))} \sqrt{\sum_{n \leq x} a_n \cdot \eta(h(n), g(n))} = \\ &= \left(\mathbb{D}_A(f, h, x) + \mathbb{D}_A(h, g, x) \right)^2, \end{aligned}$$

where in the last inequality, we used the Cauchy-Schwarz inequality in the form

$$(x_1 y_1 + \dots x_k y_k) \leq \sqrt{x_1^2 + \dots x_k^2} \sqrt{y_1^2 + \dots y_k^2}.$$

Taking square roots gives us the result. $\square$

As it is natural to multiply multiplicative functions together, there is a suitable other form of the triangle inequality.

Proposition 5.9 (Triangle inequality 2). *Let f_1, f_2, g_1, g_2 be multiplicative functions with values in the unit disc. Let $x > 0$ or $x = \infty$ and let $A = (a_n)$ be a sequence of non negative real numbers. Then*

$$\mathbb{D}_A(f_1 f_2, g_1 g_2, x) \leq \mathbb{D}_A(f_1, g_1, x) + \mathbb{D}_A(f_2, g_2, x).$$

Proof. Because

$$\mathbb{D}_A(f, g, x) = \sum_{n \leq x} a_n \left(1 - \operatorname{Re}(f(n)\overline{g(n)}) \right) = \mathbb{D}_A(1, \bar{f}g, x)$$

for any two multiplicative functions f, g , we can assume that $f_1 = f_2 = 1$.

For $j = 1, 2$, let us denote by $r_j(n) := \operatorname{Re}(\overline{g_j(n)})$.

Then by the Cauchy-Schwarz inequality, we have

$$\begin{aligned}
& (\mathbb{D}_A(1, g_1, x) + \mathbb{D}_A(1, g_2, x))^2 = \\
& = \sum_{n \leq x} a_n (1 - r_1(n) + 1 - r_2(n)) + 2 \sqrt{\sum_{n \leq x} a_n (1 - r_1(n))} \sqrt{\sum_{n \leq x} a_n (1 - r_2(n))} \geq \\
& \geq \sum_{n \leq x} a_n \left(1 - r_1(n) + 1 - r_2(n) + 2 \sqrt{1 - r_1(n)} \sqrt{1 - r_2(n)} \right).
\end{aligned} \tag{5.3}$$

For $j = 1, 2$, let us further denote by $i_j(n) := \operatorname{Im}(\overline{g_j(n)})$. Because $|g_i(n)| \leq 1$, we have

$$\begin{aligned}
|i_1(n)i_2(n)|^2 & \leq (1 - r_1(n)^2)(1 - r_2(n)^2) = \\
& = (1 - r_1(n))(1 + r_1(n))(1 - r_2(n))(1 + r_2(n)) \leq \\
& \leq 4(1 - r_1(n))(1 - r_2(n)),
\end{aligned}$$

so

$$|i_1(n)||i_2(n)| \leq 2\sqrt{1 - r_1(n)}\sqrt{1 - r_2(n)}.$$

Substituting this into the last inequality of (5.3), we have

$$(\mathbb{D}_A(1, g_1, x) + \mathbb{D}_A(1, g_2, x))^2 \geq \sum_{n \leq x} a_n (1 - r_1(n) + 1 - r_2(n) + |i_1(n)||i_2(n)|). \tag{5.4}$$

We can now compute that

$$\begin{aligned}
& 1 - r_1(n) + 1 - r_2(n) + r_1(n)r_2(n) + |i_1(n)||i_2(n)| = \\
& = 1 + (1 - r_1(n))(1 - r_2(n)) + |i_1(n)||i_2(n)| \geq 1 + |i_1(n)||i_2(n)|,
\end{aligned}$$

so

$$\begin{aligned}
& 1 - r_1(n) + 1 - r_2(n) + |i_1(n)||i_2(n)| \geq 1 - r_1(n)r_2(n) - i_1(n)i_2(n) = \\
& = 1 - \operatorname{Re}(\overline{g_1(n)g_2(n)}).
\end{aligned}$$

Substituting this into (5.4), we obtain

$$(\mathbb{D}_A(1, g_1, x) + \mathbb{D}_A(1, g_2, x))^2 \geq \sum_{n \leq x} a_n \left(1 - \operatorname{Re}(\overline{g_1(n)g_2(n)}) \right) = \mathbb{D}_A^2(1, g_1g_2, x),$$

as we wanted. □

The following is an important corollary of the second form of the triangle inequality.

Corollary 5.10. *Let f, g be multiplicative functions with values in the unit disc. Let $x > 0$ or $x = \infty$ and let $A = (a_n)$ be a sequence of non negative real numbers. Then for any $k \in \mathbb{N}$, we have*

$$\mathbb{D}_A(f^k, g^k, x) \leq k \cdot \mathbb{D}_A(f, g, x).$$

Proof. The Corollary follows from Proposition 5.9 by induction over k after setting $f_1 = f_2 = f$ and $g_1 = g_2 = g$. $\square$

To demonstrate the usefulness of the triangle inequality, we will show how it can be applied to derive Merten's formula.

Lemma 5.11. *If $s = \sigma + it$ and $\sigma > 1$, then*

$$\log |\zeta(s)| = \sum_{p^m} \frac{1}{mp^{m\sigma}} \operatorname{Re}(p^{-it}).$$

Proof. Let $s = \sigma + it$ with $\sigma > 1$. Using Euler's product and the Taylor series of logarithm, we have

$$\begin{aligned} \log |\zeta(s)| &= \operatorname{Re}(\log \zeta(s)) = \operatorname{Re} \left(- \sum_p \log \left(1 - \frac{1}{p^s} \right) \right) = \operatorname{Re} \left(\sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{ms}} \right) = \\ &= \sum_{p^m} \frac{1}{mp^{m\sigma}} \operatorname{Re}(p^{-imt}). \end{aligned}$$

$\square$

Theorem 5.12. *If $s = \sigma + it$ with $\sigma > 1$, then*

$$|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| \geq 1.$$

Proof. By the previous lemma, we have

$$\log |\zeta(\sigma) \cdot \zeta(s)| = \log |\zeta(\sigma)| + \log |\zeta(s)| = \sum_{p^m} \frac{1}{mp^{m\sigma}} \operatorname{Re}(1 + p^{-imt}).$$

The last term looks similar to our definition of the distance function. Indeed, if we let $A = (a_n)$ be the sequence defined by $a_n = 0$ whenever n is not a prime power and $a_{p^m} = \frac{1}{mp^{m\sigma}}$ otherwise, then

$$\log |\zeta(\sigma) \cdot \zeta(s)| = \mathbb{D}_A^2(1, -n^{it}, \infty).$$

Similarly,

$$\log \left| \frac{\zeta(\sigma)}{\zeta(\sigma + 2it)} \right| = \sum_{p^m} \frac{1}{mp^{m\sigma}} \operatorname{Re}(1 - p^{-2imt}) = \mathbb{D}_A^2(1, n^{2it}, \infty).$$

By the triangle inequality, we have

$$\mathbb{D}_A(1, n^{2it}, \infty) \leq 2\mathbb{D}_A(1, -n^{it}, \infty).$$

Squaring both sides yields

$$\mathbb{D}_A^2(1, n^{2it}, \infty) \leq 4 \cdot \mathbb{D}_A^2(1, -n^{it}, \infty),$$

so

$$\log \left| \frac{\zeta(\sigma)}{\zeta(\sigma + 2it)} \right| \leq 4 \cdot \log |\zeta(\sigma) \cdot \zeta(s)|.$$

Exponentiating and rearranging the inequality, we obtain the result

$$1 \leq |\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)|.$$

$\square$

5.4 Halász's Theorem and a different proof of the prime number theorem

In this section, we state Halász's Theorem, which characterizes multiplicative functions whose mean value is zero. The statement is slightly more technical as it provides more information how fast does the mean value tend to zero. It will be proved in the next chapter, now we only demonstrate how to use it to prove the prime number theorem.

Definition. Let f be a multiplicative function with $|f(n)| \leq 1$. Then we define the value $M_f(x, T)$ by

$$M_f(x, T) := \min_{|t| \leq T} \mathbb{D}^2(f, n^{it}, x).$$

We are now ready to state Halász's Theorem:

Theorem 5.13 (Halász). Let f be a multiplicative function with $|f(n)| \leq 1$. Then for any $x > 2$ and $2 < T \leq (\log x)^2$, we have

$$\frac{1}{x} \sum_{n \leq x} f(n) \ll (1 + M_f(x, T)) e^{-M_f(x, T)} + \frac{1}{T} + \frac{\log \log x}{\log x}.$$

Especially important for us is the following corollary, which we will also refer to as Halász's Theorem.

Corollary 5.14. Let f be a multiplicative function with $|f(n)| \leq 1$. Suppose that f is not n^{it} -pretentious for any real t . Then

$$\frac{1}{x} \sum_{n \leq x} f(n) = o(1).$$

Proof. Let $\varepsilon > 0$ be arbitrary. We will show that if x is large enough, then

$$\left| \frac{1}{x} \sum_{n \leq x} f(n) \right| < C \cdot \varepsilon$$

for a constant C which does not depend on ε .

First we pick T large enough so that $\frac{1}{T} < \frac{\varepsilon}{3}$. Because the interval $[-T, T]$ is compact and we assume that for every $t \in [-T, T]$ it holds that $\mathbb{D}^2(f, n^{it}, x) \rightarrow \infty$ as $x \rightarrow \infty$, we have that $M_f(x, T) \rightarrow \infty$ as $x \rightarrow \infty$. Therefore we can choose x large enough such that $(1 + M_f(x, T))e^{-M_f(x, T)} < \frac{\varepsilon}{3}$, $\frac{\log \log x}{\log x} < \frac{\varepsilon}{3}$ and $(\log x)^2 \geq T$. By Halász's Theorem, if we denote by C the implied constant, we conclude that

$$\left| \frac{1}{x} \sum_{n \leq x} f(n) \right| < C \left((1 + M_f)e^{-M_f} + \frac{1}{T} + \frac{\log \log x}{\log x} \right) < C \cdot \varepsilon.$$

□

The proof of Halász's Theorem will be given in the next chapter. Let us first show how it can be used to deduce the prime number theorem. It suffices to use Theorem 3.11 and show that μ is not n^{it} -pretentious for any t .

We will prove a slightly more general theorem.

Theorem 5.15. *Let f be a multiplicative function with values in the unit disc such that there exists a $k \in \mathbb{N}$ for which $f(p)^k = 1$ for every prime p . Then $\mathbb{D}(f, n^{it}, \infty) = \infty$ for every non-zero real t .*

Proof. Let f and k satisfy the assumptions and suppose for contradiction that there exists a non-zero real t for which $\mathbb{D}(f, n^{it}, \infty) < \infty$. Then by the corollary of the second version of triangle inequality, we have

$$\mathbb{D}(1, n^{kit}, \infty) = \mathbb{D}(f^k, n^{kit}, \infty) \leq k \cdot \mathbb{D}(f, n^{it}, \infty) < \infty.$$

For $x > 1$, let $s = 1 + \frac{1}{\log x} + kit$. Then by Proposition 5.6, we have

$$\log |\zeta(s)| = \log \log x - \mathbb{D}^2(1, n^{kit}, x) + O(1) = \log \log x + O(1),$$

so $\zeta(s) \gg \log x$. Sending x to infinity, we get a contradiction with the fact that $\zeta(1 + kit)$ is analytic, because $t \neq 0$. $\square$

It is now easy to deduce the prime number theorem.

Corollary 5.16. *As $x \rightarrow \infty$, we have $\psi(x) \sim x$.*

Proof. By Theorem 3.11, it suffices to show that $\sum_{n \leq x} \mu(n) = o(x)$. By Halász's Theorem, this is equivalent to the fact that μ is not n^{it} -pretentious for any real t . By Theorem 5.15, μ is not n^{it} -pretentious for any non-zero t . If $t = 0$, $n^{it} = 1$, so noting that $\mu(p) = -1$ for every prime number p , we see directly from the definition and Theorem 3.7 that μ does not pretend to be 1, so the proof is completed. $\square$

There is, however, a significant issue with this proof. Theorem 3.11 can be easily modified to give an error term in the prime number theorem depending on the rate of decay of the mean of $\mu(n)$. Let $M(x) := \sum_{n \leq x} \mu(n)$. We can see that the best Halász's Theorem can give is

$$M(x) \ll x \frac{\log \log x}{\log x},$$

which is far from what we believe is the truth. For example the Riemann Hypothesis is equivalent to

$$M(x) = o\left(x^{1/2+\varepsilon}\right)$$

for any $\varepsilon > 0$. For many applications, one needs at least the estimate

$$M(x) \ll \frac{x}{(\log x)^A}$$

for any A . On the other hand, we know that the error term in Halász's Theorem is sharp in some cases (see [GS1]), and therefore the theorem cannot be improved while preserving its generality.

Koukoulopoulos [Ko2] found a way to overcome this problem by combining pretentious techniques with sieve methods and obtained the prime number theorem in its strongest known form.

6. Proof of Halász's Theorem

In this chapter, we give a proof of Halász's Theorem. We will give an exposition of the proof given by Granville, Harper, and Soundararajan in [GHS1] with more details worked out.

Throughout this section, let f be a multiplicative function with values in the unit disc, and denote by

$$\begin{aligned} S(x) &:= \sum_{n \leq x} f(n), \\ F_x(s) &:= \prod_{p \leq x} \left(1 + \sum_{k=1}^{\infty} \frac{f(p^k)}{p^{ks}} \right), \\ L(x)^2 &= \sum_{|N| \leq (\log x)^2 + 1} \frac{1}{N^2 + 1} \sup_{|t-N| \leq \frac{1}{2}} |F_x(1+it)|^2. \end{aligned}$$

We will prove the following form of Halász's Theorem:

Theorem 6.1. *Let f , F_x , $S(x)$ and $L(x)$ be as above. Then it holds that*

$$|S(x)| \ll x \cdot \frac{L(x)}{\log x} \log \left(100 \cdot \frac{\log x}{L(x)} \right) + x \cdot \frac{\log \log x}{\log x}.$$

This is a little different version than the statement in the previous chapter. To see how these statements are related, first recall the connection between the size of $F(s)$ and the distance $\mathbb{D}^2(f, n, n^{it})$ given in Proposition 5.6. Note that in the definition of $M_f(x, T)$, we take the minimum over the whole interval, which corresponds to taking the maximum of $F_x(1+it)$ over the whole interval $|t| \leq T$. Since the large values of F_x tend to be clustered together on short intervals, this might appear to be wasteful. This is therefore taken into account in the definition of $L(x)$, where we take the maxima over intervals of length 1.

Theorem 5.13 is more convenient to work with and can be deduced from 6.1 by using Proposition 5.6 to obtain

$$L(x) \ll \max_{|t| \leq (\log x)^2 + 1} |F_x(1+it)| \asymp \log x \cdot e^{-M_f(x, (\log x)^2)}.$$

As in the article [GHS1], the proof will be divided into two parts. The first part is combinatorial, where we first express $S(x)$ as a triple convolution, and then divide the range of summation into smaller intervals. As we will see, the contribution of large primes will be easily estimated, but we will need to use analytic methods to obtain proper bounds on the contribution of the smaller primes. This will be done in the second, analytic part of the proof.

6.1 The combinatorial part

In the first part of the proof, our initial goal is to express $S(x)$ as a triple multiplicative convolution. We will start by writing it as a double convolution and iterate it to obtain a triple convolution.

Proposition 6.2. *It holds that*

$$S(x) = \frac{1}{\log x} \sum_{\substack{mp \leq x, \\ p \text{ prime}}} f(m)f(p) \log p + O\left(\frac{x}{\log x}\right). \quad (6.1)$$

Proof. Let us start by writing

$$S(x) = \frac{1}{\log x} \sum_{n \leq x} f(n) \left(\log n + \log \frac{x}{n} \right).$$

Using the assumption that $|f(n)| \leq 1$ and Corollary 1.8, we obtain

$$S(x) = \frac{1}{\log x} \sum_{n \leq x} f(n) \log n + O\left(\frac{x}{\log x}\right).$$

Substituting $\log n = \sum_{d|n} \Lambda(d)$, we obtain

$$S(x) = \frac{1}{\log x} \sum_{n \leq x} f(n) \sum_{d|n} \Lambda(d) = \frac{1}{\log x} \sum_{d \leq x} \Lambda(d) \sum_{m \leq x/d} f(md) + O\left(\frac{x}{\log x}\right).$$

Now we show that the contribution of higher powers of primes is small. Indeed, we have

$$\begin{aligned} \sum_{d \leq x} \Lambda(d) \sum_{m \leq x/d} f(md) &= \sum_{p \leq x} \log p \sum_{m \leq x/p} f(mp) + \sum_{\substack{p^k \leq x \\ k \geq 2}} \log p \sum_{m \leq x/p^k} f(mp^k) = \\ &= \sum_{p \leq x} \log p \sum_{m \leq x/p} f(mp) + O(x), \end{aligned}$$

since

$$\left| \sum_{\substack{p^k \leq x \\ k \geq 2}} \log p \sum_{m \leq x/p^k} f(mp^k) \right| \leq \sum_{\substack{p^k \leq x \\ k \geq 2}} \log p \cdot \frac{x}{p^k} \leq x \sum_{k \geq 2} \sum_{n=1}^{\infty} \frac{\log n}{n^k} \ll x.$$

Combining our results so far, we get

$$S(x) = \frac{1}{\log x} \sum_{p \leq x} \log p \sum_{m \leq x/p} f(mp) + O\left(\frac{x}{\log x}\right).$$

For those m not divisible by p , we can write $f(mp) = f(m)f(p)$, so it suffices to deal with the terms with $p|m$. But we have

$$\begin{aligned} \left| \sum_{p \leq x} \log p \sum_{\substack{m \leq x/p \\ p|m}} f(mp) \right| &\ll x \sum_{p \leq x} \frac{\log p}{p^2} \ll x, \\ \left| \sum_{p \leq x} \log p \sum_{\substack{m \leq x/p \\ p|m}} f(m)f(p) \right| &\ll x \sum_{p \leq x} \frac{\log p}{p^2} \ll x, \end{aligned}$$

so we can subtract the first sum and add the second without introducing a bigger error term than allowed. $\square$

In Theorem 6.1, the acceptable error term is $O\left(x \cdot \frac{\log \log x}{\log x}\right)$, which allows us to throw out from the sum those primes which are too small or too big. We do so in the following proposition:

Proposition 6.3. *It holds that*

$$S(x) = \frac{1}{\log x} \sum_{(\log x)^4 < p \leq x/2} f(p) \log p \sum_{m \leq x/p} f(m) + O\left(x \cdot \frac{\log \log x}{\log x}\right).$$

Proof. By Proposition 6.3, it suffices to show that the contribution of small and big primes is $O\left(x \cdot \frac{\log \log x}{\log x}\right)$. We have:

$$\frac{1}{\log x} \sum_{p \leq (\log x)^4} f(p) \log p \sum_{m \leq x/p} f(m) \ll \frac{x}{\log x} \sum_{p \leq (\log x)^4} \frac{\log p}{p} \ll x \cdot \frac{\log \log x}{\log x}$$

by Proposition 3.6. Similarly, we have

$$\begin{aligned} \frac{1}{\log x} \sum_{x/2 < p \leq x} f(p) \log p \sum_{m \leq x/p} f(m) &\ll \frac{x}{\log x} \sum_{x/2 < p \leq x} \frac{\log p}{p} \ll \\ &\ll \frac{x}{\log x} (\log x - \log x/2 + O(1)) \ll \frac{x}{\log x}. \end{aligned}$$

□

To obtain a triple convolution, it suffices to use the expression (6.1) again on $\sum_{m \leq x/p} f(m)$. The following lemma gives us a good estimate of the error term.

Lemma 6.4. *It holds that*

$$\sum_{p \leq x/2} \frac{\log p}{p \log(x/p)} \ll \log \log x.$$

Proof. Let $Y > 2$ be arbitrary. Then by Proposition 3.6, we have

$$\sum_{Y/2 \leq p \leq Y} \frac{\log p}{p} = \log Y - \log(Y/2) + O(1) = O(1),$$

where the implied constant does not depend on Y . Therefore for any $A \geq 2$, we have

$$\sum_{\frac{x}{2A} \leq p \leq \frac{x}{A}} \frac{\log p}{p \log(x/p)} \ll \frac{1}{\log A} \sum_{\frac{x}{2A} \leq p \leq \frac{x}{A}} \frac{\log p}{p} \ll \frac{1}{\log A}.$$

Hence

$$\sum_{p \leq x/2} \frac{\log p}{p \log(x/p)} \ll \sum_{1 \leq k \leq \log x} \sum_{\frac{x}{2^{k+1}} \leq p \leq \frac{x}{2^k}} \frac{\log p}{p \log(x/p)} \ll \sum_{1 \leq k \leq \log x} \frac{1}{k} \ll \log \log x.$$

□

We are finally ready to express $S(x)$ as a triple convolution.

Proposition 6.5. *It holds that*

$$S(x) = \frac{1}{\log x} \sum_{(\log x)^4 < p \leq x/2} \frac{f(p) \log p}{\log(x/p)} \sum_{\substack{nq \leq x/p, \\ q \text{ prime}}} f(n)f(q) \log q + O\left(x \cdot \frac{\log \log x}{\log x}\right). \quad (6.2)$$

Proof. By Proposition 6.2, we have for all primes $p \leq x/2$ that

$$\sum_{m \leq x/p} f(m) = S(x/p) = \frac{1}{\log(x/p)} \sum_{q \leq x/p} f(q) \log q \sum_{n \leq \frac{x}{pq}} f(n) + O\left(\frac{x}{p \log(x/p)}\right).$$

Plugging this into Proposition 6.3, we get

$$\begin{aligned} S(x) &= \frac{1}{\log x} \sum_{(\log x)^4 < p \leq x/2} \frac{f(p) \log p}{\log(x/p)} \sum_{\substack{nq \leq x/p, \\ q \text{ prime}}} f(n)f(q) \log q + \\ &+ O\left(\frac{x \log \log x}{\log x} + \frac{x}{\log x} \cdot \sum_{(\log x)^4 < p \leq x/2} \frac{\log p}{p \log(x/p)}\right). \end{aligned}$$

By Lemma 6.4, the error term is $O\left(\frac{x \log \log x}{\log x}\right)$, which gives the result. $\square$

In the sum (6.2), if p is large, the range of possible q 's is small, which would lead to bigger error terms. Therefore we split the sum depending on the size of p .

Let $\mathcal{P}$ be the set of primes in the interval $((\log x)^4, x/2]$, and for an integer k , let $\mathcal{P}_k := \mathcal{P} \cap (x^{1-e^{1-k}}, x^{1-e^{-k}}]$. Since

$$x^{1-e^{-(\log \log x - \log \log 2)}} = x^{1-\frac{\log 2}{\log x}} = x \cdot e^{-\log 2} = \frac{x}{2},$$

k runs from 1 to $\lceil \log \log x - \log \log 2 \rceil = \log \log x + O(1)$, where the $O(1)$ lies in the interval $[-1, 1]$.

Now define $S_k(x)$ by

$$S_k(x) := \sum_{\substack{pqn \leq x, \\ p, q \text{ primes}, \\ p \in \mathcal{P}_k}} \frac{f(p) \log p}{\log(x/p)} f(n)f(q) \log q.$$

By Proposition 6.5, we have

$$S(x) \ll \frac{1}{\log x} \sum_{k=1}^{\log \log x + O(1)} |S_k(x)| + \frac{x \log \log x}{\log x}.$$

Proposition 6.6. *for $1 \leq k \leq \log \log x + O(1)$, we have*

$$|S_k(x)| \ll e^{-k} x \log x.$$

Proof. We have

$$\begin{aligned}
|S_k(x)| &\leq \sum_{\substack{pqn \leq x, \\ p, q \text{ primes}, \\ p \in \mathcal{P}_k}} \frac{\log p \log q}{\log(x/p)} \leq x \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} \sum_{q \leq x/p} \frac{\log q}{q} = \\
&= x \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} (\log(x/p) + O(1)) \ll \\
&\ll x \sum_{x^{1-e^{1-k}} < p \leq x^{1-e^{-k}}} \frac{\log p}{p} \ll x \left((1 - e^{-k}) \log x - (1 - e^{1-k}) \log x \right) \ll \\
&\ll x \log x (e^{1-k} - e^{-k}) \ll e^{-k} x \log x,
\end{aligned}$$

as we wanted to prove. $\square$

The next proposition shows that the contribution of $S_k(x)$ with $k > \log\left(\frac{100 \log x}{L(x)}\right)$ is smaller than the allowed error term in Theorem 6.1.

Proposition 6.7. *We have*

$$\frac{1}{\log x} \sum_{k=\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil}^{\log \log x + O(1)} |S_k(x)| \ll \frac{x}{\log x}.$$

Proof. By Proposition 6.6, we have

$$\frac{1}{\log x} \sum_{k=\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil}^{\log \log x + O(1)} |S_k(x)| \ll x \sum_{k=\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil}^{\log \log x + O(1)} e^{-k}.$$

Using the fact that the $O(1)$ in the range of k is at most 1, we get

$$\begin{aligned}
&\log \log x + O(1) - \left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil = \\
&= \log \log x - \log 100 - \log \log x + \log L(x) + O(1) \leq \log L(x).
\end{aligned}$$

Therefore, we can bound the geometric series as

$$\sum_{k=\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil}^{\log \log x + O(1)} e^{-k} \ll \frac{L(x)}{100 \log x} \sum_{k=0}^{\log L(x)} e^{-k} \ll \frac{L(x)}{100 \log x} \cdot \frac{1}{L(x)} \ll \frac{1}{\log x}.$$

Combining this with the first inequality gives the result. $\square$

To finish the proof of Halász's Theorem, it suffices to bound the contribution of $S_k(x)$ with $k \leq \log\left(\frac{100 \log x}{L(x)}\right)$, in particular to show that $|S_k(x)| \ll xL(x) + x$ (because $L(x) \gg 1$, so $x \cdot \log\left(\frac{100 \log x}{L(x)}\right) \ll x \log \log x$).

Because in the definition of $S_k(x)$, we sum over $q \leq \frac{x}{p}$, for any integer $1 \leq k \leq \log\left(\frac{100 \log x}{L(x)}\right)$, we have $q \leq \frac{x}{x^{1-e^{1-k}}} = x^{e^{1-k}}$. Let us denote by g_k, h_k the arithmetic functions defined as

$$g_k(n) = \begin{cases} \frac{f(n) \log n}{\log(x/n)} & \text{if } n \in \mathcal{P}_k, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$h_k(n) = \begin{cases} f(n) \log n & \text{if } n \leq x^{e^{1-k}} \text{ is a prime,} \\ 0 & \text{otherwise.} \end{cases}$$

Then we have

$$S_k(x) = \sum_{lmn \leq x} g_k(l) h_k(m) f(n) = \sum_{n \leq x} (g_k * h_k * f)(n),$$

which allows us to use Perron's formula and express the Dirichlet series in terms of the Dirichlet series of h_k , g_k and f , which can be individually estimated. A similar technique is used in the circle method invented by Hardy and Littlewood, which served as a motivation for this proof.

6.2 The analytic part

In the analytic part, we begin by using Perron's formula to transform the triple convolution into an integral. We will then split the integral into two parts and estimate them separately.

Proposition 6.8. *For any $1 \leq k \leq \log\left(\frac{100 \log x}{L(x)}\right)$, we have*

$$|S_k(x)| \ll x \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} \right| \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right| \left| \frac{F_x(s)}{s} \right| ds + x$$

Proof. We have

$$S_k(x) = \sum_{n \leq x} (h_k * g_k * f)(n) = \sum_{n \leq x} (g_k * h_k * f_x)(n),$$

where $f_x(n) = f(n)$ if all primes $p|n$ are $\leq x$ and 0 otherwise. Then by Proposition 2.5, we have

$$\begin{aligned} L(g_k * h_k * f_x, s) &= L(g_k, s) L(h_k, s) L(f_x, s) = \\ &= \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} \cdot \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \cdot F_x(s). \end{aligned}$$

Using the truncated Perron's formula, setting $c = 1$ and $T = (\log x)^2$, it holds that

$$\begin{aligned} |S_k(x)| &\ll \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} \right| \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right| |F_x(s)| \left| \frac{x}{s} \right| ds + E(x) = \\ &= x \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} \right| \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right| \left| \frac{F_x(s)}{s} \right| ds + E(x), \end{aligned}$$

where the error term $E(x)$ satisfies

$$E(x) \ll \sum_{p \in \mathcal{P}_k} \frac{\log p}{\log(x/p)} \sum_{q \leq x^{e^{1-k}}} \log q \sum_{\substack{n \\ r|n \Rightarrow r \leq x}} \frac{x}{pqn} \cdot \frac{1}{|T| \log \frac{x}{pqn} + 1},$$

where all the variables p, q, r are primes.

To prove the Proposition, our goal is to show that $E(x) \ll x$. We will proceed similarly as in the proof of Proposition 4.12.

We will split the sum into two parts, in the first part summing over those triples (p, q, n) with $\frac{1}{2} \leq \frac{x}{pqn} \leq 2$, and in the second part over the rest.

Let the sum over these (p, q, n) with $\frac{1}{2} \leq \frac{x}{pqn} \leq 2$ be denoted by $E_1(x)$ and the rest $E_2(x)$.

In the first case, we have $\left| \log \frac{x}{pqn} \right| = \left| \log \left(1 + \frac{x-pqn}{x} \right) \right| \gg \frac{|x-pqn|}{x}$ by the Taylor expansion. Moreover, we must have $q \leq \frac{2x}{p}$ and $\frac{x}{2pq} \leq n \leq \frac{2x}{pq}$. Hence

$$E_1(x) \ll \sum_{p \in \mathcal{P}_k} \frac{\log p}{\log(x/p)} \sum_{q \leq \frac{2x}{p}} \log q \sum_{\frac{x}{2pq} \leq n \leq \frac{2x}{pq}} \frac{x}{T|x-pqn|+x}.$$

Let us first estimate the last sum over n . We have

$$\begin{aligned} x \sum_{\frac{x}{2pq} \leq n \leq \frac{2x}{pq}} \frac{1}{T|x-pqn|+x} &= \frac{x}{pq} \sum_{\frac{x}{2pq} \leq n \leq \frac{2x}{pq}} \frac{1}{T|\frac{x}{pq}-n|+\frac{x}{pq}} \ll \\ &\ll \frac{x}{pq} \sum_{0 \leq m \leq \frac{x}{pq}} \frac{1}{Tm+\frac{x}{pq}} = \frac{x}{Tpq} \sum_{0 \leq m \leq \frac{x}{pq}} \frac{1}{m+\frac{x}{pqT}} = \\ &= \frac{x}{Tpq} \sum_{\frac{x}{pqT} \leq m \leq \frac{x}{pq}+\frac{x}{pqT}} \frac{1}{m} = \frac{x}{pqT} \left(\log \left(\frac{x}{pq} + \frac{x}{pqT} \right) - \log \frac{x}{pqT} + O\left(\frac{pqT}{x}\right) \right) = \\ &= \frac{x}{pqT} \log(1+T) + O(1) \ll 1 + \frac{x \log T}{pqT}, \end{aligned}$$

so

$$E_1(x) \ll \sum_{p \in \mathcal{P}_k} \frac{\log p}{\log(x/p)} \sum_{q \leq \frac{2x}{p}} \log q \left(1 + \frac{x \log T}{pqT} \right).$$

Let us first bound

$$\sum_{p \in \mathcal{P}_k} \frac{\log p}{\log(x/p)} \sum_{q \leq \frac{2x}{p}} \log q.$$

For $p \in \mathcal{P}_k$, we have $\log(x/p) \geq \log\left(\frac{x}{x^{1-e^{-k}}}\right) = e^{-k} \log x$, so using Chebyshev's estimates and Corollary 3.6, we obtain

$$\sum_{p \in \mathcal{P}_k} \frac{\log p}{\log(x/p)} \sum_{q \leq \frac{2x}{p}} \log q \ll \frac{xe^k}{\log x} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p} \ll x.$$

Similarly, for the other part of the sum, we have

$$\begin{aligned} \frac{x \log T}{T} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} \sum_{q \leq \frac{2x}{p}} \frac{\log q}{q} &\ll \frac{x \log T}{T} \sum_{p \in \mathcal{P}_k} \frac{\log p \cdot \log(2x/p)}{p \log(x/p)} \ll \\ &\ll \frac{x \log T}{T} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} \ll \frac{xe^k \log T}{T \log x} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p} \ll x, \end{aligned}$$

so

$$E_1(x) \ll x.$$

Let us now estimate $E_2(x)$. In this case, we have $|\log \frac{x}{pqn}| \gg 1$, so we have

$$\begin{aligned} E_2(x) &= x \sum_{p \in \mathcal{P}_k} \sum_{q \leq x^{e^{1-k}}} \sum_{\substack{n \geq 1 \\ r|n \Rightarrow r \leq x \\ |\log \frac{x}{pqn}| \geq \log 2}} \frac{\log p \log q}{pq \log(x/p)} \cdot \frac{1}{T |\log \frac{x}{pqn}| + 1} \cdot \frac{1}{n} \ll \\ &\ll \frac{x}{T} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} \sum_{q \leq x^{e^{1-k}}} \frac{\log q}{q} \sum_{\substack{n \geq 1 \\ r|n \Rightarrow r \leq x}} \frac{1}{n}. \end{aligned}$$

Note that by Corollary 5.5, we have

$$\sum_{\substack{n \geq 1 \\ r|n \Rightarrow r \leq x}} \frac{1}{n} = \zeta_x(1) \asymp \zeta \left(1 + \frac{1}{\log x} \right) \ll \int_1^\infty \frac{1}{t^{1+\frac{1}{\log x}}} dt = \log x.$$

We can now finish the prove similarly as in the estimates of $E_1(x)$:

$$E_2(x) \ll \frac{x}{\log x} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p \log(x/p)} \sum_{q \leq x^{e^{1-k}}} \frac{\log q}{q} \ll \frac{e^{1-k} x}{e^{-k} \log x} \sum_{p \in \mathcal{P}_k} \frac{\log p}{p} \ll x,$$

and we are done. $\square$

Let us now split the integral into to parts and denote by

$$\begin{aligned} I_1 &:= \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} \right|^2 ds, \\ I_2 &:= \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right|^2 \left| \frac{F_x(s)}{s} \right|^2 ds. \end{aligned}$$

Recall that the sum over q in the integral I_2 runs over primes.

Proposition 6.9. *For any $1 \leq k \leq \log \left(\frac{100 \log x}{L(x)} \right)$, we have*

$$S_k(x) \ll x \sqrt{I_1 I_2} + x.$$

Proof. It suffices to use Proposition 6.8 and the Cauchy-Schwarz inequality. $\square$

To bound the integrals I_1 and I_2 , we will use the following general lemma:

Lemma 6.10. *For any sequence of complex numbers a_n and $T \geq 1$, it holds that*

$$\int_{-T}^T \left| \sum_{T^2 \leq n \leq x} \frac{a_n \Lambda(n)}{n^{1+it}} \right|^2 dt \ll \sum_{T^2 \leq n \leq x} \frac{|a_n|^2 \Lambda(n)}{n},$$

where the implied constant does not depend on the sequence a_n .

Proof. Let $\Phi(t) := \frac{1}{(\sin 1)^2} \left(\frac{\sin t}{t} \right)^2$. This function is even, non-negative, $\Phi(t) \geq 1$ for $-1 \leq t \leq 1$ and its Fourier transform $\hat{\Phi}(t)$ is supported in $[-1, 1]$. Therefore we have

$$\begin{aligned}
\int_{-T}^T \left| \sum_{T^2 \leq n \leq x} \frac{a_n \Lambda(n)}{n^{1+it}} \right|^2 dt &= \int_{-T}^T \left(\sum_{T^2 \leq n \leq x} \frac{a_n \Lambda(n)}{n^{1+it}} \right) \cdot \left(\sum_{T^2 \leq m \leq x} \frac{\overline{a_m} \Lambda(m)}{m^{1-it}} \right) dt \leq \\
&\leq \int_{-\infty}^{\infty} \left(\sum_{T^2 \leq n \leq x} \frac{a_n \Lambda(n)}{n^{1+it}} \right) \cdot \left(\sum_{T^2 \leq m \leq x} \frac{\overline{a_m} \Lambda(m)}{m^{1-it}} \right) \Phi\left(\frac{t}{T}\right) dt = \\
&= \sum_{T^2 \leq m, n \leq x} \frac{a_n \overline{a_m} \Lambda(n) \Lambda(m)}{nm} \left| \int_{-\infty}^{\infty} n^{-it} m^{it} \cdot \Phi\left(\frac{t}{T}\right) dt \right| \leq \\
&\leq \sum_{T^2 \leq m, n \leq x} \frac{|a_n a_m| \Lambda(n) \Lambda(m)}{nm} \left| \int_{-\infty}^{\infty} e^{-it \log \frac{n}{m}} \cdot \Phi\left(\frac{t}{T}\right) dt \right| = \\
&= \sum_{T^2 \leq m, n \leq x} \frac{|a_n a_m| \Lambda(n) \Lambda(m)}{nm} \left| \int_{-\infty}^{\infty} T \cdot e^{-iu(T \log \frac{n}{m})} \cdot \Phi(u) du \right| = \\
&= \sum_{T^2 \leq m, n \leq x} \frac{|a_n a_m| \Lambda(n) \Lambda(m)}{nm} \left| T \cdot \hat{\Phi}(T \log(n/m)) \right|.
\end{aligned}$$

Because $\hat{\Phi}$ is an even function, the last expression is symmetric in m and n . Using this symmetry and the inequality $2 \left| \frac{a_n a_m}{nm} \right| \leq \left| \frac{a_n}{n} \right|^2 + \left| \frac{a_m}{m} \right|^2$, we obtain that the last expression is

$$\leq \sum_{T^2 \leq m \leq x} \frac{|a_m|^2 \Lambda(m)}{m^2} \cdot T \cdot \sum_{T^2 \leq n \leq x} \left| \Lambda(n) \cdot \hat{\Phi}(T \log(n/m)) \right|.$$

Now we use the Brun-Titchmarsh theorem and the fact, that $\hat{\Phi}$ is supported in $[-1, 1]$ to bound the later sum over n .

The summands over n are zero unless $|T \log(n/m)| \leq 1$.

If $n \geq m$, this gives the condition $\log(n/m) \leq \frac{1}{T}$, so $\frac{n}{m} \leq e^{1/T}$. Expanding the exponential, we get

$$\frac{n}{m} \leq \sum_{k=0}^{\infty} \frac{1}{k! \cdot T^k},$$

so

$$\frac{n-m}{m} \leq \frac{1}{T} \sum_{k=1}^{\infty} \frac{1}{k! \cdot T^{k-1}} \ll \frac{1}{T}.$$

Therefore for $n \leq m$, the summands are zero unless $n-m \ll \frac{m}{T}$.

Similarly if $m > n$, we get $\log(n/m) \geq -\frac{1}{T}$, so after exponentiating and expanding, we obtain

$$\frac{n}{m} \geq \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \cdot T^k}.$$

Hence

$$\frac{n-m}{m} \geq -\frac{1}{T} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k! \cdot T^{k-1}} \gg -\frac{1}{T}.$$

Therefore in this case the summands are zero unless $m-n \ll \frac{m}{T}$.

Altogether, we see that the later sum runs over these n with $|m-n| \ll \frac{m}{T}$. Hence, using the Brun-Titchmarsh theorem (Theorem 3.8), and the fact that $\hat{\Phi}(t) \ll 1$, the sum over n is $\ll \frac{m}{T}$, which gives the result $\square$

Note that with an exactly same proof, by only summing over primes instead of the prime powers, we would get the following corollary:

Corollary 6.11. *For any sequence of complex numbers a_n and $T \geq 1$, it holds that*

$$\int_{-T}^T \left| \sum_{T^2 \leq p \leq x} \frac{a_p \log p}{p^{1+it}} \right|^2 dt \ll \sum_{T^2 \leq p \leq x} \frac{|a_p|^2 \log p}{p},$$

where the implied constant does not depend on the sequence a_n .

We are finally ready to bound the two integrals I_1 and I_2 .

Proposition 6.12. *For any $1 \leq k \leq \log \left(\frac{100 \log x}{L(x)} \right)$, we have*

$$I_1 \ll \frac{e^k}{\log x}.$$

Proof. We will use Corollary 6.11 with $T = (\log x)^2$ and the sequence $a_p = \frac{f(p)}{\log(x/p)}$ for $p \in \mathcal{P}_k$ and $a_n = 0$ otherwise. Then we have

$$I_1 = \int_{-T}^T \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^{1+it} \log(x/p)} \right|^2 dt \ll \sum_{p \in \mathcal{P}_k} \frac{\log p}{p (\log(x/p))^2}.$$

Because for $p \in \mathcal{P}_k$, we have $p \geq x^{1-e^{1-k}}$, we can bound

$$(\log(x/p))^{-2} \leq \left(\log \left(\frac{x}{x^{1-e^{1-k}}} \right) \right)^{-2} \leq \frac{e^{2k-2}}{(\log x)^2}.$$

Hence by Corollary 3.6, we have

$$I_1 \ll \frac{e^{2k-2}}{(\log x)^2} \cdot \sum_{x^{1-e^{1-k}} \leq p < x^{1-e^{-k}}} \frac{\log p}{p} \ll \frac{e^k}{\log x}.$$

$\square$

Proposition 6.13. *For any $1 \leq k \leq \log \left(\frac{100 \log x}{L(x)} \right)$, we have*

$$I_2 \ll L(x)^2 e^{-k} \log x.$$

Proof. First we split the interval of integration in I_2 into intervals of length 1. This gives us

$$I_2 \leq \sum_{|h| \leq (\log x)^{2+1}} \int_{1+i(h-1/2)}^{1+i(h+1/2)} \left| \sum_{q \leq x^{1-k}} \frac{f(q) \log q}{q^s} \right|^2 \frac{|F_x(s)|^2}{|s|^2} ds.$$

For t in the interval $[h - 1/2, h + 1/2]$, we have $|1 + it|^2 = 1 + t^2 \gg 1 + h^2$, so $\frac{1}{|s|^2} \ll \frac{1}{1+h^2}$. Furthermore, bounding trivially the contribution of $F_x(s)$ to the integral gives us

$$I_2 \ll \sum_{|h| \leq (\log x)^2 + 1} \frac{1}{h^2 + 1} \sup_{|t-h| \leq 1/2} |F_x(1+it)|^2 \int_{1+i(h-1/2)}^{1+i(h+1/2)} \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right|^2 ds.$$

In order to be able to apply Corollary 6.11, for $t \in [h - 1/2, h + 1/2]$, we write $s = 1 + it = (1 + ih) + iu$, where $u \in [-1/2, 1/2]$. This gives us

$$\begin{aligned} I_2 &\ll \sum_{|h| \leq (\log x)^2 + 1} \frac{1}{h^2 + 1} \sup_{|t-h| \leq 1/2} |F_x(1+it)|^2 \int_{-1/2}^{1/2} \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) q^{-ih} \log q}{q^{1+iu}} \right|^2 du \ll \\ &\ll \sum_{|h| \leq (\log x)^2 + 1} \frac{1}{h^2 + 1} \sup_{|t-h| \leq 1/2} |F_x(1+it)|^2 \sum_{q \leq x^{e^{1-k}}} \frac{\log q}{q} \ll \\ &\ll L(x)^2 e^{1-k} \log x \ll L(x)^2 e^{-k} \log x, \end{aligned}$$

where we used Corollary 3.6. □

We can now put all of our results together to finally prove Halász's Theorem.

Proof of theorem 6.1. Let f be a multiplicative function satisfying $|f(n)| \leq 1$ for all positive integers n .

By Proposition 6.5 and the succeeding remarks, we have

$$\begin{aligned} |S(x)| &\ll \frac{1}{\log x} \sum_{(\log x)^4 < p \leq x/2} \frac{f(p) \log p}{\log(x/p)} \sum_{\substack{nq \leq x/p, \\ q \text{ prime}}} f(n) f(q) \log q + O\left(x \frac{\log \log x}{\log x}\right) \ll \\ &\ll \frac{1}{\log x} \sum_{k=1}^{\log \log x + O(1)} |S_k(x)| + x \frac{\log \log x}{\log x}, \end{aligned}$$

where for $1 \leq k \leq \log \log x + O(1)$, $S_k(x)$ is defined by

$$S_k(x) = \sum_{\substack{pqn \leq x, \\ p, q \text{ primes}, \\ p \in \mathcal{P}_k}} \frac{f(p) \log p}{\log(x/p)} f(n) f(q) \log q,$$

and $\mathcal{P}_k$ is the set of primes in the interval $((\log x)^4, x/2] \cap (x^{1-e^{1-k}}, x^{1-e^{-k}}]$.

We can now split the sum depending on the size of k as

$$\begin{aligned} T_1(x) &:= \frac{1}{\log x} \sum_{k=1}^{\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil - 1} |S_k(x)|, \\ T_2(x) &:= \frac{1}{\log x} \sum_{k=\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil}^{\log \log x + O(1)} |S_k(x)|. \end{aligned}$$

Define the integrals I_1, I_2 (which depend on k) as follows:

$$I_1 = \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{p \in \mathcal{P}_k} \frac{f(p) \log p}{p^s \log(x/p)} ds \right|,$$

$$I_2 = \int_{1-i(\log x)^2}^{1+i(\log x)^2} \left| \sum_{q \leq x^{e^{1-k}}} \frac{f(q) \log q}{q^s} \right|^2 \left| \frac{F_x(s)}{s} \right|^2 ds.$$

By Proposition 6.9, we have that

$$|T_1(x)| \ll \frac{x}{\log x} \sum_{k=1}^{\left\lceil \log\left(\frac{100 \log x}{L(x)}\right) \right\rceil - 1} \sqrt{I_1 I_2} + \frac{x}{\log x} \log \left(\frac{100 \log x}{L(x)} \right).$$

Because $L(x) \gg 1$, we have $\log \left(\frac{100 \log x}{L(x)} \right) \ll \log \log x$. By propositions 6.12 and 6.13, we have that $\sqrt{I_1 I_2} \ll L(x)$, so

$$|T_1(x)| \ll \frac{x L(x)}{\log x} \log \left(\frac{100 \log x}{L(x)} \right) + \frac{x \log \log x}{\log x}.$$

By Proposition 6.7, we have

$$|T_2(x)| \ll \frac{x}{\log x}.$$

Finally, we have

$$|S(x)| \ll T_1(x) + T_2(x) + \frac{\log \log x}{\log x} \ll \frac{x L(x)}{\log x} \log \left(\frac{100 \log x}{L(x)} \right) + \frac{x \log \log x}{\log x}.$$

□

7. Pretentious formulations of other classical theorems

In this chapter, we will show how pretentious methods can be used in the context of other classical theorems from multiplicative number theory. The purpose of this chapter is to give a broad overview, so there are not many proofs included. There might also be some inaccuracies whose correction would require a substantial amount of technicalities. This is, however, a good way to present the main ideas and demonstrate that the pretentious approach can be more intuitive without getting lost in many technical computations.

For a classical treatment of results in this chapter, see for example [Da].

7.1 The pretentious Riemann hypothesis

Recall that Riemann hypothesis states that all zeros of the Riemann zeta function in the critical strip $0 \leq \operatorname{Re}(s) \leq 1$ lie on the critical line $\operatorname{Re}(s) = \frac{1}{2}$. It is the case of many results in analytic number theory that the better zero-free region of $\zeta(s)$, the better estimates we are able to prove. In the pretentious approach, we would like to avoid talking about the analytic continuation, so it might seem impossible to obtain classical results in their full strength. In this section, we are going to show how to formulate the Riemann hypothesis without referring to the analytic continuation.

It is generally easier to find the poles of a complex function rather than its zeros, let us, therefore, define a function $Z(s)$ whose poles will be at the same points as the zeros of ζ . A good candidate is the function $\frac{\zeta'}{\zeta}(s)$, which already showed up in the classical proof of the prime number theorem, but we need to get rid of the pole at $s = 1$. A suitable way to do so seems to define $Z(s)$ as

$$Z(s) := \frac{\zeta'}{\zeta}(s) + \zeta(s).$$

We would now like to express the fact that $Z(s)$ does not have any poles with $\frac{1}{2} < \operatorname{Re}(s)$ without referring to the half-plane $\operatorname{Re}(s) < 1$. This can be done using Taylor's expansion of $Z(s)$. The fact that $Z(s)$ does not have a pole at x means that its Taylor's series converge at this point, and as we cannot cross the 1-line, we would like the series at each point of the half-plane $\operatorname{Re}(s) > 1$ to have a radius of convergence of at least $\frac{1}{2}$.

We can write the Taylor's expansion of $Z(s)$ at the point s_0 as

$$Z(s) = \sum_{k=0}^{\infty} (s - s_0)^k \frac{Z^{(k)}(s_0)}{k!}.$$

This series is convergent for $|s - s_0| < \frac{1}{2}$, if the following estimate holds for every $k \geq 0$ and s_0 with $\operatorname{Re}(s_0) > 1$:

$$\frac{Z^{(k)}(s_0)}{k!} \ll_{s_0} 2^k.$$

We can even uniformly bound the implicit constant in terms of $\text{Im}(s_0)$. This gives us the following statement of the *pretentious Riemann hypothesis*:

Conjecture (Pretentious Riemann hypothesis). *For every $k \geq 0$, $\varepsilon > 0$ and $s = \sigma + it$ with $\sigma > 1$, it holds that*

$$\frac{Z^{(k)}(s)}{k!} \ll_{\varepsilon} 2^k (1 + |t|^{\varepsilon}).$$

Let us note that if the Riemann hypothesis is true, it implies a stronger estimate, in particular that for any $k \geq 1$ and $s = \sigma + it$ with $t, \sigma \geq 1$, it holds that

$$\frac{Z^{(k)}(s)}{k!} \ll 2^k \log t.$$

We will show how an estimate on the derivatives of $Z(s)$ can give us a good error term in the prime number theorem, thus proving that the pretentious Riemann hypothesis implies the classical Riemann hypothesis.

Proposition 7.1. *If the Pretentious Riemann hypothesis holds, then for any $\varepsilon > 0$, we have*

$$\sum_{n \leq x} \Lambda(n) = x + O_{\varepsilon}(x^{\frac{1}{2} + \varepsilon}).$$

Proof. The function $Z(s)$ is the Dirichlet series of the arithmetic function $f(n) := -\Lambda(n) + 1$, therefore $Z^{(k)}(s)$ is the Dirichlet series of the function

$$f_k(n) := (-1)^{k+1} (\log n)^k (\Lambda(n) - 1).$$

Note that estimating the error term in the prime number theorem is the same as estimating the sum

$$\left| \sum_{n \leq x} f(n) \right| = \sum_{n \leq x} |\Lambda(n) - 1|.$$

We will estimate the sum $\left| \sum_{n \leq x} f_k(n) \right|$ using Perron's formula, use partial summation to obtain an estimate for $\left| \sum_{n \leq x} f(n) \right|$ and pick an optimal value of k to minimize the error term.

By Perron's formula, we have

$$\left| \sum_{n \leq x} f_k(n) \right| = \frac{1}{2\pi} \left| \int_{c-i\infty}^{c+i\infty} Z^{(k)}(s) \cdot \frac{x^s}{s} ds \right|.$$

Setting $c = 1 + \frac{1}{\log x}$, truncating the formula at T and using the classical error term (as in Theorem 4.3.1. in [Ko1]), we obtain

$$\left| \sum_{n \leq x} f_k(n) \right| = \frac{1}{2\pi} \left| \int_{1+\frac{1}{\log x}-iT}^{1+\frac{1}{\log x}+iT} Z^{(k)}(s) \cdot \frac{x^s}{s} ds \right| + O\left(\frac{x(\log x)^{k+2}}{T} + (\log x)^{k+1} \right).$$

We can now use the assumption on $|Z^{(k)}(s)|$ to get

$$\left| \sum_{n \leq x} f_k(n) \right| \ll_{\varepsilon} x 2^k k! \int_{-T}^T \frac{1 + |t|^{\varepsilon}}{1 + |t|} dt + \frac{x(\log x)^{k+2}}{T} + (\log x)^{k+1}.$$

We can estimate the integral as

$$\ll_{\varepsilon} \int_1^T |t|^{\varepsilon-1} \ll_{\varepsilon} T^{\varepsilon}.$$

Using Proposition 1.7, exponentiating the estimate for $\log(k!) = \sum_{n \leq k} \log n$, we obtain $k! \ll \left(\frac{k}{e}\right)^k$. Hence

$$\left| \sum_{n \leq x} f_k(n) \right| \ll_{\varepsilon} x \cdot T^{\varepsilon} \left(\frac{2k}{e}\right)^k + \frac{x(\log x)^{k+2}}{T} + (\log x)^{k+1}.$$

By partial summation, we have

$$\left| \sum_{n \leq x} f(n) \right| \ll_{\varepsilon} x \cdot T^{\varepsilon} \left(\frac{2k}{e \log x}\right)^k + \frac{x(\log x)^2}{T} + \log x + k \int_2^x \frac{2^k k! T^{\varepsilon}}{(\log t)^{k+1}} + \frac{\log t}{T} + \frac{1}{t} dt.$$

The integral is asymptotically smaller than the other terms, so we have

$$\sum_{n \leq x} |f(n)| \ll_{\varepsilon} x \cdot T^{\varepsilon} \left(\frac{2k}{e \log x}\right)^k + \frac{x(\log x)^2}{T} + \log x.$$

It now suffices to choose an optimal k to minimize the first error, and then choose a suitable T . Setting $k = \left\lfloor \frac{\log x}{2} \right\rfloor$, we have

$$\left(\frac{2k}{e \log x}\right)^k \leq \left(\frac{1}{e}\right)^{\left\lfloor \frac{\log x}{2} \right\rfloor} \ll x^{-\frac{1}{2}},$$

so we obtain

$$\left| \sum_{n \leq x} f(n) \right| \ll_{\varepsilon} x^{\frac{1}{2}} \cdot T^{\varepsilon} + \frac{x(\log x)^2}{T} + \log x.$$

Setting $T = x$ gives us the final estimate

$$\left| \sum_{n \leq x} f(n) \right| \ll_{\varepsilon} x^{\frac{1}{2}+\varepsilon}.$$

□

Using this idea of bounding the derivatives of $\zeta(s)$, Koukoulopoulos [Ko2] was able to obtain a pretentious proof of the prime number theorem in its strongest known form (see Section 7.4), thus avoiding the difficulties arising from the poor error in Halász's Theorem. The best error known is $O\left(x \exp\left(-C \frac{(\log x)^{\frac{3}{5}}}{(\log \log x)^{\frac{1}{5}}}\right)\right)$ for some positive constant C , is due to Korobov and Vinogradov and was obtained from their results on exponential sums. It might seem strange that such a bizarre formula arises in two different proofs. This happens because the two proofs are fundamentally similar, both use the results about exponential sums, with the difference that Koukoulopoulos avoids using the analytic continuation.

7.2 Primes in arithmetic progressions and Dirichlet characters

An important topic in analytic number theory is the distribution of prime numbers in arithmetic progressions. It was Dirichlet who first showed that whenever $(a, q) = 1$, then there are infinitely many primes which are $\equiv a \pmod{q}$. A stronger form of this theorem states that the primes are evenly distributed among different arithmetic progressions with the same modulus, so if we denote by

$$\pi(x; q, a) := \{p \leq x : p \text{ is a prime and } p \equiv a \pmod{q}\},$$

then

$$\pi(x; q, a) \sim \frac{1}{\varphi(q)} \cdot \frac{x}{\log x}.$$

The idea of Dirichlet was to replicate the proof of Euler and show that

$$\sum_{p \equiv a \pmod{q}} \frac{1}{p} = \infty.$$

Since the sum runs over primes in an arithmetic progression, it is very hard to work with it. To deal with this problem, Dirichlet invented Dirichlet Characters and the theory of L -functions, which is often considered as the beginning of analytic number theory. These characters enabled Dirichlet to transform sums over arithmetic progressions into sums over all prime numbers.

A function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ which is not constantly 0 is called a Dirichlet character modulo q if it is periodic with period q , completely multiplicative and $\chi(n) = 0$ whenever $(n, q) > 1$. As we will see, the characteristic function of each arithmetic progression can be expressed as their linear combination, and due to their multiplicativity, it is easier to work with them.

Because of periodicity, we can consider a character χ modulo q as a homomorphism $(\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}$. If q is a prime power, the group $(\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic of order $\varphi(q)$, so each character modulo q is determined by its value at a generator, and that value must be a $\varphi(q)^{th}$ root of unity. There are therefore $\varphi(q)$ characters modulo q . Point-wise multiplication of two characters modulo q gives us another character modulo q and it can be easily showed that the set of characters with this operation forms a group denoted by $\widehat{(\mathbb{Z}/q\mathbb{Z})^\times}$, with unit element the so-called trivial character χ_0 defined as $\chi_0(n) = 1$ if $(n, q) = 1$ and 0 otherwise. The inverses then coincide with complex conjugation, so $(\chi)^{-1}(n) = \overline{\chi(n)} := \overline{\chi(n)}$. It is also easy to prove that $\widehat{(\mathbb{Z}/q\mathbb{Z})^\times} \simeq (\mathbb{Z}/q\mathbb{Z})^\times$. Using the Chinese Remainder Theorem, these results can be analogically extended to all moduli q .

For every $q > 2$, there is a unique non-trivial character modulo q which takes only real values. This character is called quadratic.

Note that whenever χ is a non-trivial character, there always exists an integer a such that $\chi(a) \neq 0, 1$. On the other hand, if $n \not\equiv 1 \pmod{q}$ and $(n, q) = 1$, then we can find a character χ modulo q such that $\chi(n) \neq 0, 1$.

The following proposition is referred to as the orthogonal relations of characters.

Proposition 7.2 (Orthogonal relations). 1. Let χ be a Dirichlet character modulo q . Then

$$\sum_{n=0}^{q-1} \chi(n) = \begin{cases} \varphi(q) & \text{if } \chi = \chi_0 \text{ is trivial,} \\ 0 & \text{otherwise.} \end{cases}$$

2. Let $q, n \in \mathbb{Z}$. Then

$$\sum_{\chi \pmod{q}} \chi(n) = \begin{cases} \varphi(q) & \text{if } n \equiv 1 \pmod{q}, \\ 0 & \text{otherwise,} \end{cases}$$

where the sum runs over all characters χ modulo q .

Proof. We begin by the first assertion.

If χ is trivial, then $\chi(n) = 1$ if $(q, n) = 1$ and 0 otherwise, so the equality holds.

Suppose that χ is not trivial and pick an arbitrary integer a such that $\chi(a) \neq 0, 1$. Then, using the complete multiplicativity and periodicity, we have

$$\chi(a) \sum_{n=0}^{q-1} \chi(n) = \sum_{n=0}^{q-1} \chi(an) = \sum_{(an) \pmod{q}} \chi(an) = \sum_{n \pmod{q}} \chi(n) = \sum_{n=0}^{q-1} \chi(n).$$

Since $\chi(a) \neq 0, 1$, the only possibility is that

$$\sum_{n=0}^{q-1} \chi(n) = 0.$$

Now we prove the second assertion.

If $n \equiv 1 \pmod{q}$, then by periodicity, $\chi(n) = 1$ for every character χ modulo q . Because there are $\varphi(q)$ such characters, the claim follows.

Now suppose that $n \not\equiv 1 \pmod{q}$. If $(n, q) > 1$, then $\chi(n) = 0$ for every character χ modulo q , so we may further assume that $(n, q) = 1$. In that case, there exists a character ψ modulo q such that $\psi(n) \neq 0, 1$. Then we have

$$\psi(n) \sum_{\chi \pmod{q}} \chi(n) = \sum_{\chi \pmod{q}} (\psi\chi)(n) = \sum_{\chi' \pmod{q}} \chi'(n) = \sum_{\chi \pmod{q}} \chi(n),$$

where the second equality holds because multiplication by ψ is an isomorphism of the group $(\widehat{\mathbb{Z}/q\mathbb{Z}})^\times$. Because $\psi(n) \neq 0, 1$, the only possibility for the equality to hold is that

$$\sum_{\chi \pmod{q}} \chi(n) = 0.$$

□

We are now going to show how to express the characteristic function of arithmetic progressions as a linear combination of characters.

Proposition 7.3. Let $q, a \in \mathbb{Z}$ be coprime integers. Then

$$\frac{1}{\varphi(q)} \sum_{\chi \pmod{q}} \chi(n) \overline{\chi}(a) = \begin{cases} 1 & \text{if } n \equiv a \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. First suppose that $n \equiv a \pmod{q}$. Then $\chi(n) = \chi(a)$ for any character χ modulo q , so we have

$$\sum_{\chi \pmod{q}} \chi(n) \bar{\chi}(a) = \sum_{\chi \pmod{q}} \chi(n) \bar{\chi}(n) = \sum_{\chi \pmod{q}} \chi_0(a) = \varphi(q),$$

where we used Proposition 7.2 and the fact that $\chi, \bar{\chi}$ are inverse to each other in the group of characters.

Now suppose that $n \not\equiv a \pmod{q}$. If $(n, q) > 1$, $\chi(n) = 0$ for each character χ modulo q so we may suppose that $(n, q) = 1$. Then there is an integer $b \not\equiv 1 \pmod{q}$ which is coprime to q and such that $n \equiv ab \pmod{q}$. Then we have

$$\sum_{\chi \pmod{q}} \chi(n) \bar{\chi}(a) = \sum_{\chi \pmod{q}} \chi(ab) \bar{\chi}(a) = \sum_{\chi \pmod{q}} \chi(a) \chi(b) \bar{\chi}(a) = \sum_{\chi \pmod{q}} \chi(b) = 0$$

by Proposition 7.2. □

One can think of the last proposition as an analogy of Fourier transform on finite groups.

We can now use Euler's strategy and use the last proposition to obtain

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \frac{1}{p} = \sum_{p \leq x} \frac{1}{p \cdot \varphi(q)} \sum_{\chi \pmod{q}} \chi(p) \bar{\chi}(a).$$

The second sum looks more complicated, but we do not sum over an arithmetic progression anymore! To work with this sum, Dirichlet introduced L -functions defined for a character χ as

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

which is, for a non-trivial character, convergent when $\operatorname{Re}(s) > 0$ and absolutely convergent when $\operatorname{Re}(s) > 1$. Due to the total multiplicativity of χ , we can express $L(\chi, s)$ as an Euler product when $\operatorname{Re}(s) > 1$ as

$$L(\chi, s) = \prod_p \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right).$$

It shows up that a crucial and surprisingly difficult step in the proof of Dirichlet Theorem is to prove that $L(\chi, 1) \neq 0$ for each character χ . This is not hard to establish when χ is complex (by which we mean that $\chi(n)$ is not real for some n), because if $L(\chi, 1) = 0$, then also $L(\bar{\chi}, 1) = 0$, but we can show that $L(\chi, 1) = 0$ for at most one character χ . This phenomenon with “at most one exceptional character” arises often in number theory, we will see this for example in our discussion about Siegel's zero.

Proving that $L(\chi, 1) \neq 0$ for any real character χ is substantially more complicated. Dirichlet gave several proofs, the first one contained the famous class number formula, which gives a relation between $L(\chi, 1)$ for a real character χ , the class number and other invariants of a quadratic field.

We already know that the distribution of zeros of Riemann zeta function and the distribution of prime numbers are closely related. Similarly, when we

want to find out something about the distribution of prime numbers in arithmetic progressions, such as whether they are evenly distributed among different residue classes, we need to investigate the distribution of zeros of Dirichlet L -functions. The so-called Generalized Riemann Hypothesis states that all zeros of Dirichlet L -functions in the critical strip have their real part equal to $1/2$. Because the Euler's product converges when $\operatorname{Re}(s) > 1$, we see that there are no zeros in this half-plane. Similarly to the case of Riemann zeta function, one can prove that $L(\chi, s) \neq 0$ when $\operatorname{Re}(s) = 1$. This leads to the stronger form of Dirichlet's Theorem that the primes are evenly distributed among different residue classes.

Once the Dirichlet characters are defined, it is possible to prove Dirichlet Theorem (in its stronger form) using pretentious methods similarly as the prime number theorem. First, we can restate it using the Von Mangoldt's function, so our goal is to show that

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n) \sim \frac{x}{\varphi(q)}.$$

Similarly to the prime number theorem, we can show that the primes are equidistributed modulo q if and only if

$$\frac{1}{x} \sum_{n \leq x} \mu(n) \chi(n) = 0$$

for every Dirichlet character χ modulo q . This allows us to use Halász's Theorem to obtain a pretentious proof. The goal is therefore to show that the function $\mu(n)\chi(n)$ is not n^{it} -pretentious for any real t . For $t \neq 0$, we have already established this in Proposition 5.15, so it suffices to show that $\mathbb{D}(\chi\mu, 1, \infty)$ diverges. If χ is complex, this can be done, because then $\operatorname{Re}(\chi(n)\mu(n))$ is often small. In the case of a real character χ , this turns out to be equivalent to $L(\chi, 1) \neq 0$, so this approach does not allow us to circumvent the difficulty of the classical proof.

7.3 Pretentiousness is repulsive

To verify the Riemann hypothesis, researchers computed the first several billion zeros of $\zeta(s)$ and confirmed that all of them really lie on the critical line. A potential counterexample would, therefore, need to be very high in the critical strip. Such a conclusion is not possible to be done in the case of the Generalized Riemann hypothesis, as it would be necessary to compute the zeros of infinitely many possible L -functions.

Indeed, we cannot so far even eliminate the possibility that there is a point $\beta \in (0, 1)$ very near to 1 such that $L(\chi, \beta) = 0$ for some character χ . Such a hypothetical point is referred to as a *Siegel zero* and is an important topic in analytic number theory. Even though mathematicians have so far not been able to rule out their existence, they were able to obtain some so-called *repulsive* results, which state that if $L(\chi, \rho) = 0$ for some ρ close to 1 and character χ modulo q , then the function $L(\chi, s)$ or even $L(\psi, s)$ for a different character ψ modulo q' close to q cannot have any zeros near ρ . Similarly as in the case of Dirichlet Theorem, we can conclude that a Siegel zero can exist only in the case of quadratic characters. The first such result is due to Landau and it shows that

if Siegel zeros exist, it is very rare. We will show how one can prove these results by the pretentious techniques and we will see that the proofs separated of the technical computations are very intuitive.

Let us be a little more precise. Fix a (small) constant $c > 0$. If χ is a character modulo q , we say that ρ is a Siegel zero of $L(\chi, s)$ if $\rho \geq 1 - \frac{c}{\log q}$ and $L(\chi, \rho) = 0$. Because we can bound the derivative $L'(\chi, s)$, this is equivalent to $L(\chi, 1)$ being small, in particular $L(\chi, 1) \ll_c \frac{1}{\log q}$, so elimination of the Siegel zero is equivalent to obtaining strong lower bounds on $L(\chi, 1)$.

We already know from the previous section that Dirichlet proved that $L(\chi, 1) \neq 0$ for any χ . From his class number formula, one can obtain a lower bound

$$L(\chi, 1) \gg \frac{1}{\sqrt{q}}.$$

Siegel improved this estimate in 1936 and showed that for every $\varepsilon > 0$, we have

$$L(\chi, 1) \gg_{\varepsilon} q^{-\varepsilon}.$$

Let us now show how to prove that there is at most one character χ which has a Siegel zero.

Because

$$L(\chi, 1) = \prod_p \left(\frac{1}{1 - \frac{\chi(p)}{p}} \right),$$

this value being small is equivalent to $\chi(p) = -1$ for many primes p . This can be made precise and one can show that this happens if and only if χ pretends to be μ (since $\mu(p) = -1$ for *every* prime p). Therefore if we had two such characters χ, ψ , then we can use the triangle inequality and obtain

$$\mathbb{D}(\chi\bar{\psi}, 1, \infty) = \mathbb{D}(\chi, \psi, \infty) \leq \mathbb{D}(\chi, \mu, \infty) + \mathbb{D}(\mu, \psi, \infty) \ll 1.$$

However, if $\chi \neq \psi$, then $\chi\bar{\psi}$ is not trivial and one can easily show from the orthogonality relations of characters that $\sum_{n \leq x} \chi\bar{\psi}(n) = O(1)$. This would be a contradiction, so there can be at most one such exceptional character – the real character modulo q . One can analogically extend this result to show that for any Q , there is at most one exceptional character modulo q with $Q < q \leq 2Q$.

7.4 Other deep results about the distribution of prime numbers

The existence of exceptional zeros is a delicate problem and its solution would have many applications. It is related to many other well-known questions in number theory, for example, Heath-Brown [He] proved in 1983 that the twin primes conjecture holds unless there are infinitely many Siegel zeros. Even though most researchers believe in the Generalized Riemann hypothesis which excludes the exceptional characters, one can sometimes obtain easier or even stronger results if one assumes their existence.

The proofs of deep theorems are sometimes divided into two cases, depending on whether the exceptional zero exists or not. One of these is Siegel's lower

bound $L(\chi, 1) \gg_{\varepsilon} q^{-\varepsilon}$ mentioned in the previous section. It is also the reason we are not able to determine the implied constant with the current methods because if the Generalized Riemann hypothesis is false, we can only find the implied constant once we have a counter-example. It therefore seemed preposterous to expect that such a result and its consequences could be proved by the pretentious techniques, until Koukoulopoulos [Ko2] gave a proof together with the strongest version of the prime number theorem. He proved a “converse” to Halász’s Theorem, that if the mean value of a multiplicative function f in the unit disc tends to zero faster than $x/(\log x)^A$ for any A , as in the case of μ or $f = 0$, then $f(p)$ must be in average equal to 0 or -1 . This result allows one to bound mean values more effectively than only using Halász’s Theorem.

Siegel’s estimate allows one not only to show that the primes are equidistributed among different residue classes but to bound the error uniformly over all arithmetic progressions with moduli bounded by a fixed power of $\log x$, a result which is known as the Siegel-Walfisz Theorem.

Theorem 7.4. *Let $A > 0$. Then there exists a positive constant C_A such that for every $q \leq (\log x)^A$ and a such that $(a, q) = 1$, we have*

$$\sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n) = \frac{x}{\varphi(q)} + O\left(xe^{-C_A(\log x)^{3/5}(\log \log x)^{-1/5}}\right)$$

Based on numerical computations, researchers believe that the equidistribution in Siegel-Walfisz theorem holds for all $q \leq x^{1-\varepsilon}$ (with a much stronger error term). This is in fact even stronger than what we would be able to obtain from the Generalized Riemann hypothesis, which only gives $q \leq x^{1/2-\varepsilon}$.

It seems that such results are out of reach with our current knowledge. However, there is the great theorem of Bombieri and Vinogradov which gives us that the above estimate holds in average for *almost all* $q \leq x^{1/2-\varepsilon}$. This result can often serve as a useful substitution for the Generalized Riemann hypothesis and thanks to the proof of Koukoulopoulos and a pretentious formulation of the large sieve, it is possible to be proved using purely pretentious techniques. We can, therefore, often show that the strong estimates proven under the assumption of Generalized Riemann hypothesis at least hold in most cases.

Another deep theorem in analytic number theory is Linnik’s Theorem, which states that there are two positive constants c, L such that every arithmetic progression with $(a, q) = 1$ contains a prime which is $\leq c \cdot q^L$. The current record on the size of L is $L = 5$, even though it is conjectured that one can take $L = 2$ and $c = 1$.

This result was proved by Linnik in 1944 and was considered to be one of the hardest (if not *the* hardest) theorems in the classical analytic number theory. Friedlander and Iwaniec [FI] found a new technically elementary proof, which inspired Granville and Soundararajan to develop their own pretentious proof which ended up being even simpler and shorter. A further refinement of their approach can be found in [GHS2], where they generalize the method we saw in chapter 6 to be applicable for arithmetic progressions and small intervals.

7.5 New results

The purpose of developing the pretentious approach was of course not to give different proofs of classical theorems, but to obtain techniques which are strong and flexible enough to give some new results.

The first such result is the improvement of the estimates of large character sums (see [GS3] or [BGGK]). If χ is a character modulo q , we define $M(\chi)$ as

$$M(\chi) := \max_x \left| \sum_{n \leq x} \chi(n) \right|.$$

The size of $M(\chi)$ is related to many other problems and good bounds on its size would have many interesting consequences. Pólya and Vinogradov showed independently in 1918 that

$$M(\chi) \ll \sqrt{q} \log q$$

and this bound was not improved (beside the size of the implied constant) until 2005, when Granville and Soundararajan showed that for χ of odd order, it holds that

$$M(\chi) \ll \sqrt{q} (\log q)^{c+o(1)}$$

for some explicit constant $0 < c < 1$ depending on the order of χ . Moreover, the authors used the pretentious language to reveal more structure in the character sums, which was previously unnoticed.

The case of characters of even order is closely tied to the problem of finding least quadratic non-residue, which was recently investigated and improved in [XL], where the author builds upon other results about the structure of character sums of Granville, Soundararajan and others.

Matomäki and Radziwiłł [MR] used pretentious techniques in their treatment of sign changes of coefficients of holomorphic Hecke cusp forms.

Soundararajan [So1],[So2] and Holowinsky [HS] applied the pretentious methods to resolve Arithmetic Quantum Unique Ergodicity, a famous open problem.

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